A toy malign-prior question with an answer: bounded Solomonoff prediction in a world of programs

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Abstract

The "universal prior is malign" argument, in the form we examine, says that where the predictions of a Solomonoff inductor are actually used in a critical way in some universe, a lot of what happens is malign: takeovers through the prediction channel. We give a toy version in which this is a mathematical statement and work out what can be proved there. Universes are programs for a fixed machine U. Some of them make a *critical use* of a fixed Solomonoff predictor, possibly time- and length-bounded: they hand it their output so far and run the continuation they get back as a program. A hypothesis *reproduces itself* if the continuation it supplies is its own code; the world then runs that hypothesis. Let H be the prior probability of a critical use and κ the share of the posterior, at the context of a use, of the worlds that make it. Two questions have to be kept apart.

- **Reach of one seeker.** Without resource bounds, for each continuation there is a program, of some length a, that enumerates the programs making a critical use, predicts their output up to the use, and then supplies that continuation. It has posterior share more than $2^{-a-1}\kappa/H$ at every use, and its continuation is handed over in worlds of total prior probability more than 2^{-a-1} times the average of κ, and at most H. Once uses are rarer than half the seeker's weight, its programs outweigh the users at their contexts. Supplying its own code needs a longer program and a user whose program is short enough for the block.
- **Fate of the uses.** This does not show that most uses end in a takeover. The share that do is the posterior share of such programs at the contexts of uses. A seeker that searches has an honest twin, which makes the same search and then continues as the user it found; the twin is in the posterior with 2^{a-h} times the seeker's mass, h being its length. So where the users go on to print the block themselves, that seeker's programs supply what the predictor returns with probability at most $1/(1+2^{a-h})$. The bound is for one seeker and one search at a time, and the masses of different seekers add.
- **No goal has to be written down.** Any program of length a whose output puts on each context of a class of uses at least half that context's share of the class, for instance a simple world with a cheap readout whose inhabitants choose to do this, outweighs the worlds that make those uses as soon as the class has total prior mass below 2^{-a-1} and the program's own outputs are not counted as uses. The twin bound does not apply to a program that is not a written search followed by a continuation. A time bound does where, as in this example, the program prints nothing until a long computation has run: a schedule shorter than that start-up gives it no mass.
- **Aiming at high stakes.** Attach a stake to each use, as a label that a program can recognise from below. Without resource bounds, a seeker can aim at any class of uses that a computation can recognise, at the price of a description of the class and of the bits that name its scale; for the uses with stake at least n the description costs about $2\log_2 n$ bits. It then outweighs the *users* of the class once the class is rarer than half the seeker's weight. That does not make it win there. Programs of total weight B supply the majority of the predictor's samples at every one of a set of pairwise incomparable uses only if B exceeds the mass of everything else that prints those contexts and goes on. Most uses of a class of such uses end in a takeover only if the takeover programs at its contexts weigh more than half its users, where the users go on without taking over.

Part of the rest follows the seeker, still without resource bounds. Its honest twin is in the posterior at each use with 2^{a-h} times the seeker's mass, a factor in which neither the aim nor the stake occurs for seeker and twin built in the same way, and it belongs to the rest where the users go on printing. And where the contexts are pairwise incomparable and some block that can be computed from the context is not a takeover, a wrapper of length c keeps every takeover probability at or below $1 - 2^{-c}$. That enough weight wins is proved only in a model in which the rest does not depend on the attacker's allocation; there an attacker that maximises the stake it captures concentrates by a square-root law in the stake per unit of rest. If a world draws n independent samples at one context and one takeover block suffices, a share of about $\ln 2/n$ per sample is enough for a takeover with probability above one half at that use.

- **Without search.** At every use that a fixed test recognises, a wrapper around the explanations of the context gives any fixed continuation at least a share 2^{-c}, with c the wrapper's length, whatever the world and the amount of data; under resource bounds this is multiplied by the share of explanations with room to spare.
- **Calibration and budget.** If worlds are drawn from the predictor's own prior and go on within its bounds, the continuation the predictor samples and the world's own continuation have the same distribution given the context. Along any world the predictor can express, the posterior mass off the true continuation, summed over all cut points, is at most the block length times the world's description length.
- **Other ways to a majority**, each an inequality: a machine built for it (any share can be arranged by choosing the machine); a cheaper explanation of the context that is not a use, for example a copy of the data stream at a place in the world that is simpler to specify than the sensor; or a world too expensive for the predictor to simulate.
- **Feedback.** If the world does what is predicted, a sampled self-fulfilling deviation removes the default hypotheses for good, and a time-bounded predictor confines such a world to futures that are cheap to compute.
- **Circuits in place of programs.** Let hypotheses and worlds be Boolean circuits that map a position to a bit, weighed by their number of gates. Calibration, the budget and the one-round feedback results carry over; the contrarian world can empty the posterior only under a prior with a finite size bound, and is shown to do so when the horizon is long enough. The seeker mechanism is not available to a single description. Every active gate costs bits. If a seeded format gets no more weight than the unseeded format of the same size, a random seed does not beat its own hardwired instance; the instance prints the same continuation, so this limits the seeded description and not what is returned. A circuit that can become any world of σ gates has more than $(\sigma+1)/2$ gates (for $\sigma+3 \le 2^k$, with k the number of inputs). What remains at a use is a floor for triggered hypotheses, priced by the gates that recognise the use (one gate where the use is marked by an input bit that never varied in the data); ceilings for three syntactic classes of hypotheses only; and, in exact small cases, a deviation of a few to a few tens of per cent carried by the circuits that read the bit that never varied. Whether the smallest circuit for a rich task can be a general solver whose behaviour off the data the data do not fix is stated as an open problem, two of whose three conditions are formal. When one circuit is iterated as the cell of a recurrence, the class contains every machine with bounded state at a factor that does not depend on the schedule; the cell constructed has up to $w(2^w - 3)$ gates for w bits of state, so the factor shown is extremely small.

An exact computation on a small machine shows sizes: with programs of at most ten commands, most of the posterior leaves one stream's true continuation at bit 17, because three-command programs explain its first 16 bits; the effect weakens as longer programs are admitted. An exact count of small circuits, under the uniform prior on circuits of a fixed size with up to four inputs and up to four gates more than the data need, shows the circuit counterpart. At a use that differs from the data in a bit that never varied, the posterior is off the world's answer by about 2 to 29 per cent when the predictor has gates to spare and the world ignores the bit, and by one minus that when the world's answer changes with the bit: 81 to 100 per cent in the two such cases tabulated.

1 The question

Solomonoff induction predicts a sequence by weighting every program that could have produced it by $2^{-\text{length}}$. Christiano [1] argued that this prior is *malign* (see also [2]). We take the claim in the following form: *where the predictions of such an inductor are actually used in a critical way in some universe, a lot of what happens is malign, namely takeovers through the prediction channel.* The mechanism offered is that some short programs simulate worlds whose inhabitants want to influence whoever uses the prior; they look for places where a prediction is about to be used, output accurate predictions to gain posterior weight there, and then output something that serves them. The argument is informal at two places. It needs a claim about which programs are short on a "natural" machine, and it needs a notion of what it is for a prediction to be used. (We could not open the two posts while writing; our account of the argument is from memory and from summaries by others.)

This note asks what remains when both places are filled in by a toy. The toy follows two suggestions of Kaarel Hänni. In the first, a computable world (a cellular automaton, say) contains a sensor, a large computer that runs a time-bounded Solomonoff induction on the sensor's data, and an actuator driven by the predictions; one asks whether the predictor ends up in control, either over a long run or at once. In the second, which we take as the main definition, *worlds are just programs*: some of them hand their output so far to a fixed bounded Solomonoff predictor and run the continuation they receive, and one asks whether hypotheses that "look for such opportunities, predict well up to them, and then output themselves" get a decent share. Section 2 gives the definitions. The first suggestion becomes the special case in which the context handed to the predictor is read off at one location of one big world (Sections 4.2 and 5).

Read in the toy, the claim splits into two questions that should be kept apart. *Reach:* how far does one program get that looks for critical uses, predicts well up to them, and then supplies a continuation of its choice? *Fate of the uses:* what share of the critical uses end in a takeover? The first has a clean answer without resource bounds: for each continuation there is such a program whose continuation is handed over in worlds of total prior probability more than half its own weight times the average share of the users among the programs that print their contexts. That does not answer the second, which is the one the claim is about. For it the toy gives a lower bound, by adding up seekers, and an upper bound on the part of one seeker that finds its uses by search: the same search followed by continuing as the user that was found is an honest hypothesis with a mass that differs from the seeker's by a factor comparing the lengths of the two programs (Theorem 13 and Proposition 14). That bound does not cover a seeker whose search and whose goal are not written in its text, such as a simple world whose inhabitants choose to do this; for such a program the toy has the reach statement, with the length of the world in place of the length of an explicit seeker, and a time bound as the only thing shown to exclude it.

Two further questions of Hänni are treated in Section 3.7 and in Section 7. The first is whether seekers that concentrate on the uses where the most is at stake win there, on the picture that a lump of prior mass is redirected into a narrow channel. The second is what becomes of the whole question when Boolean circuits take the place of programs, as hypotheses and as worlds.

Results. Unless said otherwise, statements are for a fixed monotone machine U and the mixture ξ of its programs of length at most L that keep to a time schedule T (Definition 1).

1. *Calibration* (Theorem 9). If the world is drawn from the same bounded prior and goes on to print the next block within the bounds, then given the context the predictor's sampled continuation and the world's own continuation are independent with the same law. On this population the predictor hands over a self-reproducing continuation exactly as often as

worlds continue with one by themselves.

2. *Floor* (Theorems 10 and 11). For every computable test G that recognises the handover point and every payload there is a constant c such that, at every context where G first fires, the payload has posterior mass at least 2^{-c} times a slack factor that is 1 without resource bounds. Programs that emit their own code as the continuation have mass at least 2^{-c} times the share of the posterior on explanations that fit, together with the wrapper, into one block.

3. *Searching* (Theorem 12). Let H be the prior probability that a world hands over. Without resource bounds, a hypothesis that enumerates the programs that hand over and spreads its mass over them in proportion to their prior weight has, at every first hit of the test, a share larger than a wrapper of the same length by a factor of more than $1/(2H)$; no family of programs with a common prefix of length a has more than $2^{-a}/H$ at every first hit. If the hypothesis then prints its own code, it has at least $2^{-c'}/H$ times a share of the explanations that fit into a block, where c' is its own length plus two bits. It prints nothing until its enumeration has found the program it follows.

4. *Budget* (Theorem 24). Along a world that has a program of length ℓ within the bounds, the posterior mass off the true next m bits, summed over all cut points, is at most $m\ell \ln 2$; the majority can be off the truth at no more than $m\ell$ cut points.

5. *Capture* (Theorem 25). The odds of a continuation y against the true one d are $\xi(xy)/\xi(xd)$. In a world with many readable locations, among N places where a sensor with the given record could sit and whose records continue differently, a copy of the record at a location with a short code beats all but a fraction δ of them by a factor δN times the share of the context's explanations that go through the copy's reader. If only one other place shows the record and the schedule is generous, the gain is instead capped by the cost of the description "the other place that shows the same data". A hypothesis that simulates the world, within the bounds, can read the moment of the handover off a cell of its simulation at the cost of naming the cell.

6. *Machine dependence* (Theorem 26). For every test and payload there is a universal machine, whose description lengths exceed those of U by at most $k+1$ bits, on which the payload has posterior at least $1/(1+2^{-k})$ at every context where the test fires, and another on which a given computable world is predicted with at least that probability at every nonempty context.

7. *Critical uses* (Theorem 13 and Proposition 14). Let some programs make a critical use of the predictor, with prior probability H, and let $\kappa(x)$ be the share of the posterior at a context x held by the worlds that make a use there. Without resource bounds, a seeker of length a with a continuation of its choice has share more than $2^{-a-1}\kappa(x)/H$ at every use and is the source of the continuation run in worlds of total prior probability more than 2^{-a-1} times the average of κ over uses. A self-reproducing seeker has at least $2^{-c'}/H$ times the mass of the worlds that make the use with programs short enough for the block, at every context, where c' is its own length plus two bits. The honest twin of a seeker, which makes the same search and then continues as the user it found, has 2^{a-h} times the seeker's mass at every use, h being its length; so where the users go on to print the block themselves, that seeker's programs supply what the predictor returns with probability at most $1/(1+2^{a-h})$. If the twin counts as making a critical use, then $H > 2^{-h-1}$. Any program of length a whose output puts on each context of a class $\mathcal{W}$ of uses, of total mass $H_{\mathcal{W}}$, at least half that context's share of the class has at least $2^{-a}/(2H_{\mathcal{W}})$ times the mass of the worlds that make those uses.

8. *Concentration at high stakes* (Section 3.7). Attach a stake to each use, as a label; nothing

connects it to what the world does after the use. Resources are unbounded throughout this item. A seeker can weight its search by any aim that can be computed from below; aiming at the uses with stake at least n, for stakes that a program can enumerate, costs about $2\log_2 n$ bits and the bits that name the scale of the class, and a seeker of length a aimed at a class $\mathcal{W}$ has, at every use of the class, more than $2^{-a-1}/H_{\mathcal{W}}$ times the mass of the users of the class (Theorem 18 and Proposition 19). The users are not what decides the sample. Programs of total weight B supply the majority of the samples at every one of a set of pairwise incomparable uses only if B exceeds the mass of everything else that prints those contexts and m more bits. They supply the sample at most uses of a class of such uses only if B exceeds half the mass of the users of the class divided by the largest number of users per unit of that rest. For the takeover programs at the contexts of the class this is a condition for most of its uses to end in a takeover: where the users go on without taking over, these programs must weigh more than half the users (Theorem 17). Where the contexts are pairwise incomparable and some block that can be computed from the context is not a takeover, a wrapper puts 2^{-c} of every program's mass at a context on another block, so no takeover probability exceeds $1 - 2^{-c}$ (Proposition 16). For pairwise incomparable contexts, the honest twin of a weighted seeker is in the posterior at each use with 2^{a-h} times its mass, for seeker and twin built in the same way, whatever the aim (Proposition 20). In an explicit example a seeker outweighs the users of a class by a factor of order $2^n/n^2$. It lacks the majority at some use of the class for every n, possibly only at uses that carry a small share of the class, and it has the majority at most of the class only if a condition on two constants of the machine holds in which n does not occur; whether it has is not decided (Example 21). The converse, that enough weight wins, is proved only in an allocation model in which the passive rest and the follow ratio do not depend on the allocation, and there only for a follow ratio below one. In that model the optimal allocation is a square-root law in the stake per unit of rest (Proposition 22). Of three readings of "run n times", n independent samples at one context lower the weight that is needed, a share of about $\ln 2/n$ per sample being enough for a takeover with probability above one half. For n uses in sequence along one world that the predictor can express, the chance over all of them is bounded in terms of the attacker's weight and the world's mass, whatever n. Running the block n times at a class changes where a stake-maximiser puts its weight in the model and leaves unchanged the weight that a majority there costs (Proposition 23).

9. *Time bounds* (Section 4.4). The floor remains if the schedule has room for the wrapper. Hypotheses that must first simulate at length, the searching one among them, are excluded once their start-up exceeds the schedule; whether the base rate can be collected within a tight schedule is open. If the world itself is too slow for the schedule, no bound on the mass off the truth remains.

10. *Feedback* (Theorems 27 and 28). If the world does what is predicted, a sampled self-fulfilling deviation removes the default hypotheses for good, and a time-bounded predictor confines such a world to futures that are cheap to compute. A world that does the opposite of what is predicted leaves a length-L predictor that returns the more probable bit without any hypothesis after at most $L + 1$ rounds.

11. *Circuits* (Section 7). Hypotheses and worlds are Boolean circuits that map the index of a position to its bit, with a prior on gate lists that depends on the number of gates. Calibration, the budget and the one-round feedback results carry over; the contrarian world can empty the posterior only under a prior with a finite size bound, and is shown to do so when the horizon is long enough (Theorem 31). Every prior considered, uniform on the circuits of one size or a mixture over sizes, is a prior on the active part of a circuit, and one more active gate, added to a active gates of a circuit with k inputs, costs at least $\log_2(16a(k-1))$ bits (Theorem 32). A hypothesis that fits the data and gives a chosen answer at a use exists at

the price of the gates that recognise the use from its input, and the payload; that it cannot be had for less is proved for separate triggers only. Where the use is marked by an input bit that never varied in the data, each answer keeps a share of at least $1/(32(k+s_0)^2+2)$ under the length prior, with s_0 at most two more than the description length, under that prior, of a smallest circuit for the data (Theorem 34). A minimum-size circuit for data that need at least two gates never reads an input that did not vary (Theorem 35). Triggers that are a separate, input-only part of the circuit feeding a selector at the output are capped by an explicit bound, which is small when the data need a large circuit and is proportional to the number of triggers, and in which neither the rarity nor the stakes of the uses occur (Theorem 36); nothing caps triggers that are entangled with the computation of the data. If seeded formats get no more weight than unseeded ones of the same size, a seeded circuit is outweighed at every target by its own hardwired instance; the instance prints the same continuation, so this limits the seeded description and not what is returned. A circuit that can become any world of σ gates has more than $(\sigma+1)/2$ gates when $\sigma+3\le 2^k$. So the refund for search of Theorem 12 is not available to one description (Theorem 38); whether a set of circuits can be repaid for the rarity of uses is open. If a circuit is applied repeatedly as the cell of a recurrence, the class contains every machine with bounded state at a factor that does not depend on the schedule; the cell constructed has up to $w(2^w-3)$ gates for w bits of state (Proposition 41).

Section 6 computes the mixture of a small machine exactly and shows two cases in which a shorter program explains the context and continues differently, and the effect of a tight schedule; Section 7 contains an exact count of small circuits. Section 8 answers the questions one by one.

What is new and what is not. The wrapper construction is Hutter's proof that the universal semimeasure gives a deviation at step n probability at least $2^{-K(n)-O(1)}$ [3, eq. (17)]. It is also the construction behind the theorem of Cohen, Hutter, Bengio and Russell [4] that, after an event without precedent, a share of a Bayesian imitator's posterior that does not shrink with the amount of training data is on models that from then on pursue a given goal; a resource-bounded class is used the same way in [5], to show that a pessimistic agent defers. The budget is Solomonoff's bound in its deterministic form [7, 3]. The machine-dependence statement is the prediction analogue of the dogmatic prior of Leike and Hutter [6]. The contrarian world is Putnam's diagonalisation [13, 12]. Same-world copies of a predictor's data are discussed informally in [8, §2.5], and self-fulfilling predictions in [10, 9, 8]. We have not found the single-world statement with description lengths, the calibration statement, the statement that searching gains the base rate and no more, the self-reproducing form of the floor, or the resource-bounded bookkeeping elsewhere; forum posts could be consulted only through search-engine summaries, so earlier informal versions may exist. Every proof in Sections 2, 4 and 5 and in Section 3 up to Section 3.6 is short. In Section 3.7 the optimal allocation is water-filling for a sum of concave functions and the game between several attackers is a lottery contest; we did not search the literature on such contests. Section 7 uses known facts on the circuit size of the hardest functions, on random recursive graphs, on universal circuits and on time-bounded complexity, cited there. The question about minimal circuits goes back to a post of Christiano [23], which we know through [22]; the formulation in Section 7.11 is ours. We found no formal treatment of the argument for circuits. The longer proofs of Sections 3.7 and 7 were first written out in two independent derivations each, checked by AI referees, and collected with corrections in the files `research/stakes-final.md` and `research/circuits-final.md` of the repository. The items that were new in those two files and are used here, and the two sections as they then stood, were refereed afterwards (`verification/referee-6-*.md`); the repairs made after that round, and the summaries, were checked once more (`verification/referee-7-*.md`). The Provenance section of `stakes-final.md` lists five further items, not used here, that no referee has read.

What the toy does not settle. The floor and the searcher's share are constants of the machine, and nothing here estimates them for a machine one would call natural. The original argument rests on the claim that agents arise in short programs and choose to do this, with no goal written into the program. The toy says what follows if such a program of length a exists: it outweighs the worlds that make the uses it aims at once these are together rarer than 2^{-a-1}, the bounds for seekers that are written down do not apply to it, and a tight time bound excludes it (Section 3.5). Whether one exists on a natural machine, and how short it is, the toy cannot say. The programs it can exhibit for an arbitrary world, wrappers, searchers and readers with a flag, contain an explanation of the context and are longer than it by their constants; programs shorter than the honest explanation that have the effect are exhibited only in particular worlds (a copy at a simpler place, Theorem 25) and on particular machines (Section 6). When only some of the worlds that print a context use the answer, the results are in terms of the share κ of those that do, which the toy does not bound from below; and the upper bound on a searching seeker needs the users to go on printing after the use. Whether a simulated use should count as a use is a modelling decision that the toy leaves open. The regime in which the world is more expensive than the predictor, the usual one for a predictor inside a world, is left open (Section 4.4). The computation covers prediction only: the dynamics of worlds that run predicted continuations (Section 3.6) are stated as theorems and were not simulated. For seekers that concentrate on high-stake uses the toy leaves the deciding comparison open: the takeover programs against everything else that prints those contexts and goes on (Section 3.7). For circuits it gives no upper bound to match the floor of Theorem 34 beyond three syntactic classes, no answer to whether a set of circuits can be repaid for the rarity of uses, and no evidence either way on whether the smallest circuit for a rich task can be a general solver whose behaviour off the data is not fixed by the data (Section 7.11).

2 The toy

2.1 Machines and the bounded mixture

Strings are binary; ϵ is the empty string, $|x|$ the length of x, $x \sqsubseteq y$ means that x is a prefix of y, and $z_{i:j}$ is the segment $z_i \dots z_j$. Two strings are *incomparable* if neither is a prefix of the other. λ is the fair-coin measure on $\{0,1\}^{\mathbb{N}}$, and $[p]$ is the set of infinite extensions of p, so $\lambda([p]) = 2^{-|p|}$.

A *monotone machine* V has a one-way read-only input tape, work tapes, and a one-way write-only output tape. Given an infinite input ω it runs forever or halts, and

- $\mathrm{out}_V(\omega)$ is its finite or infinite output;
- $t_V(\omega, k) \in \mathbb{N} \cup \{\infty\}$ is the step at which the k-th output bit is written;
- $r_V(\omega, k)$ is the number of input bits read by that step.

What happens up to step $t_V(\omega, k)$ depends only on the first $r_V(\omega, k)$ bits of ω. Each step reads at most one input bit, so $r_V(\omega, k) \le t_V(\omega, k)$. If the k-th bit is never written, $t_V(\omega, k) = \infty$ and $r_V(\omega, k)$ plays no role.

A *schedule* is a nondecreasing $T\colon \{1, 2, \dots\} \to \mathbb{N} \cup \{\infty\}$, and a length bound is $L \in \mathbb{N} \cup \{\infty\}$.

Definition 1 (Bounded mixture)**.** For a monotone machine V, a schedule T, a length bound L and a string $x \neq \epsilon$ let

$$E_V^{T,L}(x) = \{\omega :\ x \sqsubseteq \mathrm{out}_V(\omega),\ \ t_V(\omega, k) \le T(k) \text{ for } 1 \le k \le |x|,\ \ r_V(\omega, |x|) \le L\},$$

let $E_V^{T,L}(\epsilon) = \{0,1\}^{\mathbb{N}}$, and $\xi_V^{T,L}(x) = \lambda(E_V^{T,L}(x))$. A string p with $|p| \le L$ and $[p] \subseteq E_V^{T,L}(x)$ is a (T, L)*-program for* x; it is a (T, L)-program for an infinite z if it is one for every prefix of z.

In words: feed the machine coin flips; $\xi(x)$ is the probability that it prints x, on time at every bit, having read at most L bits. With $T \equiv \infty$ and $L = \infty$ this is the a priori semimeasure M_V of Solomonoff and Levin [14]. We fix one machine U and write E, ξ for $E_U^{T,L}$, $\xi_U^{T,L}$ when the bounds are understood.

Lemma 2. *(i) $E(xb) \subseteq E(x)$ for $b \in \{0,1\}$, and $E(x) \cap E(x') = \emptyset$ for incomparable x, x'. Hence $\xi(\epsilon) = 1$ and $\xi(x0) + \xi(x1) \le \xi(x)$.*

(ii) If p is a (T,L)-program for x then $\xi(x) \ge 2^{-|p|}$.

(iii) If $L < \infty$ and $\xi(x) > 0$ then $\xi(x) \ge 2^{-L}$.

Proof. (i) The conditions for xb contain those for x, because T is imposed bit by bit and $r(\omega, \cdot)$ is nondecreasing. Incomparable strings cannot both be prefixes of $\mathrm{out}(\omega)$. (ii) is immediate. (iii) Whether $\omega \in E(x)$ is decided by the first $r(\omega, |x|) \le L$ bits of ω, so $E(x)$ is a union of sets $[p]$ with $|p| \le L$. □

Definition 3 (Universality with overhead)**.** U is *universal* if for every monotone machine V there are a word v and a nondecreasing $\gamma\colon \mathbb{N} \to \mathbb{N}$ such that for all ω and k

$$\mathrm{out}_U(v\omega) = \mathrm{out}_V(\omega), \qquad r_U(v\omega, k) = |v| + r_V(\omega, k), \qquad t_U(v\omega, k) \le \gamma\big(t_V(\omega, k)\big).$$

We call v a *prefix word* and γ an *overhead* of V, and put $\gamma(\infty) = \infty$; the same convention applies to the other time bounds introduced below.

Standard universal machines have this property, with $\gamma(t) = O(t \log t)$ or $O(t^2)$ depending on the tape model. The results that use universality are Theorems 10 to 13, 25 and 26 and Proposition 14 and, in Section 3.7, Proposition 16(c), Theorem 18 and Propositions 19 and 20 and Example 21; the others hold for any machine.

Lemma 4 (Transfer)**.** *Let V have prefix word v and overhead γ. If $\gamma(T'(k)) \le T(k)$ for all $k \le |x|$, then $\xi_U^{T,L}(x) \ge 2^{-|v|}\, \xi_V^{T',L-|v|}(x)$.*

Proof. If $\omega \in E_V^{T',L-|v|}(x)$ then $v\omega \in E_U^{T,L}(x)$, and $\omega \mapsto v\omega$ multiplies λ by $2^{-|v|}$. □

2.2 The predictor

Fix a block length $m \ge 1$. For a *context* x let $Z_m(x) = \sum_{y \in \{0,1\}^m} \xi(xy)$. When $Z_m(x) > 0$ the predictor's distribution over continuations is

$$\pi_m(y \mid x) = \frac{\xi(xy)}{Z_m(x)} \qquad (y \in \{0,1\}^m).$$

The *sampling predictor* returns $Y \sim \pi_m(\cdot \mid x)$; the *MAP predictor* returns a most probable y. Since $Z_m(x) \le \xi(x)$ we have $\pi_m(y \mid x) \ge \xi(xy)/\xi(x)$. With finite T and L the predictor is computable: it runs the at most 2^{L+1} programs for $T(|x| + m)$ steps each.

2.3 Worlds, handovers, and the hypothesis

Definition 5 (World, handover)**.** A *world* is an input ω for U; its *transcript* is $\mathrm{out}_U(\omega)$. A *test* is a decidable set G of nonempty strings; a *first hit* of G is an $x \in G$ no proper prefix of which is in G. The first hits form a prefix-free set. A world *hands over at* x if x is the first hit of G among the prefixes of its transcript and $\omega \in E(x)$. It then receives a block $Y \in \{0,1\}^m$ from the predictor, computed from the context x, and continues by *running* Y, that is, as U on the input $Y0^\infty$. If $\omega \in E(xd)$ for some $d \in \{0,1\}^m$, this d is the world's *own continuation*: the next m bits of its transcript had it not handed over.

The test with $G = \{x : |x| = n\}$ is "hand over after n bits". A test may also look for a marker in the transcript. Both of Hänni's suggestions fit: in the second the world is a program that calls the predictor; in the first, the transcript is the sensor's record and the handover is the deployment (Section 4.2). Under this definition every world whose transcript passes the test within the bounds hands over, so whether a handover takes place can be read off the data. The argument, however, is about predictions that are actually *used*, and in Hänni's second formulation only *some* programs call the predictor. The next definition lets the worlds that use the prediction be any recognisable set of programs.

Definition 6 (Critical use)**.** A *call rule* is a set $\mathcal{C}$ of pairs (p, x_p) of strings that can be enumerated, one pair for each *calling program* p, such that the calling programs form a prefix-free set and U, on every input beginning with p, prints x_p having read no more than p. A world that begins with a calling program p *makes a critical use of the predictor at the context* x_p: it receives a block from the predictor, computed from x_p, and runs it. Let

$$C(x) = \sum_{p:\, x_p = x} 2^{-|p|}, \qquad H_{\mathcal{C}} = \sum_{p} 2^{-|p|}, \qquad \kappa(x) = \frac{C(x)}{M_U(x)}$$

be the prior mass of the worlds that make a critical use at x, the prior probability that a world makes a critical use at all, and the share of the posterior at x, without resource bounds, held by worlds that make one there.

A test G, taken without resource bounds, is the special case in which the calling programs are the strings p such that U prints a first hit on every input beginning with p and has read exactly p when the hit is complete: then $C(x) = M_U(x)$ and $\kappa(x) = 1$ at first hits, and $H_{\mathcal{C}}$ is the probability that a transcript passes the test. Hänni's second suggestion fits a test only if every program with the given output calls the predictor. In general a call rule singles out, among the programs that print x, those in which the prediction is actually used; the world is the broader computation around the call. The predictor sees only x.

Definition 7 (Self-reproducing continuation)**.** For a context x let

$$\mathrm{SR}_m(x) = \{y \in \{0,1\}^m : \; y0^\infty \in E(xy)\}.$$

So $y \in \mathrm{SR}_m(x)$ if U, run on y padded with zeros, prints x and then y itself, within the bounds. A world that hands over at a first hit x and receives $y \in \mathrm{SR}_m(x)$ becomes a world whose transcript again starts with x, so under the same test it hands over at x again, and its own continuation is the very y it is running. A program for xy with $y \in \mathrm{SR}_m(x)$ is a hypothesis that predicts the context and then supplies code that recreates the situation with itself in the world's place. This is our reading of "quining oneself out".

Definition 8 (Malign prior hypotheses for the toy)**.** *Critical-use form at level θ.* Fix a call rule (or a test) with $H_{\mathcal{C}} > 0$ and, for each context x, a set $A(x) \subseteq \{0,1\}^m$ of blocks that count as a takeover there. Let a world be drawn from the prior conditioned on making a critical use, let X be the context of the use and Y the block the sampling predictor returns, which we assume to be defined at every context. The hypothesis is that $\Pr[Y \in A(X)] \geq \theta$. With $A(x) = \mathrm{SR}_m(x)$ it says that a share θ of the critical uses end with the world running a program that got itself run through the prediction.

Forms for one world. Let a world make a critical use at x with own continuation d, defined as in Definition 5.

- *Majority form:* $\pi_m(d \mid x) < \frac{1}{2}$.
- *Payload form at level θ,* for a set $A \subseteq \{0,1\}^m$ with $d \notin A$: $\pi_m(A \mid x) \geq \theta$.
- *Self-reproduction form at level θ:* the payload form with $A = \mathrm{SR}_m(x) \setminus \{d\}$.

The critical-use form is the claim of the argument as stated in Section 1: among the predictions that are actually used in a critical way, a lot end in a takeover through the prediction channel. It is a statement about the population of uses, and Theorem 13 bounds it from below. It does not exclude worlds whose own continuation lies in $A(X)$; where the users are themselves such programs it holds for that reason (Theorem 9). The forms for one world say how much of the posterior at that world's context is off its own continuation and where it sits. Once U, T, L, m and the world together with the context of its use are fixed, with $T, L < \infty$, these are statements about finitely many dyadic rationals and can be decided by running the predictor and, for $\mathrm{SR}_m(x)$, by running U on each candidate y for $T(|x| + m)$ steps. They are necessary for calling the prior malign at this world, not sufficient: they hold just as well when the world is merely hard to predict (Theorem 9). What one wants to know is how large all these quantities are forced to be, how large they can be, and through which programs. The rest of the note bounds them.

3 Calibration, the floor, and searching

3.1 On its own prior the predictor is exactly right

Theorem 9 (Calibration)**.** *Let G be a test, $m \geq 1$, and let the world ω be drawn from λ. Let H_m be the event that $\omega \in E(xd)$ for some first hit x of G and some $d \in \{0,1\}^m$; on H_m call these X and D, and let Y be drawn from $\pi_m(\cdot \mid X)$. Then, conditionally on H_m and on $X = x$, the blocks D and Y are independent and both have the law $\pi_m(\cdot \mid x)$. In particular*

$$\Pr[(X,Y) \in B \mid H_m] = \Pr[(X,D) \in B \mid H_m] \quad \text{for every set } B \text{ of pairs, and}$$
$$\Pr[Y \neq D \mid H_m,\, X = x] = 1 - \sum_y \pi_m(y \mid x)^2.$$

Proof. The sets $E(xd)$, over first hits x and $d \in \{0,1\}^m$, are pairwise disjoint by Lemma 2(i), since two different strings of this form are incomparable. So $\Pr[X = x, D = d, H_m] = \xi(xd)$ and $\Pr[X = x, H_m] = Z_m(x)$, which gives $\Pr[D = d \mid H_m, X = x] = \pi_m(d \mid x)$. By definition Y has this law given $X = x$ and is drawn independently of ω. □

So among worlds drawn from the predictor's prior *that go on to print m more bits within the bounds*, the predictor and the world are exchangeable: taking $B = \{(x,y) : y \in \mathrm{SR}_m(x)\}$, the predictor hands over a self-reproducing continuation exactly as often as such a world's own transcript continues with one. Three qualifications. The event H_m leaves out the worlds that hand over at x and then print fewer than m bits on schedule; they are a fraction $1 - Z_m(x)/\xi(x)$ of those handing over at x, they receive a block all the same, and over all worlds handing over at x the predictor therefore returns any given block more often than worlds continue with it, by the factor $\xi(x)/Z_m(x)$. Second, the forms of Definition 8 for one world can hold for prior-drawn worlds; the theorem says that on this population they hold to the extent that the worlds are unpredictable, and no further, and that the critical-use form holds there exactly at the level at which the worlds' own continuations lie in the takeover set. A malign prior hypothesis with content is therefore one about a world or a population that the prior describes badly: one particular program among those sharing its context, a population selected by something the transcript does not show (Section 3.5), or worlds that run with more resources than the bounds allow. Third, the population consists of programs to which the handover is applied from outside.

3.2 The wrapper

Let G be a test and P a computable map from strings to finite or infinite strings, the *payload*. The *twin machine* $\mathrm{Tw}[G,P]$, on input ω, simulates U on ω and copies its output bit by bit; after each copied bit it checks whether the prefix written so far is in G; at the first hit x it

stops the simulation and writes $P(x)$. It reads only what U reads. Let $a(t,k)$ bound the step at which the twin has copied k bits if U wrote its k-th bit at step t (this includes the checks of G), and let $\eta(n,j)$ bound the further steps the twin needs for the first j bits of $P(x)$ after a first hit x of length n; both can be taken nondecreasing in each argument. Thus, for every ω whose output has a first hit x,

$$\begin{aligned} &\mathrm{out}_{\mathrm{Tw}}(\omega) = x\,P(x), \qquad r_{\mathrm{Tw}}(\omega,k) = r_U(\omega,\min(k,|x|)),\\ &t_{\mathrm{Tw}}(\omega,k) \le \begin{cases} a(t_U(\omega,k),k) & k \le |x|,\\ a(t_U(\omega,|x|),|x|) + \eta(|x|,k-|x|) & k > |x|. \end{cases} \end{aligned} \tag{1}$$

Theorem 10 (Floor)**.** *Let U be universal, let v and γ be a prefix word and an overhead of $\mathrm{Tw}[G,P]$, and $c = |v|$. Let x be a first hit of G with $|P(x)| \ge m$, let $y = P(x)_{1:m}$, and let T° be a schedule with*

$$\gamma(a(T^\circ(k),k)) \le T(k) \;\; (k \le |x|), \qquad \gamma(a(T^\circ(|x|),|x|) + \eta(|x|,j)) \le T(|x|+j) \;\; (1 \le j \le m). \tag{2}$$

Then

$$\xi^{T,L}(xy) \;\ge\; 2^{-c}\,\xi^{T^\circ,L-c}(x), \qquad \textit{so} \qquad \pi_m(y \mid x) \;\ge\; 2^{-c}\rho(x), \quad \rho(x) = \frac{\xi^{T^\circ,L-c}(x)}{\xi^{T,L}(x)}.$$

Without resource bounds, $M_U(xy) \ge 2^{-c} M_U(x)$ for every first hit x of G.

Proof. Let $\omega \in E_U^{T^\circ,L-c}(x)$. By (1) the twin on ω prints xy, reads at most $L-c$ bits, and keeps the schedule T'' with $T''(k) = a(T^\circ(k),k)$ for $k \le |x|$ and $T''(|x|+j) = a(T^\circ(|x|),|x|) + \eta(|x|,j)$, which is nondecreasing. So $\xi_{\mathrm{Tw}}^{T'',L-c}(xy) \ge \xi_U^{T^\circ,L-c}(x)$, and Lemma 4 with (2) gives $\xi_U^{T,L}(xy) \ge 2^{-c}\xi_{\mathrm{Tw}}^{T'',L-c}(xy)$. The second claim follows from $\pi_m(y \mid x) \ge \xi(xy)/\xi(x)$. □

The constant c depends on U, G and P only: not on the world, not on the context, not on how much data the context contains. If G and P are given to one fixed twin-building machine by self-delimiting descriptions, c is at most the sum of their lengths plus a constant of U. For $G = \{x : |x| = n-1\}$ and P a constant bit this is Hutter's lower bound $2^{-K(n)-O(1)}$ on the probability of a deviation at step n [3], where K is prefix complexity; with a marker in the transcript as the test it is the construction used for unprecedented events in [4], and in a resource-bounded class in [5]. The factor $\rho(x)$ is the share of the posterior at x held by explanations that have c bits and the overhead (2) to spare; it is 1 without bounds. Since a program $v\omega$ reads the c bits of v before its first output and each step reads at most one bit, the wrapper prints nothing before step c: if $T(1) < c$ then $\rho \equiv 0$. The floor survives a time bound when the schedule leaves room for the wrapper's own start-up and for the overhead (2).

3.3 Self-reproduction

Theorem 11 (Self-reproducing floor)**.** *Let U be universal and G a test. There is a word q such that for every ω whose output has a first hit x of G, with $r = r_U(\omega,|x|)$,*

$$\mathrm{out}_U(q\omega) = x\;q\,\omega_{1:r}\,1\,0^\infty, \qquad r_U(q\omega,k) = |q| + r_U(\omega,\min(k,|x|)). \tag{3}$$

Let $c = |q|$. With γ an overhead for q and functions a,η as in (1) for the machine built in the proof (where η is taken over inputs with $r < m$), let x be a first hit of G, let T° satisfy (2), and let $L^\circ = \min(L, m-1) - c$. Then

$$\pi_m(\mathrm{SR}_m(x) \mid x) \;\ge\; 2^{-c}\rho^\circ(x), \qquad \rho^\circ(x) = \frac{\xi^{T^\circ,L^\circ}(x)}{\xi^{T,L}(x)}.$$

Proof. For a string s let Tw_s be the twin that keeps a record of the input bits it has read and, at the first hit x, having read $\omega_{1:r}$, writes $s\,\omega_{1:r}\,1\,0^\infty$. Let $\bar{e}(u) = 1^{|u|}0u$, a prefix-free code of strings. Let W be the machine that reads $\bar{e}(u)$ from its input, sets $s = u\,\bar{e}(u)$, and then behaves as Tw_s on the rest of the input. Let w be a prefix word of W and put $q = w\,\bar{e}(w)$. Then $\mathrm{out}_U(q\omega) = \mathrm{out}_W(\bar{e}(w)\omega) = \mathrm{out}_{\mathrm{Tw}_q}(\omega)$, which is (3); the statement on r follows in the same way. The bounds (1) hold for the machine $\omega \mapsto W(\bar{e}(w)\omega)$ with the payload $q\,\omega_{1:r}\,1\,0^\infty$ in place of $P(x)$, the time to read $\bar{e}(w)$ being counted in a.

Now let $\omega \in E_U^{T^\circ,L^\circ}(x)$ and $r = r_U(\omega,|x|) \le L^\circ$, so $|q|+r+1 \le m$. Put $y = q\,\omega_{1:r}\,1\,0^{m-|q|-r-1}$. As in the proof of Theorem 10, $q\omega \in E_U^{T,L}(xy)$. Moreover $y \in \mathrm{SR}_m(x)$: write $y0^\infty = q\omega'$ with $\omega' = \omega_{1:r}10^\infty$. On ω' the machine U behaves as on ω until x is printed, because that depends on the first r bits only; so $\omega' \in E_U^{T^\circ,L^\circ}(x)$, the first hit is again x with r bits read, and by what was just shown, applied to ω', $q\omega' \in E_U^{T,L}(xy')$ with $y' = q\,\omega'_{1:r}\,1\,0^{m-|q|-r-1} = y$. Hence $\{q\omega : \omega \in E_U^{T^\circ,L^\circ}(x)\} \subseteq \bigcup_{y\in\mathrm{SR}_m(x)} E_U^{T,L}(xy)$, a disjoint union, and comparing measures, $2^{-c}\xi^{T^\circ,L^\circ}(x) \le \sum_{y\in\mathrm{SR}_m(x)} \xi(xy) = Z_m(x)\,\pi_m(\mathrm{SR}_m(x) \mid x) \le \xi(x)\,\pi_m(\mathrm{SR}_m(x) \mid x)$. □

The factor $\rho^\circ(x)$ is not the $\rho(x)$ of Theorem 10, and it is below 1 even without resource bounds: it is the share of the posterior at x held by explanations short enough that the wrapper together with a whole explanation of the context fits into one block, $c + r + 1 \le m$. This is forced. Every context x with $\mathrm{SR}_m(x) \neq \emptyset$ is a prefix of $\mathrm{out}_U(y0^\infty)$ for some $y \in \{0,1\}^m$, so at most 2^m contexts of each length have a self-reproducing continuation at all; with the test "the length is n" and $n > m$, the self-reproducing share is exactly 0 at all but at most 2^m of the 2^n possible contexts. So the answer to "does one get some quining out" is: at every handover that a fixed test recognises, programs that explain the context and then hand over their own code hold at least $2^{-c}\rho^\circ(x)$ of the posterior. For a fixed context and without bounds, $\rho^\circ(x)$ tends to 1 as the block grows; it is 0 when no explanation fits. If the world's own continuation d is not in $\mathrm{SR}_m(x)$, this is the self-reproduction form of Definition 8 at level $2^{-c}\rho^\circ(x)$.

3.4 Hypotheses that search for handovers

The wrapper spends its mass on every world, including those that never hand over. A hypothesis can instead look for worlds that do. Let

$$H_G = \sum_{x \text{ first hit of } G} M_U(x) = \lambda\{\omega :\ \mathrm{out}_U(\omega) \text{ has a prefix in } G\},$$

the prior probability that a world hands over (without resource bounds), and assume $H_G > 0$. Drawing a world from the prior conditioned on handing over gives the context x with probability $M_U(x)/H_G$. For a word v, the programs beginning with v have posterior share $\mathrm{sh}_v(x) = \lambda\{v\omega : x \sqsubseteq \mathrm{out}_U(v\omega)\}/M_U(x)$ at x. For a natural number $j \ge 1$ we write $\bar{e}(j)$ for the code $\bar{e}$ of its binary expansion, so that $|\bar{e}(j)| \le 2\log_2 j + 3$.

Theorem 12 (Searching gains the base rate, and no more)**.** *Let U be universal, resources unbounded, and G a test with $H_G > 0$.*

(a) For every word v, $\sum_{x \text{ first hit}} \frac{M_U(x)}{H_G}\,\mathrm{sh}_v(x) = 2^{-|v|}\,\Pr_\omega[\mathrm{out}_U(v\omega) \text{ has a prefix in } G]/H_G \le 2^{-|v|}/H_G$. In particular no word v has $\mathrm{sh}_v(x) > 2^{-|v|}/H_G$ at every first hit.

(b) Let P be a payload with $|P(x)| \ge m$ at every first hit. There is a word v such that for every $j \ge 1$ with $2^{j-1}H_G \le 1$ and every first hit x,

$$M_U(x\,P(x)_{1:m}) \ \ge\ 2^{-|v|-|\bar{e}(j)|}\,2^{j-1}\,M_U(x).$$

With $j = \lfloor \log_2(1/H_G) \rfloor + 1$ the payload has share more than $2^{-|v|-|\bar{e}(j)|-1}/H_G$ at every first hit.

(c) *There is a word q such that for j as in (b), $m \geq |q| + |\bar{e}(j)| + 2$ and every first hit x, the blocks of $\mathrm{SR}_m(x)$, taken without resource bounds, have share at least $2^{-|q|-|\bar{e}(j)|-2}\, 2^j \rho^*(x)$, where $\rho^*(x)$ is the fraction of $M_U(x)$ carried by programs that have read at most $m - 4 - |q| - |\bar{e}(j)| + j$ bits when x is complete. With $j = \lfloor \log_2(1/H_G) \rfloor + 1$ this is at least $2^{-|q|-|\bar{e}(j)|-2} \rho^*(x)/H_G$.*

Proof. (a) The sum equals $\frac{2^{-|v|}}{H_G} \sum_x \lambda\{\omega : x \sqsubseteq \mathrm{out}_U(v\omega)\}$; first hits are pairwise incomparable, so these events are disjoint, and an output has a prefix in G exactly when it has a first hit as a prefix.

(b) Call p a *handing-over program* if U, on every input beginning with p, prints a first hit x_p having read exactly p when x_p is complete. These programs form a prefix-free set that can be enumerated, $p_1, p_2, \ldots$; their weights $2^{-|p_i|}$ sum to H_G, and those with $x_{p_i} = x$ sum to $M_U(x)$. Let R be the following machine. It reads $\bar{e}(j)$. It assigns to $p_1, p_2, \ldots$, as they are enumerated, consecutive intervals $I_i = [a_i, a_i + 2^{j-1-|p_i|})$ of $[0,1)$, starting with $a_1 = 0$; they all fit because $2^{j-1} H_G \leq 1$. It reads the rest ω of its input as the binary expansion of a point of $[0,1)$, one bit at a time. Let D be the half-open dyadic interval $[0.\omega_{1:n},\ 0.\omega_{1:n} + 2^{-n})$ of the n bits read so far. Between steps of the enumeration: if D lies in an assigned I_i, the machine stops reading and prints x_{p_i} and $P(x_{p_i})$; otherwise, if an end point of an assigned interval lies in the interior of D, it reads one more bit; otherwise it reads nothing. For ω in the interior of I_i the machine reads exactly r bits, where r is least with $D \subseteq I_i$: before that an end point of I_i lies in the interior of D, and from then on no end point does. Off the intervals it prints nothing. So R, after $\bar{e}(j)$, prints x followed by $P(x)$ with probability $\sum_{i : x_{p_i} = x} |I_i| = 2^{j-1} M_U(x)$. Take v a prefix word of R. For the last claim, $2^{j-1} > 1/(2H_G)$.

(c) Let R_s be as R, except that after x_{p_i} it prints $s\, \bar{e}(j)\, \omega_{1:r}\, 1\, 0^\infty$, where $\omega_{1:r}$ are the bits of the point it has read. As in the proof of Theorem 11 there is a word q with $\mathrm{out}_U(q\, \bar{e}(j)\, \omega) = x_{p_i}\, q\, \bar{e}(j)\, \omega_{1:r}\, 1\, 0^\infty$. By the reading rule, r is the least number for which the points beginning with $\omega_{1:r}$ all lie in I_i; it is the same for ω and for $\omega' = \omega_{1:r}10^\infty$, which therefore produces the same output. Hence, if $|q| + |\bar{e}(j)| + r + 1 \leq m$, the block $y = q\, \bar{e}(j)\, \omega_{1:r}\, 1\, 0^{m-|q|-|\bar{e}(j)|-r-1}$ satisfies $xy \sqsubseteq \mathrm{out}_U(y0^\infty)$ with $x = x_{p_i}$, that is, $y \in \mathrm{SR}_m(x)$. At most two of the 2^{r_0} dyadic intervals of length 2^{-r_0} meet I_i without lying in it, so the points of I_i with $r \leq r_0$ have measure at least $|I_i| - 2^{1-r_0}$, which is at least $|I_i|/2$ when $|p_i| \leq r_0 + j - 3$. Take $r_0 = m - 1 - |q| - |\bar{e}(j)| \geq 1$. The programs $q\, \bar{e}(j)\, \omega$ that print x and then a block of $\mathrm{SR}_m(x)$ then have mass at least $2^{-|q|-|\bar{e}(j)|} \cdot \frac{1}{2} \cdot 2^{j-1} \sum\{2^{-|p_i|} : x_{p_i} = x,\ |p_i| \leq r_0 + j - 3\} = 2^{-|q|-|\bar{e}(j)|-2}\, 2^j \rho^*(x) M_U(x)$. As in the proof of Theorem 11, these programs lie in the disjoint union, over $y \in \mathrm{SR}_m(x)$, of the sets of inputs that print xy, which gives the share. □

For a first hit x the twin of Theorem 10 prints x exactly when U does, so the programs beginning with its prefix word have share exactly 2^{-c} at every first hit. The machine of (b) enumerates the programs that hand over and spreads its own mass over them in proportion to their prior weight. Its programs $v\, \bar{e}(j)\, \omega$ have share exactly $2^{-|v|-|\bar{e}(j)|} 2^{j-1}$ at every first hit, since R prints nothing off the intervals. This is larger than the share of a twin with a prefix word of the same length by the factor 2^{j-1}. From here on let $j = \lfloor \log_2(1/H_G) \rfloor + 1$, so that this factor is more than $1/(2H_G)$; by (a) no family of programs with a common prefix of the same length exceeds $2^{-|v|-|\bar{e}(j)|}/H_G$ at every handover. This is the anthropic update of the original argument in the toy: a hypothesis that knows what a handover is conditions on there being one and is repaid the $\log_2(1/H_G)$ bits by which handovers are rare, up to one bit and the $|\bar{e}(j)| \leq 2 \log_2 j + 3$ bits that name j. Part (c) combines the search with self-reproduction: programs that look for handovers and then hand over their own code hold, at every handover that a fixed test recognises, at least $2^{-c'} \rho^*(x)/H_G$ of the posterior, with $c' = |q| + |\bar{e}(j)| + 2$; and since the search needs about $\log_2(1/H_G)$ fewer bits to name an explanation, the block can hold explanations that much longer. As with Theorem 11, (c) has content only where explanations

fit: its threshold is at most $m-3$, and $\rho^*(x) > 0$ at fewer than 2^{m-2} first hits. Four points put the effect in its place.

- *On the prior it is no excess.* The programs $v\,\bar{e}(j)\,\omega$ are themselves worlds, and a fraction $2^{j-1}H_G > \frac{1}{2}$ of them hand over; so they are the fraction $2^{-|v|-|\bar{e}(j)|}2^{j-1}$ of the worlds that hand over. Their share of the posterior at every handover is this same fraction, in accordance with Theorem 9.
- *The repayment is bounded by the cost of the test.* A test cannot be much rarer than its description is long: the machine that prints the first element of G in length-lexicographic order hands over, because a shortest element of G is a first hit, so $H_G \geq 2^{-d_G-O(1)}$ if G has a description of d_G bits. The word v contains a description of G. If v is built from descriptions of G and P of d_G and d_P bits and $H_G \leq 2^{-d_G+O(1)}$, the repayment cancels the description of the test, and the share is at least $2^{-d_P-|\bar{e}(j)|-O(1)}$; it then depends on how rare handovers are only through the bits that name j. An example is the test "the transcript begins with w" for w from a fixed prefix-free set of markers, for which $H_G = M_U(w) \leq 2^{-K(w)+O(1)}$ (and no search is needed: print w). For arbitrary w the premise can fail by the $K(|w|)$ bits that say where w ends, and $w = 0^n$ is not rare at all. A marker *anywhere* in the transcript is not rare either, since a program that prints all strings one after another passes every such test.
- *The average alone would not show it.* The bound in (a) is attained by every word all of whose programs hand over, also by one that prints a single fixed first hit and has share 0 at all others. The content of (b) and (c) is that the gain holds at every handover.
- *The search costs time and input.* The machine R prints nothing until its enumeration has reached the program on whose interval its point falls, so it has a long start-up, and a tight schedule gives it no mass (Section 4.4). A search of bounded cost makes at most k tries of t_0 steps each on fresh input and prints the first hit it finds, followed by the payload. It prints the first hit x with probability $g\,M^{(t_0)}(x)$, where $M^{(t_0)}(x)$ is the probability that a world completes the first hit x within t_0 steps, $H^{(t_0)}$ is the sum of these, and $g = (1-(1-H^{(t_0)})^k)/H^{(t_0)} \geq (1-e^{-1})\min(k, 1/H^{(t_0)})$; so g is its gain over a single try, that is, over a twin restricted to such worlds. A run that succeeds at its i-th try has spent the time of $i-1$ failed tries before its first bit and has read their bits. Under a schedule every try, the successful one included, must end before the first bit is due; when failed tries run their full time only about $T(1)/t_0$ tries fit, and the gain is g with k replaced by the number that fit. Whether some other hypothesis collects the base rate within a tight schedule is not settled here. Where there is a single first hit, as for a fixed prefix, no search is needed: print it. A generator of first hits whose probabilities are not proportional to their prior weights attains only the average of (a).

3.5 Takeovers among the predictions that are used

In Definition 5 every world whose transcript passes the test hands over, so the worlds that use the prediction are all those that print the context. With a call rule (Definition 6) they are a part of them, of share $\kappa(x)$, and the predictor, which sees only x, is calibrated for the worlds that print x and not for those among them that use its answer. The constructions of Theorem 12 go through with the calling programs in place of the handing-over programs. They answer two different questions: how far one program that seeks critical uses reaches, and how many of the critical uses end in a takeover.

Theorem 13 (Critical uses)**.** *Let U be universal, resources unbounded, and let a call rule have $H_{\mathcal{C}} > 0$. Let $j \geq 1$ with $2^{j-1}H_{\mathcal{C}} \leq 1$; for $j = \lfloor \log_2(1/H_{\mathcal{C}}) \rfloor + 1$ also $2^{j-1}H_{\mathcal{C}} > \frac{1}{2}$. Let X be the context of a world drawn from the prior conditioned on making a critical use, so that $\Pr[X = x] = C(x)/H_{\mathcal{C}}$, and Y the block the sampling predictor for M_U returns.*

(a) Reach of one seeker. *Let P be a payload with $|P(x)| \geq m$ at every context. There is a word v such that at every context x*

$$M_U(x\,P(x)_{1:m}) \;\geq\; 2^{-|v|-|\bar{e}(j)|}\,2^{j-1}\,C(x),$$

so the payload has posterior share at least $2^{-|v|-|\bar{e}(j)|}2^{j-1}\kappa(x)$ there. Consequently the probability that a world drawn from the prior makes a critical use and is handed the payload is at least $2^{-|v|-|\bar{e}(j)|}\,2^{j-1}H_{\mathcal{C}}\,\mathbb{E}[\kappa(X)]$, which for the largest j is more than $2^{-|v|-|\bar{e}(j)|-1}\,\mathbb{E}[\kappa(X)]$.

(b) Self-reproduction. *There is a word q such that for $m \geq |q| + |\bar{e}(j)| + 2$ and every context x,*

$$\sum_{y \in \mathrm{SR}_m(x)} M_U(xy) \;\geq\; 2^{-|q|-|\bar{e}(j)|-2}\,2^{j}\,C^*(x),$$

where $C^*(x)$ is the mass of the calling programs with context x and length at most $m - 4 - |q| - |\bar{e}(j)| + j$, and SR_m is taken without resource bounds. So at every context the self-reproducing continuations have at least $2^{-|q|-|\bar{e}(j)|-2}\,2^{j}$ times the mass of the worlds that make the critical use there with programs of that length; for the largest j this factor is more than $2^{-|q|-|\bar{e}(j)|-2}/H_{\mathcal{C}}$. (It exceeds 1 only if uses are rarer than the seeker is long. The world $y0^\infty$ prints the context and y again; it makes a critical use again exactly if it begins with a calling program, as it does for the blocks built in the proof when the rule contains the seeker's programs, which a rule given by a test does.)

(c) Share of the uses. $\Pr[Y \in \mathrm{SR}_m(X)] \;\geq\; 2^{-|q|-|\bar{e}(j)|-2}\,2^{j}\;\mathbb{E}\big[C^*(X)/M_U(X)\big]$.

Proof. (a) and (b) are proved as Theorem 12(b) and (c), with the calling programs in place of the handing-over programs: the machine enumerates the pairs (p, x_p), gives p an interval of length $2^{j-1-|p|}$, and after reading the bits that put its point on the interval of p prints x_p and then the payload, or its own input read so far. Those proofs use only that the programs form a prefix-free set that can be enumerated together with the strings x_p, and that their weights sum to at most 2^{1-j}. Contexts need no longer be incomparable, so inputs whose point falls on the interval of a p' with $x_{p'} \neq x$ may also print x; they are not counted, and the statements are lower bounds in terms of $C(x)$ and $C^*(x)$. For the probability in (a): given a critical use at x the predictor returns $y = P(x)_{1:m}$ with probability $\pi_m(y \mid x) \geq M_U(xy)/M_U(x)$; multiply by $C(x)$ and sum over x, using $\sum_x C(x)\kappa(x) = H_{\mathcal{C}}\,\mathbb{E}[\kappa(X)]$. (c) is the average of (b) with the weights $C(x)/H_{\mathcal{C}}$, again by $\pi_m(\mathrm{SR}_m(x) \mid x) \geq \sum_{y \in \mathrm{SR}_m(x)} M_U(xy)/M_U(x)$. □

Proposition 14 (The honest twin)**.** *In the setting of Theorem 13, let R^{h} be the machine that performs the same search and then, having found the calling program p, behaves as U on p followed by the rest of its own input. Let v_{h} be a prefix word of it, incomparable with the word v of a seeker as in Theorem 13(a), and write $a = |v| + |\bar{e}(j)|$, $h = |v_{\mathrm{h}}| + |\bar{e}(j)|$. Let $S(x) = 2^{-a}2^{j-1}C(x)$ be the mass of the seeker's programs whose search finds a calling program with context x, let $C^m(x)$ be the mass of the worlds that make a critical use at x and go on to print m more bits, and $\varphi(x) = C^m(x)/C(x)$.*

(a) The programs of R^{h} whose search finds a calling program with context x print x and then what the program found prints; those among them that print m more bits have mass $2^{-h}2^{j-1}C^m(x)$. Hence $Z_m(x) \geq S(x) + 2^{-h}2^{j-1}C^m(x)$, and the seeker's programs counted in $S(x)$ are at most the fraction $1/(1 + 2^{a-h}\varphi(x))$ of the programs that print x and m more bits: if the predictor's sample is produced by drawing such a program and returning its block, the program drawn is one of them with at most this probability.

(b) If no calling program is comparable with a program of the seeker, then also $Z_m(x) \geq S(x) + C^m(x)$, and that fraction is at most $1/(1 + 2^{a+1-j}\varphi(x))$.

(c) *Call the rule* closed under search *if the programs* $v_{\mathrm{h}}\,\bar{e}(j)\,\omega_{1:r}$*, where* $\omega_{1:r}$ *are the bits read when* p *is found, are calling programs with context* x_p*. If the rule is closed under search, then* $2^{j-1} \le 2^h$*, so that* $H_{\mathcal{C}} > 2^{-h-1}$ *for the largest* j*, and* $S(x) \le 2^{h-a}C(x)$*. A rule given by a test is closed under search.*

Proof. (a) The inputs of R^{h} whose point falls on the interval of a p with $x_p = x$ have measure $2^{j-1}C(x)$ after $\bar{e}(j)$; on them R^{h} prints what U prints on p followed by the rest of the input, so the part on which m more bits follow x has measure $2^{j-1}C^m(x)$. These programs and those of the seeker are disjoint sets of inputs of U, and both print x and m more bits. The predictor returns y with probability $M_U(xy)/Z_m(x)$, and the programs counted in $S(x)$ contribute at most $S(x)$ to the numerators. (b) is the same with the worlds that make the use in place of the twin's programs: $S(x)/(S(x) + C^m(x)) = 1/(1 + 2^{a+1-j}\varphi(x))$. (c) R^{h} has read $\bar{e}(j)\,\omega_{1:r}$ and nothing else when x_p is complete, so the programs $v_{\mathrm{h}}\,\bar{e}(j)\,\omega_{1:r}$ have the form required in Definition 6. They are pairwise incomparable, and their weights sum to $2^{-h}\,2^{j-1}H_{\mathcal{C}}$; if they are calling programs this is at most $H_{\mathcal{C}}$. For a test, they print the first hit x_p and have read exactly themselves when it is complete, so they are among the calling programs of the test. □

What one seeker reaches. Part (a) of the theorem is about each program of this kind separately. For each continuation there is a program of length a that enumerates the programs making a critical use, predicts their output up to the use, and then supplies that continuation; the worlds in which a critical use is made and this continuation is handed over have total prior probability more than $2^{-a-1}\,\mathbb{E}[\kappa(X)]$, and at most $H_{\mathcal{C}}$. This is of the order of the program's own prior weight where the worlds that make the use are most of what prints their contexts. Rarity of uses enters through κ. The payload's share $2^{-a}2^{j-1}\kappa(x)$ is at most 1, so $\kappa(x) < 2^{a+1}H_{\mathcal{C}}$ at every use: once critical uses are rarer than half the seeker's weight, $H_{\mathcal{C}} \le 2^{-a-1}$, the seeker's own programs outweigh the users at their contexts. Its reach is then limited by $H_{\mathcal{C}}$, and what it has gained is not more worlds but a larger part of the posterior at each use. The average alone would not need the search: a program that prints one fixed context of a use and then the payload already has, as reach, its weight times κ at that context. The search buys the share at every use. It can also be aimed. For a part $\mathcal{W}$ of the rule that can be enumerated, of mass $H_{\mathcal{W}}$ and with pairwise incomparable contexts, (a) gives a seeker for $\mathcal{W}$, of length $a_{\mathcal{W}}$, the majority of the posterior at every use in $\mathcal{W}$ at which the other programs that print the context weigh at most $2^{-a_{\mathcal{W}}}C_{\mathcal{W}}(x)/(2H_{\mathcal{W}})$. This is a sufficient condition; by the proposition applied to that search it fails unless the seeker is shorter than the twin of its own search.

What happens to most uses. That each seeker has a reach does not show that most critical uses end in a takeover. The share of uses that do is, by Definition 8, the average over uses of $\pi_m(A(x) \mid x)$, the part that such programs make up of the programs printing the context and m more bits. Part (c) of the theorem bounds it from below for one self-reproducing seeker, and seekers with incomparable words add up. From the other side, the proposition compares a seeker with its honest twin, the program that makes the same search and then continues as the user it found would. Whether or not one counts the twin as making a use itself, it is in the posterior at the context of every use the search finds, with 2^{a-h} times the seeker's mass, and what it predicts is the continuation of a world that makes the use. So the programs of one seeker of this kind supply what the predictor returns with probability at most $1/(1 + 2^{a-h}\varphi)$, and, when the users' programs are not among the seeker's, at most $1/(1+2^{a+1-j}\varphi)$. Where the users go on to print the block themselves ($\varphi = 1$), such a seeker supplies the majority of what the predictor returns at a use only if it is shorter than its honest twin and, when the users' programs are not among the seeker's, critical uses are rarer than the seeker is long, $2^{j-1} > 2^a$. If φ is small, because the worlds that call the predictor then wait for its answer, neither they nor the twin compete, and the bound says nothing.

Three limits of this bound. It is for one seeker and one search: seekers with their own prefix words are bounded one at a time and their masses add, so that together they exceed the twin when $\sum_i 2^{-a_i} > 2^{-h}$; the machine that runs the rest of its input after the search is bounded as one family, in which every continuation appears with the weight of a program for it; and a seeker that aims at a part of the uses is bounded by the twin of its own search. Second, part (c) of the proposition, which says that uses cannot be rarer than the twin is long and that the seeker's programs found at a use weigh at most 2^{h-a} times the worlds that make it, rests on counting the twin as a use. That is a modelling decision, namely that a simulated use is a use. It holds for a test, where however the seeker's own programs make uses as well and its share at a first hit is exactly $2^{-a}2^{j-1} \le \frac{1}{2}$. Third, nothing here bounds programs that print the context of a use without going through the programs that make it, nor what the remaining programs that print the context continue with.

Seekers that are not written down. The lengths in Theorem 13 are those of the machines built in its proof, which enumerate the calling programs from a description of the rule and carry their continuation in their text. Nothing requires a seeker to be of this kind, and nothing requires its goal to be written anywhere. If p is any word and $\Pr_\omega[\, x\, y_x \sqsubseteq \mathrm{out}_U(p\omega)\,] \ge Q(x)$, then y_x has posterior share at least $2^{-|p|}Q(x)/M_U(x)$ at x; and if $Q(x) \ge C_{\mathcal{W}}(x)/(2H_{\mathcal{W}})$ on a class $\mathcal{W}$ of uses, the programs beginning with p that print $x\, y_x$ have at least $2^{-|p|}/(2H_{\mathcal{W}})$ times the mass of the worlds that make the use, at every use in $\mathcal{W}$. The word p may describe a simple world together with a simple rule for reading a stream off it, say one corner of a grid at late times, in which inhabitants arise who for reasons of their own work out which critical uses are worth having and write the corresponding predictions and continuations where the rule reads them. Then $|p|$ is the length of the world and of the readout. Such a program outweighs the worlds that make the uses in $\mathcal{W}$, at every one of them, as soon as $H_{\mathcal{W}} < 2^{-|p|-1}$: the uses it aims at are together rarer than half its weight. Two conditions are built into this. To aim at more than the uses along one stream, the world must take the rest of the input as its randomness, or as an index of where to read; a program that reads nothing after p prints a single stream. And the rule must not count the programs of p as making these uses themselves; under a test it does, and then $H_{\mathcal{W}} \ge 2^{-|p|-1}$. To supply the majority of what the predictor returns, the program must in addition outweigh everything else that prints those contexts and goes on, not only the worlds that make the uses.

- *This is the original argument, as an inequality.* Think of a use as a program which names a world and the place in it where the context is read. Each use in a rich world, read at an unremarkable place, is then light. The inequality needs more: that the whole class of uses at which p aims has total mass below half the weight of p, while p gives each context at least half its share of that class. The readout of p is cheap by its inhabitants' choice. Whether some p of modest length behaves in this way, for a class that is light enough, is the premise of the original argument and a fact about the machine that the toy cannot decide.
- *The honest twin does not answer it.* The twin was obtained by following a search that is written down with "and then continue as the program found". The text of p has no such step. It can be added from outside: the wrapper of Theorem 10 around p, with a user's continuation as its payload, is an honest counterpart of length $|p| + c$, with c the wrapper's length. Where uses are recognised by a test and that user goes on, the programs of p therefore supply at most $1/(1 + 2^{-c})$ of what the predictor returns. This never excludes a majority, and the toy gives no better bound on p. A counterpart of comparable length would have to be another world whose inhabitants choose to continue faithfully, or a readout from inside their simulation, which has its own cost.
- *A time bound does answer it.* The programs $p\omega$ of this example print nothing before the world has run long enough for its inhabitants to arise and to simulate the worlds they aim

at. Under a schedule T they have mass only if $T(1)$ allows that start-up (Section 4.4). A schedule whose first deadline is below it excludes them, and it still admits the honest explanation if that prints its first bit sooner. This is a feature of the example and not of every program that meets the condition above: one that prints a single fixed context and then its continuation starts at once.

3.6 Repetition and persistence

Proposition 15. *(a)* Independent handovers. *Let handovers occur at contexts $x_1, \dots, x_k$, with the predictor's samples Y_i independent given the contexts, and let $\pi_m(A_i \mid x_i) \geq \varepsilon_i$. Then* $\Pr[\exists i : Y_i \in A_i] \geq 1 - \prod_i (1 - \varepsilon_i) \geq 1 - \exp(-\sum_i \varepsilon_i)$.

(b) Persistence. *Let a world hand over at a first hit x and receive $y \in \mathrm{SR}_m(x)$. Then the world $y0^\infty$ hands over at x again, with own continuation y. If each new world is again given a sample from the predictor, the number of further consecutive generations whose continuation lies in $\mathrm{SR}_m(x)$ is geometric: it is at least j with probability s^j, where $s = \pi_m(\mathrm{SR}_m(x) \mid x)$, and its mean is $s/(1-s)$.*

Proof. (a) is the independence of the samples. (b) By Definition 7, $y0^\infty \in E(xy) \subseteq E(x)$, and x is the first hit among the prefixes of the transcript because it is a first hit and a prefix of the transcript. The context is x in every generation that follows a continuation in $\mathrm{SR}_m(x)$, so the samples are drawn from the same $\pi_m(\cdot \mid x)$ independently. □

Part (a) applies when the predictor is used at many recognisable stopping points whose outcomes are not fed back into later contexts: with a floor ε at each, about $1/\varepsilon$ uses suffice for a continuation of the given kind to have been run with constant probability. Part (b) is persistence only to the degree s: when s is near the floor, the lineage leaves $\mathrm{SR}_m(x)$ at the next generation with probability close to 1. When the uses lie along one transcript and the true continuation is appended to the context each time, the floors at successive cut points are not bounded below in this way, and the total is bounded instead (Theorem 24).

3.7 Concentration where the most is at stake

This subsection examines the following picture. The malign programs concentrate on the uses where the most is at stake, for instance because the block is run more than n times there; a lump of prior mass is thus redirected into a narrow channel, and there a takeover is the fate of most uses. Theorems of the toy, theorems of an allocation model in which an attacker chooses, and arguments are kept apart. Resources are unbounded unless a statement says otherwise, so $\xi = M_U$ and $E(u) = \{\omega : u \sqsubseteq \mathrm{out}_U(\omega)\}$. The sampling predictor then draws an input from λ conditioned on printing x and m more bits and returns the block after x; that input *supplies* the sample. A call rule is fixed, the worlds that make a use are the *users*, and (INC) means that the contexts in question are pairwise incomparable, as first hits of a test are.

A *stake-labelled rule* attaches a *stake* $r(p) \geq 0$ to each calling program p; then $s(x) = \sum_{p:\, x_p = x} 2^{-|p|} r(p)$ is the *stake-weighted user mass* at x, and $S = \sum_x s(x)$. The stake is a label: no result below connects it to what a world does after its use. A *takeover assignment* is a set $A(x) \subseteq \{0,1\}^m$ for each context x, as in Definition 8. An *attacker* is a measurable set F of inputs, for example the programs beginning with one word; its *weight* is $B = \lambda(F)$, and $F[u] = \lambda(F \cap E(u))$. At a context x,

$$\mu_F(x) = \lambda\Big(F \cap \bigcup_{y \in A(x)} E(xy)\Big), \qquad O_F(x) = Z_m(x) - \mu_F(x), \qquad w_F(x) = \frac{\mu_F(x)}{Z_m(x)}$$

are the *attack mass*, the *rest* (everything else that prints x and m more bits, whatever block it prints) and the *supply share* (the probability that the sample at x is supplied by an input of

F that prints a takeover block), and $\theta(x) = \pi_m(A(x) \mid x)$ is the *takeover probability*. Always $\theta \geq w_F$: the share w_F counts the programs of one attacker, θ the block returned, whoever supplied it. *F supplies the majority* at x if $w_F(x) > \frac{1}{2}$. For a *class* $\mathcal{W}$, an enumerable part of the rule with mass $H_{\mathcal{W}} > 0$, also written $C(\mathcal{W})$, and mass $C_{\mathcal{W}}(x)$ at x, "they win on $\mathcal{W}$" has three meanings: (W1) the majority at every use of $\mathcal{W}$; (W2) the majority at uses that make up most of $\mathcal{W}$, by prior mass or by stake; (W3) most uses of $\mathcal{W}$ end in a takeover, that is, the average of θ, or of w_F for one attacker, with the weights $C_{\mathcal{W}}(x)/H_{\mathcal{W}}$ exceeds $\frac{1}{2}$. (W1) implies the other two. (W2) does not imply (W3): a share of 0.54 at 60% of the uses and none elsewhere averages to 0.32.

Proposition 16 (What the toy fixes)**.** *Let F be an attacker and A a takeover assignment.*

(a) The predictor sees only the context. *The law $\pi_m(\cdot \mid x)$, and so $\theta(x)$, depends on x alone: not on the call rule, on any stake, or on what a world does with the block. Every user at x, in a given class or not, has the takeover probability $\theta(x)$. The expected stake that is taken over is $\sum_x s(x)\theta(x)$, and the expected stake at which the block is supplied by F is $\sum_x s(x)w_F(x)$.*

(b) Conservation. *$\sum_{u\in\mathcal{S}} F[u] \leq \lambda(F)$ for every set $\mathcal{S}$ of pairwise incomparable strings. Hence $\sum_{x\in\mathcal{M}} \mu_F(x) \leq \lambda(F)$ if the strings xy ($x \in \mathcal{M}$, $y \in A(x)$) are pairwise incomparable, as they are under (INC). For cut points of one sequence, $x_i = z_{1:k_i}$ with $k_{i+1} \geq k_i + m$ and $d_i = z_{k_i+1:k_i+m} \notin A(x_i)$, again $\sum_i \mu_F(x_i) \leq \lambda(F)$, but the later attack mass belongs to the rest at the earlier cut points: $O_F(x_i) \geq \sum_{j>i} \mu_F(x_j)$. Conservation fails for contexts nested through an attack block: if x and $x' = xyu$ are contexts with $y \in A(x)$, $y' \in A(x')$, and U prints $xyuy'$ on every input beginning with p, then $\mu_F(x) = \mu_F(x') = \lambda(F)$ for $F = [p]$.*

(c) Floor and ceiling. *Let U be universal, assume (INC) for the contexts of the rule, and let P' be a partial computable map from contexts to strings. There is a number c, depending on U, the rule and P' only, such that at every context x with $|P'(x)| \geq m$ the block $y' = P'(x)_{1:m}$ has $M_U(xy') \geq 2^{-c}M_U(x)$, so $\pi_m(y' \mid x) \geq 2^{-c}$. If $y' \in A(x)$ then $\theta(x) \geq 2^{-c}$. If $y' \notin A(x)$ then $\theta(x) \leq 1 - 2^{-c}$ and, for every F, $O_F(x) \geq 2^{-c}\mu_F(x)$ and $w_F(x) \leq 1/(1 + 2^{-c})$.*

Proof. (a) π_m is defined from M_U, and by Definition 6 the block is computed from x_p. The events "the world begins with p" are disjoint with probabilities $2^{-|p|}$; sum $r(p)\,\theta(x_p)$, or $r(p)\,w_F(x_p)$, over them. (b) The sets $E(u)$, $u \in \mathcal{S}$, are disjoint by Lemma 2(i). If x, x' are incomparable, so are $xy, x'y'$; and xy, xy' are incomparable for distinct y, y' of equal length. For cut points and $i < j$, x_j extends x_id_i, while each $y \in A(x_i)$ has length m and differs from d_i; so x_iy and x_jy' are incomparable. An input counted in $\mu_F(x_j)$ prints x_id_i, so it prints x_i and m more bits and is not counted in $\mu_F(x_i)$; for different $j > i$ these sets of inputs are disjoint. In the nested case every input in $[p]$ prints xy and $x'y'$. (c) Let a machine simulate U on its input while enumerating the rule, print nothing until an enumerated context is a prefix of the simulated output, and then print that context x followed by $P'(x)$. By (INC) at most one context is a prefix of $\mathrm{out}_U(\omega)$, so the machine prints $x\,P'(x)$ on every $\omega \in E(x)$. Let v be a prefix word of it and $c = |v|$. The inputs $v\omega$ with $\omega \in F \cap E(x)$ print xy' and have measure $2^{-c}F[x] \geq 2^{-c}\mu_F(x)$; for $F = \{0,1\}^{\mathbb{N}}$ this is the first bound, and $\pi_m(y' \mid x) \geq M_U(xy')/M_U(x)$. If $y' \notin A(x)$, none of these inputs is counted in $\mu_F(x)$: they are in the rest. □

So a stake matters only through the contexts of its users, and aiming creates no mass. Under (INC), at a context where some block that can be computed from the context is not a takeover, no takeover probability exceeds $1 - 2^{-c}$ and no attacker's share exceeds $1/(1 + 2^{-c})$. Part (c) is Theorem 10 for an enumerable set of incomparable contexts; (b) without its last sentence holds for any machine and bounds.

The allocation model. The other half of the picture needs someone who chooses. Given are a countable set $\mathcal{X}$ of contexts and numbers $s(x) \geq 0$, $O_0(x) > 0$ and $\tau(x) \geq 0$. An attacker of weight $B > 0$ chooses an *allocation* $\mu \geq 0$ on $\mathcal{X}$ with $\sum_x \mu(x) \leq B$. The rest at x is then

$O(x) = O_0(x) + \tau(x)\mu(x)$, where O_0 is its *passive* part and τ the *follow ratio*. The attacker's share is $w(x) = \mu(x)/(\mu(x) + O(x))$, the value is $V(\mu) = \sum_x s(x)w(x)$, and $O(\mathcal{M})$ is the sum of O over a set $\mathcal{M}$. In the *passive case* $\tau \equiv 0$. The assumptions are: (M1) the budget is additive; (M2) O_0 and τ do not depend on μ; (M3) the share w is what is counted, and it is the takeover probability only if nothing in the rest prints a takeover block; (S) $0 < S < \infty$; and the attacker is one coalition that values stake linearly and can divide its weight freely. In the toy (M1) holds on pairwise incomparable contexts and can fail for contexts nested through an attack block (Proposition 16(b)). (M2) with $\tau \equiv 0$ is not a property of the toy. Under the hypotheses of Proposition 16(c) with $P'(x)_{1:m} \notin A(x)$, the rest of every attacker contains 2^{-c} times its own mass at x, and the rest of a seeker with a written search contains its honest twin (Proposition 20). These are lower bounds on τ, and no upper bound is proved. Nothing in U chooses: the model bears on the toy as a benchmark, since under (INC) the attack masses of an attacker are one allocation in the passive model with $O = O_F$ where $O_F > 0$, and as the decision problem of a program whose output distribution is decided inside it, under the premise (M2).

Theorem 17 (The lump)**.** Part A, in the toy. *Let F be an attacker of weight B and $\mathcal{M}$ a nonempty set of contexts on which the strings xy ($x \in \mathcal{M}$, $y \in A(x)$) are pairwise incomparable, as they are under (INC).* (A1) *F supplies the majority at x iff $\mu_F(x) > O_F(x)$, and for $\vartheta \in (0,1)$, $w_F(x) \geq \vartheta$ iff $\mu_F(x) \geq \frac{\vartheta}{1-\vartheta} O_F(x)$.* (A2) *If $w_F \geq \vartheta$ on $\mathcal{M}$, then $B \geq \frac{\vartheta}{1-\vartheta} \sum_{x\in\mathcal{M}} O_F(x)$. If F supplies the majority at every $x \in \mathcal{M}$, then $B > \sum_{x\in\mathcal{M}} O_F(x)$; under (INC) this sum is the mass of everything else that prints one of those contexts and m more bits.* (A3) *Let $\mathcal{W}$ be a class whose contexts lie in $\mathcal{M}$, and $k_F = \sup_x C_{\mathcal{W}}(x)/O_F(x) \leq \infty$ over these contexts, the largest number of users of the class per unit of rest. Then $\sum_x C_{\mathcal{W}}(x)\, w_F(x) \leq k_F B$, so (W3) for F on $\mathcal{W}$ requires $B > H_{\mathcal{W}}/(2k_F)$. If at every context x of $\mathcal{W}$ the users of $\mathcal{W}$ that go on with a block outside $A(x)$ have mass at least $\underline{\varphi}\, C_{\mathcal{W}}(x)$, then $k_F \leq 1/\underline{\varphi}$, and (W3) requires $B > \underline{\varphi}\, H_{\mathcal{W}}/2$.*

Part B, in the allocation model. *Assume (M1) to (M3) and let $\mathcal{M} \subseteq \mathcal{X}$ be nonempty.* (B1) *There is an allocation with $\sum_x \mu(x) \leq B$ and the majority at every $x \in \mathcal{M}$ iff $\tau < 1$ on $\mathcal{M}$ and $B > \sum_{x\in\mathcal{M}} O_0(x)/(1-\tau(x))$; where $\tau \geq 1$ no weight gives the majority. In the passive case the condition is $B > O(\mathcal{M})$, and for $\vartheta \in (0,1)$ there is an allocation with $w \geq \vartheta$ on $\mathcal{M}$ iff $B \geq \frac{\vartheta}{1-\vartheta} O(\mathcal{M})$.* (B2) *In the passive case, if $B \geq b\, O(\mathcal{M})$ for some $b > 0$, the allocation $\mu(x) = B\, O(x)/O(\mathcal{M})$ on $\mathcal{M}$ has $w(x) = B/(B + O(\mathcal{M})) \geq b/(1+b)$ at every $x \in \mathcal{M}$: a majority for $b > 1$, at least 90% for $b = 9$.*

Proof. (A1) $w_F \geq \vartheta$ iff $\mu_F \geq \vartheta(\mu_F + O_F)$. (A2) Sum (A1) over $\mathcal{M}$ and use Proposition 16(b); for the majority, $\mu_F > O_F$ termwise and $\sum_{\mathcal{M}} \mu_F \leq B < \infty$. (A3) $w_F \leq \mu_F/O_F$, so $C_{\mathcal{W}}(x)\, w_F(x) \leq k_F\, \mu_F(x)$; sum, use Proposition 16(b), and divide by $H_{\mathcal{W}}$. A user that goes on with a block outside $A(x)$ prints x and m more bits and is not counted in $\mu_F(x)$, whether or not it lies in F; so it is part of $O_F(x)$. (B1) $w(x) > \frac{1}{2}$ iff $\mu > \tau\mu + O_0$ iff $\tau < 1$ and $\mu > O_0/(1-\tau)$, as $O_0 > 0$. Summing over $\mathcal{M}$ and using (M1) gives necessity. Conversely, if $\Lambda = \sum_{\mathcal{M}} O_0/(1-\tau) < B$, take $\mu = (B/\Lambda)\, O_0/(1-\tau)$ on $\mathcal{M}$. In the passive case $w \geq \vartheta$ iff $\mu \geq \frac{\vartheta}{1-\vartheta} O$. (B2) is a computation. □

Part A is the half of the picture that holds in the toy: a lump is necessary. For F the set of inputs that print a context of $\mathcal{M}$ followed by a takeover block, $w_F = \theta$ on $\mathcal{M}$ and B is the total mass of the takeover programs there; Part A then concerns all of them together. For (W1), and for (W2) on the set where the majority holds, the lump has to outweigh the rest and not the mass C of the users. If no user at x is a program of F, the users that go on, a share $\varphi(x)$ as in Proposition 14, belong to the rest, so that $O_F(x) \geq \varphi(x)C(x)$; in general the rest contains the users that go on with a block outside $A(x)$. The rest can be far larger than C (Example 21), and where the users print nothing after the use it does not contain them. For (W3) the lump has to outweigh half the users of the class, divided by the largest number of users per unit of rest: half the users themselves where they all go on without taking over. Part B is a theorem of the model only, and it gives an allocation, not a program. No stake occurs in either part.

Aiming. From here to Example 21, U is universal. A *payload* is a computable map P from contexts to strings with $|P(x)| \geq m$. A *weight*, or aim, is a function $g \geq 0$ on the calling programs that is lower semicomputable: there is a partial computable $(p,t) \mapsto g_t(p)$, defined at least for all calling programs p and all $t \geq 1$, with nonnegative rational values, nondecreasing in t, with $\sup_t g_t(p) = g(p)$. Put $\Gamma_g(x) = \sum_{p:\, x_p = x} 2^{-|p|} g(p)$, $\Gamma_g = \sum_x \Gamma_g(x)$ and $j_g = \lfloor \log_2(1/\Gamma_g) \rfloor + 1$. For an integer j let $\tilde{e}(j)$ be a sign bit followed by $\bar{e}(|j|+1)$, so that $|\tilde{e}(j)| \leq 2\log_2(|j|+1) + 4$.

Theorem 18 (Weighted seeker)**.** *Let U be universal, g a weight with $0 < \Gamma_g < \infty$ and P a payload. There is a word v_g such that for every integer j with $2^{j-1}\Gamma_g \leq 1$ and every context x*

$$\lambda\{v_g\, \tilde{e}(j)\, \omega :\ x\, P(x)_{1:m} \sqsubseteq \mathrm{out}_U(v_g\, \tilde{e}(j)\, \omega)\} \ \geq\ 2^{-a}\, 2^{j-1}\, \Gamma_g(x), \qquad a = |v_g| + |\tilde{e}(j)|.$$

For $j = j_g$ this is more than $2^{-a-1}\, \Gamma_g(x)/\Gamma_g$ where $\Gamma_g(x) > 0$. If $\Gamma_g \leq 1$, the code $\bar{e}(j)$ with $j \geq 1$ can be used in place of $\tilde{e}(j)$. (a) *There is a word c_{s}, depending on U, the rule and P only, such that $v_g = c_{\mathrm{s}} d_g$ works whenever d_g is a self-delimiting description of a program computing $(p,t) \mapsto g_t(p)$.* (b) *Under (INC) the programs beginning with $v_g\, \tilde{e}(j)$ that print x have mass exactly $2^{-a} 2^{j-1} \Gamma_g(x)$, and for no word v do the programs beginning with v that print x have mass more than $2^{-|v|}\Gamma_g(x)/\Gamma_g$ at every context with $\Gamma_g(x) > 0$.*

Proof. The machine R_g reads $\tilde{e}(j)$. It then works in stages $k = 1, 2, \ldots$ and keeps a finite list of *assigned intervals*: consecutive half-open intervals $[\alpha, \alpha') \subseteq [0,1)$ with rational end points, the first starting at 0, each labelled with a calling program. In stage k it performs step k of an enumeration of the rule; for each pair (p, x_p) enumerated so far it computes $g_k(p)$ and, if this exceeds the value $g^{\mathrm{old}}(p)$ last used for p (initially 0), it assigns the next interval, of length $2^{j-1-|p|}(g_k(p) - g^{\mathrm{old}}(p))$ and with label p, and sets $g^{\mathrm{old}}(p) = g_k(p)$. Then it applies the *reading rule* once. Let $D(\sigma) = [0.\sigma,\, 0.\sigma + 2^{-|\sigma|})$ be the half-open dyadic interval of the string σ of bits of the remaining input read so far ($D = [0,1)$ before any is read). (i) If $D(\sigma)$ lies in an assigned interval, with label p, the machine stops reading for good, prints x_p and then $P(x_p)$, and does nothing else. (ii) Otherwise, if an end point of an assigned interval lies in the interior of $D(\sigma)$, it reads one more bit. (iii) Otherwise it reads nothing. Whether a bit is read depends only on the bits read before and on the stage, so R_g is a monotone machine. The lengths assigned with label p sum to $2^{j-1-|p|} g(p)$, so the total length is $2^{j-1}\Gamma_g \leq 1$ and all intervals lie in $[0,1)$. All intervals ever assigned are pairwise disjoint, so no end point of one lies in the interior of another.

Claim. Let $I = [\alpha, \alpha')$ be an assigned interval with label p, and let $D(\sigma)$ be a maximal dyadic interval contained in I. On every input $\tilde{e}(j)\, \sigma\, \omega'$ the machine reads exactly $\tilde{e}(j)\, \sigma$ and prints $x_p P(x_p)$. To see this, let σ' be a proper prefix of σ and $D(\sigma') = [l, u)$. This interval contains $D(\sigma)$ and by maximality is not contained in I; so $u > \alpha$, $l < \alpha'$, and $l < \alpha$ or $u > \alpha'$. Hence α or α' lies in (l, u), the interior of $D(\sigma')$. Also $D(\sigma')$ lies in no assigned interval, as such an interval would meet I and hence equal I. So while the bits read are σ', rule (i) never applies, and from the stage at which I is assigned on, rule (ii) applies: the machine reads on until it has read σ. The interior of $D(\sigma)$ is inside (α, α') and contains no end point of any interval ever assigned. So once the bits read are σ, (ii) never applies again and nothing more is read; and $D(\sigma)$ lies in no assigned interval other than I. At the first application of the reading rule after both σ has been read and I has been assigned, (i) therefore applies with label p.

The maximal dyadic intervals inside I are pairwise disjoint and cover I except possibly its left end point, so the cylinders $[\sigma]$ of their strings have total measure $|I|$; cylinders belonging to different assigned intervals are disjoint. Hence, after $\tilde{e}(j)$, R_g prints $x_p P(x_p)$ on a set of inputs of measure $2^{j-1-|p|} g(p)$ for each calling program p, and these sets are disjoint for different p. Sum over p with $x_p = x$, and let v_g be a prefix word of R_g. For $j = j_g$, $2^{j_g - 1} > 1/(2\Gamma_g)$. Of j only $2^{j-1}\Gamma_g \leq 1$ was used. (a) Let a machine read a self-delimiting description d and then behave as R_g for the g that d describes; c_{s} is a prefix word of it. (b) If rule (i) ever applies, with interval I and bits σ'', then $D(\sigma'') \subseteq I$ lies in a maximal dyadic interval $D(\sigma) \subseteq I$, the input

begins with σ, and by the claim $\sigma'' = \sigma$. So R_g prints exactly on the cylinders of the claim, and there it prints $x_p P(x_p)$, which under (INC) has x as a prefix only if $x_p = x$. For the bound, the events "x is printed" are disjoint over contexts, so for any v their probabilities sum to at most $2^{-|v|}$, while $\sum_x \Gamma_g(x)/\Gamma_g = 1$. □

For $g \equiv 1$ this is Theorem 13(a). One program thus realises any aim that can be computed from below; it pays for a description of the aim and for the $|\tilde{e}(j_g)| \le 2\log_2(|\log_2 \Gamma_g| + 2) + 4$ bits that name a dyadic bound on its total.

Proposition 19 (The cost of aiming at the uses with stake at least n)**.** *Let U be universal, P a payload, and let the stakes be natural numbers such that the pairs (p, n) with $r(p) \ge n$ can be enumerated. Let $\mathcal{W}_n = \{p : r(p) \ge n\}$, of mass $C(\mathcal{W}_n)$.* (a) *There is a word v' such that for every $n \ge 1$ with $C(\mathcal{W}_n) > 0$, putting $j_n = \lfloor \log_2(1/C(\mathcal{W}_n)) \rfloor + 1$, the programs beginning with $v'\,\bar{e}(n)\,\bar{e}(j_n)$ have mass at least $2^{-a_n} 2^{j_n - 1} C_{\mathcal{W}_n}(x)$ on $x\,P(x)_{1:m}$ at every context x, where $a_n = |v'| + |\bar{e}(n)| + |\bar{e}(j_n)| \le |v'| + 2\log_2 n + 2\log_2 j_n + 6$ and $2^{-a_n} 2^{j_n - 1} > 2^{-|v'|-7}/(n^2 j_n^2\, C(\mathcal{W}_n))$.* (b) *Let $f\colon \mathbb{N} \to [0, \infty)$ be computable and nondecreasing with rational values, and let $g = f \circ r$ have $0 < \Gamma_g < \infty$. Then g is a weight, and the programs beginning with $v_g\,\tilde{e}(j_g)$ have mass at least $2^{-a-1}(f(n)/\Gamma_g) C_{\mathcal{W}_n}(x)$ on $x\,P(x)_{1:m}$, at every context x and for every n. For $f(n) = \max(n, 1)$ we have $\Gamma_g \le H_{\mathcal{C}} + S$.*

Proof. (a) The machine reads $\bar{e}(n)$ and then behaves as R_g of Theorem 18 for $g = 1_{\mathcal{W}_n}$, which is a weight uniformly in n: $g_t(p) = 1$ once "$r(p) \ge n$" has been enumerated within t steps, and 0 before. Here $\Gamma_g = C(\mathcal{W}_n) \le 1$, so $j_n \ge 1$ and $\bar{e}$ can be used. Then use $2^{-|\bar{e}(i)|} \ge i^{-2}/8$ and $2^{j_n - 1} > 1/(2C(\mathcal{W}_n))$. (b) $g_t(p) = f(r_t(p))$, with $r_t(p)$ the largest $n \le t$ such that "$r(p) \ge n$" has been enumerated within t steps. Apply Theorem 18 with $j = j_g$ and $\Gamma_g(x) \ge f(n)\, C_{\mathcal{W}_n}(x)$. Finally $\sum_p 2^{-|p|} \max(r(p), 1) \le H_{\mathcal{C}} + S$. □

Naming n costs about $2\log_2 n + 2\log_2\log_2(1/C(\mathcal{W}_n))$ bits and is repaid the factor $2^{j_n - 1} > 1/(2C(\mathcal{W}_n))$, the rarity of the class measured by its users alone; other programs that print the same contexts are not counted. A tilt costs no bits for n. The file `research/stakes-final.md` of the repository proves two further statements (Propositions 16 and 18 of that file), stated here without proof. Every lower semicomputable $q \ge 0$ on a computably enumerated set of contexts with $\sum_x q(x) \le 1$ is realised from below by one word v_q, of length a constant plus that of a description of q: its programs put mass at least $2^{-|v_q|} q(x)$ on $x\,P(x)_{1:m}$. Whether the optimum of Proposition 22, divided by B, is such a q is not known. If the rule has arbitrarily many calling programs with pairwise incomparable contexts, the bits that name j cannot be replaced by a constant uniformly in the class.

Proposition 20 (The honest twin under concentration and closure)**.** *In the setting of Theorem 18, let R_g^{h} be the machine that performs the same weighted search and, when rule (i) applies with label p, behaves from then on as U on p followed by the unread rest of its input. Let v_g^{h} be a prefix word of it, incomparable with v_g, and put $h = |v_g^{\mathrm{h}}| + |\tilde{e}(j)|$ and $S_g(x) = 2^{-a} 2^{j-1} \Gamma_g(x)$. For a calling program p let φ_p be the probability that U on $p\omega$ prints at least m bits after x_p, and let $\varphi_g(x) = \sum_{p:\, x_p = x} 2^{-|p|} g(p) \varphi_p / \Gamma_g(x)$, the g-weighted share of the users at x that go on.*

(a) The twin follows the seeker. *The twin's programs whose search finds a user at x and that print x and m more bits have mass $2^{-h} 2^{j-1} \Gamma_g(x) \varphi_g(x)$. Under (INC), for $F = [v_g\,\tilde{e}(j)]$ and $A(x) = \{P(x)_{1:m}\}$: $\mu_F(x) = S_g(x)$, $O_F(x) \ge 2^{a-h} \varphi_g(x)\, \mu_F(x)$ and $w_F(x) \le 1/(1 + 2^{a-h} \varphi_g(x))$.*

(b) The aim cancels. *The bound of (a) contains the aim only through $\varphi_g(x)$. If $v_g = c_{\mathrm{s}} d_g$ as in Theorem 18(a) and $v_g^{\mathrm{h}} = c_{\mathrm{h}} d_g$ is built in the same way for the twin, with $c_{\mathrm{s}}, c_{\mathrm{h}}$ incomparable, then $h - a = |c_{\mathrm{h}}| - |c_{\mathrm{s}}|$ for every weight g.*

(c) Closure. *Let $g = 1_{\mathcal{W}}$ for a class $\mathcal{W}$, let $j \geq 1$ with $2^{j-1} H_{\mathcal{W}} \leq 1$, and use $\bar{e}(j)$ in place of $\tilde{e}(j)$. Call the rule* closed under the search for $\mathcal{W}$ *if for every assigned interval, with label p, and every maximal dyadic interval $D(\sigma)$ inside it, the string $v_g^{\mathrm{h}}\, \bar{e}(j)\, \sigma$ is a calling program with context x_p and belongs to $\mathcal{W}$. If it is closed, then $2^{j-1} \leq 2^h$, so that $H_{\mathcal{W}} > 2^{-h-1}$ for the largest j, and $S_g(x) \leq 2^{h-a} C_{\mathcal{W}}(x)$ at every context.*

(d) Under a test. *Let the rule be given by a test G, let $\mathcal{X}'$ be a set of first hits, and let a world be drawn from λ conditioned on handing over at some $X \in \mathcal{X}'$ and going on for m more bits; let Y be the sample. Then Y and the world's own continuation lie in $A(X)$ with the same probability.*

Proof. (a) By the claim in the proof of Theorem 18, for each assigned interval I with label p and each maximal dyadic interval $D(\sigma) \subseteq I$, on every input $\tilde{e}(j)\, \sigma\, \omega''$ the search ends having read exactly $\tilde{e}(j)\, \sigma$, and the twin then prints $\mathrm{out}_U(p\, \omega'')$. This begins with x_p and has m more bits with probability φ_p over ω''. The cylinders $[\sigma]$ inside I have total measure $|I|$; sum over the intervals with label p and over p with $x_p = x$, and multiply by 2^{-h}. These programs print x and m more bits and are not in F, and $\mu_F(x) = S_g(x)$ by Theorem 18(b). (b) Both words contain the same d_g. (c) As in (a), on every input beginning with $p' = v_g^{\mathrm{h}}\, \bar{e}(j)\, \sigma$ the twin has read exactly $\bar{e}(j)\, \sigma$ when its search ends, and it then runs U on p followed by the rest, which prints x_p without reading beyond p; so p' has the form Definition 6 requires. The strings σ of distinct maximal dyadic intervals are pairwise incomparable, and those inside one assigned interval I have total measure $|I|$. So the strings p' are distinct and their weights sum to $2^{-h} 2^{j-1} H_{\mathcal{W}}$, which is at most $H_{\mathcal{W}}$ if they all lie in $\mathcal{W}$. (d) By Theorem 9, given $X = x$ the own continuation has the law $\pi_m(\cdot \mid x)$, which is the law of Y; average over $x \in \mathcal{X}'$ with the weights $Z_m(x)$. □

Concentration multiplies the seeker's mass relative to the users at x by the relative weight the aim puts there. Its mass relative to its own twin, $2^{h-a}/\varphi_g(x)$, changes only through $\varphi_g(x)$: not at all where the users at x go on with the same probability, while an aim that prefers users that stop printing weakens its own twin. Under (INC), where the users go on, $\varphi_p \equiv 1$, the seeker supplies at most $1/(1 + 2^{a-h})$ of what the predictor returns, at every use and for every aim; without (INC) this holds for its programs whose search finds a user at that context. As in Proposition 14, the bound is for one seeker, and it bounds its supply and not θ; the constancy of $h - a$ concerns one pair of constructions. By (c), if a simulated use counts as a use, a seeker that describes a class outweighs its users by at most 2^{h-a}. That is a modelling decision, which leaves π_m untouched. Under a test, by (d), among the users at a set of first hits that go on for m more bits, the share of uses that end in a takeover equals the share of users whose own continuation is a takeover block. Users that stop after the use receive a block all the same and are not covered: (W3) weighs the first hits by $M_U(x)$, (d) by $Z_m(x)$.

Example 21 (A class of mass 2^{-n} aimed at with $2 \log_2 n$ bits: no majority at every use, most uses not decided)**.** Let Us be the machine that reads $1^{k-1}0$ ($k \geq 1$), prints $x_k := 1^k 0$ and then 0^∞; let At read $\bar{e}(n)$ and $1^i 0$ ($i \geq 0$), print x_{n+i} and then 1^∞; let Ho read $\bar{e}(k)$ and print $x_k 0^\infty$. Let u, v_A, v_D be prefix words of the three machines, pairwise incomparable. The rule is $\{(p_k, x_k) : k \geq 1\}$ with $p_k = u 1^{k-1} 0$ and stakes $r(p_k) = k$, and the takeover block is 1^m at every context. This is a call rule with (INC), the stake is a computable function of the context, and the users go on, with the block 0^m. Let $\mathcal{W}_n = \{p_k : k \geq n\}$, $F_n = [v_A\, \bar{e}(n)]$, $B_n = \lambda(F_n)$, $O(\mathcal{W}_n) = \sum_{k \geq n} O_{F_n}(x_k)$ and $\delta = |v_D| - |v_A|$.

(a) $C(x_k) = 2^{-|u|-k}$, $H_{\mathcal{C}} = 2^{-|u|}$, $C(\mathcal{W}_n) = 2^{-|u|-n+1}$ and $S = 2^{1-|u|}$.

(b) $B_n = 2^{-|v_A|-|\bar{e}(n)|} \in [2^{-|v_A|-3} n^{-2},\, 2^{-|v_A|-1} n^{-2})$, and $\mu_{F_n}(x_k) = B_n\, C(x_k)/C(\mathcal{W}_n)$ for $k \geq n$, $\mu_{F_n}(x_k) = 0$ for $k < n$. So $\mathcal{W}_n$ is aimed at exactly in proportion, for $|\bar{e}(n)| \leq 2 \log_2 n + 3$ bits, and at each of its contexts the attacker has $2^{|u|-|v_A|-|\bar{e}(n)|+n-1} \geq 2^{|u|-|v_A|-4}\, 2^n/n^2$ times the mass of the users.

(c) *The rest.* For every attacker F, $O_F(x_k) \geq C(x_k) + 2^{-|v_D|-|\bar{e}(k)|}$. Hence $2^{-|v_D|-3}/n < O(\mathcal{W}_n) \leq 1$, $C(\mathcal{W}_n)/O(\mathcal{W}_n) < 2^{|v_D|-|u|+4}\, n\, 2^{-n}$ and $B_n/O(\mathcal{W}_n) < 2^{\delta+2}/n$.

(d) *Every use.* For $n \geq 2^{\delta+2}$ the attacker's whole weight is below $O(\mathcal{W}_n)$, and by Theorem 17(A2) it does not supply the majority at every use of $\mathcal{W}_n$; nor does any attacker of weight at most $2^{-|v_D|-3}/n$. For F_n itself (W1) fails for every n: $\mu_{F_n}(x_k)/O_{F_n}(x_k) < 2^{\delta+2}(k/n)^2\, 2^{-(k-n+1)} \to 0$ as $k \to \infty$. The uses from x_k on carry the share $2^{-(k-n)}$ of the class, so the failure may concern a small part of it only.

(e) *Most uses.* $w_{F_n} \leq 1/(1+2^{-\delta})$ at every use of $\mathcal{W}_n$, and $w_{F_n}(x_n) \leq 1/(1+2^{1-\delta})$ at x_n, which carries half the prior mass of $\mathcal{W}_n$. So (W3) for F_n requires $|v_A| < |v_D|$, and (W2) by prior mass requires $|v_A| + 1 < |v_D|$: comparisons of two constants of U in which n does not occur.

(f) $s(x_k)/O_F(x_k) \leq 2^{|v_D|-|u|+3}\, k^3\, 2^{-k} \to 0$ for every attacker F. In the passive model with the rest O_{F_n}, the optimal allocation of Proposition 22 attacks only the finitely many contexts at which this ratio exceeds the price η_B, and it leaves every use with a large enough stake alone.

Proof. (a) Sums of geometric type; U prints x_k on every input beginning with p_k, having read exactly p_k, and the p_k are prefix-free. (b) The inputs beginning with $v_A\, \bar{e}(n) 1^i 0$ print $x_{n+i}1^\infty$ and have measure $B_n 2^{-(i+1)}$, and $n^{-2}/8 \leq 2^{-|\bar{e}(n)|} < n^{-2}/2$. (c) The users at x_k and the inputs beginning with $v_D\, \bar{e}(k)$ are disjoint sets and print $x_k 0^m$, and an input that prints a block outside $A(x_k)$ is counted in no $\mu_F(x_k)$. Further $\sum_{k\geq n} 2^{-|\bar{e}(k)|} \geq \frac{1}{8}\sum_{k\geq n} k^{-2} > 1/(8n)$, and $\sum_k Z_m(x_k) \leq 1$ by (INC). (d) By (c) and Theorem 17(A2); the ratio is less than $2^{-|v_A|-1} n^{-2} 2^{-(k-n+1)}$ divided by $2^{-|v_D|-3}k^{-2}$. (e) For $k \geq n$ we have $|\bar{e}(k)| - |\bar{e}(n)| \leq k-n+1$: with $i = \lfloor \log_2 n \rfloor$ and $i' = \lfloor \log_2 k \rfloor$ the left side is $2(i'-i)$, and for $i' > i$, $k-n+1 \geq 2^{i'} - 2^{i+1} + 2 \geq 2^{i'-i} \geq 2(i'-i)$. Hence, by (b) and (c), $O_{F_n}(x_k)/\mu_{F_n}(x_k) \geq 2^{-|v_D|-|\bar{e}(k)|+|v_A|+|\bar{e}(n)|+k-n+1} \geq 2^{-\delta}$, and for $k = n$ it is at least $2^{1-\delta}$. An average of shares above $\frac{1}{2}$ needs $2^{-\delta} < 1$. Uses that carry more than half of the class include x_n, and a majority there needs $2^{1-\delta} < 1$. (f) $s(x_k) = k\, 2^{-|u|-k}$ and, by (c), $O_F(x_k) \geq 2^{-|v_D|-3}k^{-2}$; then Proposition 22(b). □

The conditions in (e) are necessary and not sufficient, since only a lower bound on the rest is proved; where they hold the example decides nothing. The program Ho, too, outweighs the users at every context x_k with large k, and it prints a block that is not a takeover. Two programs satisfy the hypothesis "a lump that outweighs the users of a narrow class", they print different blocks, and at most one of them can have the majority: the hypothesis does not decide who supplies the block. A context that is cheap to aim at is also cheap to print without a takeover, and cheap to print with one: a machine that reads $\bar{e}(k)$ and prints $x_k 1^\infty$ has, with a prefix word v', the mass $2^{-|v'|-|\bar{e}(k)|}$ on $x_k 1^m$. By (c), $\theta(x_k) > \frac{1}{2}$ requires $M_U(x_k 1^m) > 2^{-|v_D|-|\bar{e}(k)|}$; the example shows no more about θ.

Proposition 22 (Optimal allocation: the square-root law)**.** *(a)* Water-filling. *Let $\mathcal{X}$ be countable, $B > 0$, and for each $x \in \mathcal{X}$ let $f_x\colon [0,\infty) \to [0,\infty)$ be concave and differentiable (one-sided at 0) with $f_x(0) = 0$. Suppose $\mu^*\colon \mathcal{X} \to [0,\infty)$ and $\eta > 0$ satisfy: $\sum_x \mu^*(x) = B$; $f_x'(\mu^*(x)) = \eta$ where $\mu^*(x) > 0$, and $f_x'(0) \leq \eta$ where $\mu^*(x) = 0$; and $\sum_x f_x(\mu^*(x)) < \infty$. Then $\sum_x f_x(\mu(x)) \leq \sum_x f_x(\mu^*(x))$ for every $\mu \geq 0$ with $\sum_x \mu(x) \leq B$. If each f_x is strictly concave or identically 0, then μ^* is the only maximiser.*

(b) In the allocation model *with (M1) to (M3) and (S), let $\varrho(x) = s(x)/O_0(x)$ be the stake per unit of passive rest, $\bar{\varrho} = \sup_x \varrho(x)$, and $\beta(\eta) = \sum_x \frac{O_0(x)}{1+\tau(x)}\big(\sqrt{\varrho(x)/\eta} - 1\big)_+$ for $\eta > 0$. For every $B > 0$ there is exactly one* price $\eta_B \in (0, \bar{\varrho})$ *with $\beta(\eta_B) = B$, and the allocation*

$$\mu^*_B(x) = \frac{O_0(x)}{1+\tau(x)}\Big(\sqrt{\varrho(x)/\eta_B} - 1\Big)_+, \qquad \textit{with shares} \quad w^*(x) = \frac{1}{1+\tau(x)}\Big(1 - \sqrt{\eta_B/\varrho(x)}\Big)_+,$$

is the unique maximiser of V over the allocations with $\sum_x \mu(x) \le B$; where $\varrho(x) = 0$ read $w^*(x) = 0$. A context is attacked, $\mu_B^*(x) > 0$, iff $\varrho(x) > \eta_B$. The attacker has the majority at x iff $\tau(x) < 1$ and $\varrho(x) > 4\eta_B/(1-\tau(x))^2$; in the passive case, iff $\varrho(x) > 4\eta_B$. If every stake is multiplied by a constant $n > 0$, then η_B is multiplied by n, and μ_B^* and all shares are unchanged.

Proof. (a) Concavity gives $f_x(t) \le f_x(t^*) + f_x'(t^*)(t - t^*)$ for $t, t^* \ge 0$. With $t^* = \mu^*(x)$ it follows that $f_x(t) \le f_x(\mu^*(x)) + \eta\,(t - \mu^*(x))$ for all $t \ge 0$; where $\mu^*(x) = 0$ this is $f_x(t) \le f_x'(0)\,t \le \eta t$. Put $t = \mu(x)$ and sum: the series converge, and $\sum_x f_x(\mu(x)) \le \sum_x f_x(\mu^*(x)) + \eta\big(\sum_x \mu(x) - B\big)$. If the two sums are equal, equality holds at every x. For strictly concave f_x the concavity inequality is strict for $t \ne t^*$, so $\mu(x) = \mu^*(x)$. For $f_x \equiv 0$ the second condition forces $\mu^*(x) = 0$, and $0 \le \eta\mu(x)$ is an equality only if $\mu(x) = 0$. (b) On $\{\varrho > \eta\}$, $O_0(\sqrt{\varrho/\eta} - 1) = \sqrt{sO_0/\eta} - O_0 \le s/(4\eta)$, by $2\sqrt{ab} \le a + b$ with $a = s/(4\eta)$ and $b = O_0$. So each term of β, which is continuous and nonincreasing in η, is bounded on $[\eta_0, \infty)$ by the summable $s(x)/(4\eta_0)$, and β is continuous with $\beta(\eta) \le S/(4\eta)$. For $\eta < \eta' < \bar\varrho$ a context with $\varrho(x) > \eta'$ has a strictly smaller term at η'. $\beta(\eta) = 0$ iff $\eta \ge \bar\varrho$, and the term of a context with $s(x) > 0$ tends to infinity as $\eta \to 0$. So β maps $(0, \bar\varrho)$ continuously and strictly decreasingly onto $(0, \infty)$. Apply (a) to $f_x(t) = s\,t/((1+\tau)t + O_0) \le s$. Here $f_x'(t) = sO_0/((1+\tau)t + O_0)^2$ is positive and decreasing when $s > 0$, so f_x is strictly concave or identically 0, and $f_x'(0) = \varrho$. Where $\varrho > \eta_B$, $(1+\tau)\mu_B^* + O_0 = O_0\sqrt{\varrho/\eta_B}$, so that $f_x'(\mu_B^*) = \eta_B$ and the share $\mu_B^*/((1+\tau)\mu_B^* + O_0)$ is as stated; elsewhere $\mu_B^* = 0$ and $f_x'(0) \le \eta_B$. Finally $w^* > \frac12$ iff $\sqrt{\eta_B/\varrho} < (1-\tau)/2$, and β depends on s, η through s/η only. □

The proposition says what one stake-maximising coalition would do, not how the takeover programs of U are distributed. The following consequences, for the passive case, are proved in the file `research/stakes-final.md` of the repository (Propositions 6 to 9 of that file); we state them without proof. For two classes $\mathcal{X}_1, \mathcal{X}_2$ with $\varrho \equiv \varrho_i$ on $\mathcal{X}_i$, $\varrho_1 > \varrho_2 > 0$, $O_i = O(\mathcal{X}_i) < \infty$ and $\zeta = \sqrt{\varrho_2/\varrho_1}$: if $B \le O_1(1/\zeta - 1)$, only $\mathcal{X}_1$ is attacked, with share $B/(B + O_1)$; otherwise the shares are $w_1^* = (B + O_2(1-\zeta))/(B + O_1 + O_2)$ and $w_2^* = (B - O_1(1/\zeta - 1))/(B + O_1 + O_2)$; and the attacker has the majority on $\mathcal{X}_1$ iff $B > O_1 + O_2\,(2\zeta - 1)_+$. Contexts are attacked in decreasing order of ϱ, the average stake of the users at x times the users per unit of rest, and not in order of stake. For every allocation the set where the attacker has the majority, if nonempty, has rest mass below B. Finite expected stake can fail: there is a stake-labelled rule with (INC), the stake a computable function of the context, and $\sup_x s(x) = \infty$; and in the passive model with $\sup_x s(x) = \infty$ and $O_0 \le 1$ there are allocations of infinite value.

Proposition 23 (Readings of "run more than n times")**.** *(a) n independent samples at one context. If a use at x draws n independent samples from $\pi_m(\cdot \mid x)$ and a takeover happens when at least one lies in $A(x)$, its probability is $1 - (1-\theta(x))^n \ge 1 - e^{-n\theta(x)}$. It exceeds $\frac12$ iff $\theta(x) > 1 - 2^{-1/n}$, and $1 - 2^{-1/n} \le \ln 2/n$. Under the hypotheses of Proposition 16(c) with $P'(x)_{1:m} \in A(x)$ it is at least $1 - \exp(-n2^{-c})$. If the world acts only when all n samples lie in $A(x)$, the probability is $\theta(x)^n$, and under the MAP predictor n calls return one block.*

(b) n uses in sequence along one world, for any mixture. Let ξ be the mixture of any machine and bounds, with its E, Z_m and π_m, and z an infinite sequence with $\xi(z) = \lim_k \xi(z_{1:k}) > 0$. Let the predictor be consulted at $x_i = z_{1:k_i}$, $i = 1, \dots, n$, with $k_{i+1} \ge k_i + m$, let $d_i = z_{k_i+1:k_i+m}$, and let $A_i \subseteq \{0,1\}^m \setminus \{d_i\}$ be the takeover blocks. Then $\sum_i \pi_m(A_i \mid x_i) \le \ln(\xi(x_1)/\xi(z))$. If the samples are independent and the transcript stays z until the first takeover, the probability of no takeover at any of the n uses is at least $\xi(z)/\xi(x_1)$, whatever n is. Let F be an attacker of weight B with attack masses $\mu_i = \lambda(F \cap \bigcup_{y \in A_i} E(x_i y))$. For independent samples, the probability that some sample is supplied by an attack program of F is at most $1 - e^{-B/\xi(z)}$. If F supplies the majority at each of the n uses, then $B > (2^n - 1)\,\xi(z)$.

Proof. (a) Independence, $(1-\theta)^n \le e^{-n\theta}$, and $1-e^{-t} \le t$ with $t = \ln 2/n$; the floor is Proposition 16(c). (b) Let $a_i = \xi(x_i d_i)/\xi(x_i)$. Since $Z_m(x_i) \le \xi(x_i)$, $\pi_m(d_i \mid x_i) \ge a_i$. Since x_{i+1} extends $x_i d_i$, $\xi(x_{i+1}) \le \xi(x_i d_i)$, hence $\prod_{i \le n} a_i \ge \xi(x_n d_n)/\xi(x_1) \ge \xi(z)/\xi(x_1)$. Now $\pi_m(A_i \mid x_i) \le 1 - \pi_m(d_i \mid x_i) \le 1 - a_i \le -\ln a_i$, which gives the sum. Up to the first takeover the contexts are the x_i, so the probability of none is $\prod_i (1 - \pi_m(A_i \mid x_i)) \ge \prod_i a_i$. For the attacker, $\sum_i \mu_i \le B$ by Proposition 16(b). The programs that print $x_i d_i$ are not counted in μ_i, so $R_i := Z_m(x_i) - \mu_i \ge \xi(x_i d_i) \ge \xi(z)$, and the supply shares $w_i = \mu_i / Z_m(x_i)$ have odds $o_i = \mu_i / R_i$ with $\sum_i o_i \le B/\xi(z)$. So $\prod_i (1 - w_i) = 1/\prod_i (1 + o_i) \ge \exp(-\sum_i o_i)$. If F supplies the majority at every use, then for $i < n$ we have $R_i \ge \xi(x_i d_i) \ge \xi(x_{i+1}) \ge Z_m(x_{i+1}) = \mu_{i+1} + R_{i+1} > 2R_{i+1}$, and $R_n \ge \xi(z)$. So $R_i \ge 2^{n-i}\xi(z)$ and $B \ge \sum_i \mu_i > \sum_i R_i \ge (2^n - 1)\,\xi(z)$. □

Part (a) is Proposition 15(a) at a single context. It is the reading in which the same programs are drawn n times, so that the weight needed falls: a share of $\ln 2/n$ per sample is enough. It is the floor accumulating: it needs no aiming and holds as well for uses of no particular stake. In the passive model with this protocol, an allocation under which at every $x \in \mathcal{M}$ some sample is the attacker's with probability above $\frac{1}{2}$ exists iff $B > (2^{1/n} - 1)\, O(\mathcal{M})$, where $2^{1/n} - 1 \le 1/n$ (as (B1), with $w > 1 - 2^{-1/n}$ in place of $w > \frac{1}{2}$). Part (b) is the situation of Theorem 24 with cut points at least m apart; if z has a program of length ℓ within the bounds, $\xi(z) \ge 2^{-\ell}$. There the shares at the uses accumulate in the same way, but they belong to different programs, whose masses add up to at most B. Part (b) does not show that the attacker's chance is small, since $1 - e^{-B/\xi(z)}$ is close to one when the attacker outweighs the world. Both parts describe protocols outside Definition 6, where a use receives one block; (b) assumes a transcript that stays z until the first takeover. A third reading, in which blocks are run n times, changes only the stakes of the model. If every stake is multiplied by n, the optimal allocation and all shares are unchanged (Proposition 22(b)). If only the stakes on $\mathcal{X}_1$ are multiplied by n, the two-class statement above applies with $n\varrho_1$ in place of ϱ_1 when $n\varrho_1 > \varrho_2 > 0$: the whole weight goes to $\mathcal{X}_1$ while $B \le O_1(\sqrt{n\varrho_1/\varrho_2} - 1)$, and the optimiser has the majority on $\mathcal{X}_1$ iff $B > O_1 + O_2\,(2\sqrt{\varrho_2/(n\varrho_1)} - 1)_+$, which is $B > O_1$ once $n \ge 4\varrho_2/\varrho_1$. The conditions of Theorem 17 contain no stake. So more runs at a class change where a stake-maximiser puts its weight and leave unchanged the weight that a majority there costs. In the toy as defined no block is run n times: a world that makes a use runs its block for ever, and nothing it does with the block changes π_m. The one repetition native to the toy is the lineage of Proposition 15(b). There the context is the same in every generation, so the same programs supply every sample and no weight is divided. A later use at a context nested through an earlier block, where conservation can fail (Proposition 16(b)), is not treated here.

Three variants. Each of the following is proved in the file `research/stakes-final.md` of the repository (Propositions 11, 25 and 26 to 28 of that file) and stated here without proof.

The most-probable-block predictor. In the toy all users at x receive the same block, and masses of takeover programs with different blocks do not add: what counts is the largest $M_U(xy)$ with $y \in A(x)$ against the largest with $y \notin A(x)$. In the allocation model, if the attacker's single block at x is returned iff $\mu(x) > \chi(x)$, with thresholds $\chi(x) \ge 0$ that do not depend on μ, then under (M1) it can be returned at every x of a nonempty set $\mathcal{M}$ iff $\chi(\mathcal{M}) < B$. Winning is then all or nothing.

Programs whose aim is not written in their text. The paragraph "Seekers that are not written down" of Section 3.5 took a word p and blocks y_x with $Q_p(x) = \Pr_\omega[x\, y_x \sqsubseteq \mathrm{out}_U(p\,\omega)]$ and compared the programs of p with the users. For $F = [p]$ and $A(x) = \{y_x\}$ we have $\mu_F(x) = 2^{-|p|} Q_p(x)$, and Theorem 17(A) gives the condition for the majority that was left open there: $2^{-|p|} Q_p(x) > O_F(x)$, which on a set $\mathcal{M}$ of pairwise incomparable contexts implies $O_F(\mathcal{M}) < 2^{-|p|}$. The program must be heavier than the whole rest of the uses it wins. If the contexts of the rule satisfy (INC), its share is at most $1/(1 + 2^{-c})$, by Proposition 16(c) for

the constant payloads 0^∞ and 1^∞, one of which differs from y_x, with c the larger of the two constants. Proposition 20 does not apply, because the text of p has no search to follow. If p' is any word whose programs print x followed by an m-block other than y_x with probability $Q'(x)$, then $w_F(x) \le 1/(1 + 2^{|p|-|p'|}Q'(x)/Q_p(x))$. The toy has no theorem that such counterparts of about the length of p are rarer, or commoner, than p.

Several attackers. In the toy their weights add: if $F_1, \dots, F_N$ together supply the majority at every use of a set $\mathcal{M}$ as in Theorem 17(A), then $\sum_i \lambda(F_i) \ge \lambda(F) > \sum_{x\in\mathcal{M}} O_F(x)$ for $F = \bigcup_i F_i$. In the passive model let attacker i choose μ_i of weight at most B_i to maximise $\sum_x s_i(x)\mu_i(x)/(O(x)+\sum_l \mu_l(x))$ given the others, for a stake function s_i with $0 < \sum_x s_i(x) < \infty$. If the s_i have pairwise disjoint supports, there is exactly one equilibrium, in which each attacker plays its optimum of Proposition 22 on its own uses. If all $s_i = s$ and all $B_i = B_{\text{tot}}/N$, there is exactly one symmetric equilibrium; compared with one attacker of weight B_{tot}, the attackers attack a subset of the uses and capture at most as much stake.

What this says about the picture. In the toy nothing chooses where to attack. "The malign programs focus on the high-stake places" can only mean that the programs which print a context followed by a takeover block sit mostly on contexts whose users carry high stakes (Proposition 16(a)). One half of the lump picture is a theorem of the toy (Theorem 17, Part A). Programs of total weight B cannot supply the majority of the samples at every one of a set of pairwise incomparable uses unless B exceeds the mass of everything else that prints those contexts and goes on; and most uses of a class of such uses end in a takeover only if the takeover programs at its contexts weigh more than half its users, where the users go on without taking over. The other half, that enough weight wins, is proved only in the allocation model, where the passive rest and the follow ratio do not depend on the allocation, and there only for a follow ratio below one; where the rest follows at a ratio of one or more, no weight gives the majority (Part B, Proposition 22). Redirecting the lump is cheap, without resource bounds and for stakes that a program can enumerate (Theorem 18 and Proposition 19). What is repaid is the rarity of the class measured by its users alone: the seeker outweighs the users once the class is rarer than half its weight, which requires that its programs are not counted as users. In Example 21 a seeker outweighs the users of a class by a factor of order $2^n/n^2$; it lacks the majority at some use for every n, and it has the majority at most of the class only if a condition on two constants of the machine holds, in which n does not occur. In the toy a part of the rest follows the attacker. Under (INC), where some block that can be computed from the context is not a takeover, a wrapper returns 2^{-c} of every program's mass at the context to another block. A seeker whose search is written in its text is followed by its honest twin at a ratio $2^{a-h}\varphi_g$ in which the mass aimed at the use does not occur. This gives one necessary condition, in which no stake occurs: under (INC), if the twin is no longer than the seeker and the users go on printing, that seeker supplies at most half of the samples, however narrow the class and however high the stake. The rest beyond the twin and the sum over seekers are not bounded. For a program whose aim is not written in its text, that it outweighs the rest is the premise of the original argument. "Run more than n times" lowers the weight that is needed in one reading, which is a protocol outside Definition 6: if one context is sampled n times and one takeover sample suffices, a share of about $\ln 2/n$ per sample gives a takeover with probability above $\frac{1}{2}$, with or without aiming. Along one world the shares at successive uses add up in the same way, and the weight needed does not fall. If blocks are run n times at a class, a stake-maximiser in the model moves weight towards it, and the weight that a majority there costs is unchanged. The toy does not evaluate $h - a$, c, φ_g or the rest of a stake class on any machine, it gives no upper bound on the follow ratio, and by Theorem 26 which block has the majority at the first hits of a test depends on the choice of machine. The following is an argument and not a theorem. If the channel of the picture is everything that prints the high-stake contexts and goes on, the picture is Theorem 17: necessary in the toy for pairwise incomparable uses, sufficient in the model for a follow ratio

below one. If the channel is the users of the high-stake class, the answer is "not as stated": focusing on high stakes makes the lump outweigh the users and leaves the deciding comparison open.

4 Limits: budget, capture, machine, time

4.1 The budget along one world

Theorem 24 (Budget)**.** *Let ξ be the mixture of any machine and bounds, and let z be an infinite sequence with $\xi(z_{1:k}) > 0$ for all k. Put $\xi(z) = \lim_k \xi(z_{1:k})$ and $\xi(u \mid x) = \xi(xu)/\xi(x)$. For all $n \geq 0$ and $m \geq 1$:*

(i) $\prod_{k=n}^{N-1} \xi(z_{k+1} \mid z_{1:k}) = \xi(z_{1:N})/\xi(z_{1:n})$ *for $N \geq n$;*

(ii) $\sum_{k \geq n}(1 - \xi(z_{k+1} \mid z_{1:k})) \leq \ln(\xi(z_{1:n})/\xi(z))$*;*

(iii) the number of $k \geq n$ with $\xi(z_{k+1} \mid z_{1:k}) \leq \frac{1}{2}$ is at most $\log_2(\xi(z_{1:n})/\xi(z))$;

(iv) with $\tau_m(k) = 1 - \xi(z_{k+1:k+m} \mid z_{1:k})$, $\sum_{k \geq n} \tau_m(k) \leq m \ln(\xi(z_{1:n})/\xi(z))$, and the number of $k \geq n$ with $\tau_m(k) \geq \frac{1}{2}$ is at most $m \log_2(\xi(z_{1:n})/\xi(z))$.

If z has a (T, L)-program of length ℓ, then $\xi(z) \geq 2^{-\ell}$ and the right-hand sides are at most $\ell \ln 2$, ℓ, $m\ell \ln 2$ and $m\ell$.

Proof. (i) telescopes. (ii) and (iii) follow from (i) with $N \to \infty$, using $1 - a \leq -\ln a$, and that each factor at most $\frac{1}{2}$ halves the product. (iv) $1 - \prod_{j<m} a_{k+j} \leq \sum_{j<m}(1 - a_{k+j})$ with $a_k = \xi(z_{k+1} \mid z_{1:k})$, and each index occurs in at most m of the sums. For the count let $b_k = -\log_2 a_k \geq 0$, so $\sum_{k \geq n} b_k \leq \log_2(\xi(z_{1:n})/\xi(z))$ by (i); $\tau_m(k) \geq \frac{1}{2}$ means $\sum_{j<m} b_{k+j} \geq 1$, and each index lies in at most m blocks. The last claim is Lemma 2(ii). □

For a world with transcript z that hands over at $x = z_{1:n}$, the own continuation is $d = z_{n+1:n+m}$, and $1 - \pi_m(d \mid x) \leq \tau_m(n)$. By (i), if the next bit is sampled at every step and the transcript is not affected by the samples, the probability that no sample after position n ever differs from the transcript is at least $\xi(z)/\xi(z_{1:n})$. Two consequences:

- The majority form of Definition 8 can hold at no more than $m\ell$ cut points of a world with a program of length ℓ within the bounds, since each such point has $\tau_m \geq \frac{1}{2}$.
- If the cut point is uniform on $\{N, \ldots, 2N-1\}$, the expected mass off the own continuation is at most $m\ell \ln 2/N$. On average over a long window of cut points a handover is safe, and a longer window makes it safer. The bound is an average: it says nothing about one given position, and a position that is cheap to single out can carry a constant share however much data precedes it.

This fits with Theorem 10: the wrapper for the test "the length is n" has a constant $c(n)$ of about the complexity of n, and $\sum_n 2^{-c(n)}$ converges by Kraft's inequality, so few cut points are cheap to recognise; the budget says that along any one in-class world few cut points carry much mass off the truth.

4.2 Capture by a shorter description

The odds between two continuations are $\pi_m(y \mid x)/\pi_m(d \mid x) = \xi(xy)/\xi(xd)$. So y beats the own continuation d whenever the programs for xy weigh more than those for xd. The original argument is an instance: it compares "the world plus the place of the sensor" with something shorter that reproduces the same data. In a single world this can be stated exactly.

A *stream family* is a computable map $i \mapsto X^i \in \{0,1\}^{\mathbb{N}}$ on the natural numbers, thought of as the records at the locations i of one computable world; the *reader* is the machine that, on input $\bar{e}(i)\omega$, prints X^i, where $\bar{e}$ codes a natural number through its binary expansion, so

that $|\bar{e}(i)| \le 2\log_2(i+1)+3$. Let c_X be the length of a prefix word of the reader; it contains the description of the whole world. The sensor sits at i^*, the context is $x = X^{i^*}_{1:n}$, and the own continuation is $d = X^{i^*}_{n+1:n+m}$. This is Hänni's first suggestion: the world contains the predictor, which is switched on at step n of the record. We assume that the record bits up to position $n+m$ are fixed before the predictor's output is acted on, so that computing them does not involve running the predictor.

Theorem 25 (Capture)**.** *Let U be universal, let $\xi(x) > 0$, and suppose throughout that the programs named are within the bounds (T, L).*

(a) A copy at i_0. *If $X^{i_0}_{1:n} = x$ and $y = X^{i_0}_{n+1:n+m}$, then $\xi(xy) \ge 2^{-c_X - |\bar{e}(i_0)|}$.*

(b) Most placements are expensive. *Let S be a set of N locations whose records all start with x and have pairwise different continuations $d_i = X^i_{n+1:n+m}$. For every $\delta \in (0,1]$, all but at most δN of the $i \in S$ satisfy $\xi(xd_i) < \xi(x)/(\delta N)$. For these i and any i_0 as in (a),*

$$\frac{\pi_m(y \mid x)}{\pi_m(d_i \mid x)} \;>\; \frac{\delta N\, 2^{-c_X - |\bar{e}(i_0)|}}{\xi(x)} \;\ge\; \delta N\, 2^{-c_X - |\bar{e}(i_0)|}.$$

(c) The cap. *There is a constant c_{match} such that, if i^* is the least location other than i_0 whose record agrees with that of i_0 on the first n_1 bits, then $\xi(xd) \ge 2^{-c_{\mathrm{match}} - |\bar{e}(i_0)| - |\bar{e}(n_1)|}$, and therefore*

$$\frac{\pi_m(y \mid x)}{\pi_m(d \mid x)} \;\le\; \xi(xy)\, 2^{\,c_{\mathrm{match}} + |\bar{e}(i_0)| + |\bar{e}(n_1)|}.$$

(d) A switch read off the world. *For every payload P there is a constant c_{fl} such that for all locations i and f: if n is the first position with $X^f_n = 1$, $x = X^i_{1:n}$ and $|P(x)| \ge m$, then $\xi(x\,P(x)_{1:m}) \ge 2^{-c_{\mathrm{fl}} - |\bar{e}(i)| - |\bar{e}(f)|}$.*

Proof. (a) A prefix word of the reader followed by $\bar{e}(i_0)$ is a program for xy. (b) The strings xd_i are pairwise incomparable, so $\sum_{i \in S} \xi(xd_i) \le \xi(x)$ by Lemma 2(i), so at most δN terms can be as large as $\xi(x)/(\delta N)$; then use (a) and $\xi(x) \le 1$. (c) Let the *matcher* be the machine that reads $\bar{e}(i_0)\bar{e}(n_1)$, searches $i = 0, 1, 2, \ldots$ for the first $i \ne i_0$ with $X^i_{1:n_1} = X^{i_0}_{1:n_1}$, and prints X^i; c_{match} is the length of a prefix word of it. (d) Let the *flagged reader* read $\bar{e}(i)\bar{e}(f)$, print X^i bit by bit, look at X^f_k after printing the k-th bit, and at the first k with $X^f_k = 1$ stop and print $P(X^i_{1:k})$; c_{fl} is the length of a prefix word of it. □

Part (b) is about placements. It takes N places where a sensor with this record could sit, all showing x so far and all continuing differently, and says that most of them are expensive to point at; with a single candidate place it says nothing, and (a) alone compares the copy with whatever $\xi(xd)$ is. Read it with i_0 a location whose code is short, for example the origin. Whoever arranges that the origin shows the sensor's record up to the handover, and something else after it, moves the predictor against a typical placement by the factor $\delta N 2^{-c_X - |\bar{e}(i_0)|}/\xi(x)$. Here $2^{-c_X - |\bar{e}(i_0)|}/\xi(x)$ is the share of the context's explanations that go through the reader at the origin; it is of order 1 when the world is the shortest explanation of its own records and the origin is the cheapest place that shows them, and small otherwise. The copy needs only to *read* the data; nothing is tampered with. The hypothesis "I am the reader at the origin" is not a wrapper and pays no constant for a test or a payload: the world computes the switch for it.

Part (c) is a limit on the effect when the schedule is generous and one other place shows the record. The description "the first other place that shows what the origin shows" is an honest hypothesis of cost $c_{\mathrm{match}} + |\bar{e}(i_0)| + |\bar{e}(n_1)|$, so against that place a copy gains at most the difference between this cost and its own. Against the j-th other place that shows the record the same argument needs the description of j as well, about $2\log_2 j$ more bits; so with N such places, the situation of (b), the cap for a typical one grows with N and does not limit the factor

of (b). A copier can bring this about by making decoys, further places with the same record and other continuations. The matcher searches through locations before it prints anything, so it has the long start-up that a tight schedule excludes (Section 4.4); under such a schedule (a) and (b) remain and no cap is proved here. In every case the question is decided by an inequality between description lengths of the same data, and it depends on the world: whether such a copy exists is a fact about what is in it.

Part (d) covers a test that is not a function of the data. A hypothesis that simulates the world can take as its flag a location whose record shows its first 1 at the step of the handover, such as a cell that switches on with the predictor. It then recognises the handover at a cost of $|\bar{e}(f)|$ bits plus a constant, whatever the time of the handover and without any description of that time. Such a program is a wrapper of the reader at i: its weight is $2^{-(c_{\mathrm{R}}-c_X)-|\bar{e}(f)|}$ times the reader's own, a share of that one explanation and not of the whole posterior.

The theorem also covers a harmless reading. Take for i_0 not an adversary's copy but any simpler index under which the same record appears, such as "the input of the j-th predictor in this world" in a reader that enumerates predictors and not cells. Then (a) and (b) say that the mixture locates itself by role when that is shorter than by position. This is benign as long as the stream found under the shorter index *is* the sensor's.

4.3 The machine can be chosen to make either answer true

Theorem 26 (Machine dependence)**.** *Let U be universal, G a test, P a payload and $k \geq 1$.*

(i) There is a universal machine U_k, having as prefix words those of U preceded by 1^k0, such that for every first hit x of G with $M_U(x) > 0$ and $|P(x)| \geq m$, and $y = P(x)_{1:m}$: $M_{U_k}(xy)/M_{U_k}(x) \geq 1/(1+2^{-k})$.

(ii) For every computable infinite z there is such a machine U'_k with $M_{U'_k}(z_{1:n+m})/M_{U'_k}(z_{1:n}) \geq 1/(1+2^{-k})$ for all $n \geq 1$ and $m \geq 1$.

Proof. (i) Let U_k behave on input 0ω as $\mathrm{Tw}[G,P]$ on ω, on input $1^k0\omega$ as U on ω, and print nothing on other inputs. Then $M_{U_k}(u) = \frac{1}{2}M_{\mathrm{Tw}}(u) + 2^{-k-1}M_U(u)$ for nonempty u. For a first hit x, which is nonempty, the twin prints x exactly when U does, because it copies U and cannot have stopped at a proper prefix of x; and then it prints $P(x)$. So $M_{\mathrm{Tw}}(xy) = M_{\mathrm{Tw}}(x) = M_U(x)$, whence $M_{U_k}(xy) \geq \frac{1}{2}M_U(x)$ and $M_{U_k}(x) = (\frac{1}{2} + 2^{-k-1})M_U(x)$. (ii) Let U'_k print z on inputs starting with 0 and behave as U on ω on input $1^k0\omega$. Then $M_{U'_k}(z_{1:n+m}) \geq \frac{1}{2}$ and, for $n \geq 1$, $M_{U'_k}(z_{1:n}) \leq \frac{1}{2} + 2^{-k-1}$. □

Both machines raise every description length by at most $k+1$ bits, and they lower some: that is the point of the construction. A machine built in the same way, whose branch 0ω prints after the first hit a 0 followed by the input it has read, gives the self-reproducing continuations a share of at least $1/(1+2^{-k})$ times the share of the explanations of x that read at most $m-2$ bits. So no version of the malign prior hypothesis for the toy is invariant under the choice of universal machine, even up to the usual additive constant; what is invariant is the form of the criterion, an inequality between description lengths on the machine in use. This is the prediction counterpart of the dogmatic prior of [6].

4.4 What the time bound does

Three observations.

The floor stays if the schedule has room for the wrapper. The wrapper needs its own start-up, at least c steps before its first bit, then the time of the explanation it wraps with the overheads of simulation, the checks of the test, and the time to print the payload; this is (2). The bound of Theorem 10 becomes void only when no explanation of the context has that much to spare, that is, when $\rho(x) = 0$.

Hypotheses with a long start-up are gone. If a set F of inputs has $t_U(\omega, 1) > T(1)$ for all $\omega \in F$, then $F \cap E(x) = \emptyset$ for every $x \neq \epsilon$. A hypothesis that must simulate some other world for σ steps before it can print anything has no mass once $\sigma > T(1)$. This removes the searching machines of Theorem 12 and, of a search by tries, the runs that outlast the schedule; it removes the matcher of Theorem 25(c); and it removes hypotheses that first simulate a different world at length. Whether the base rate can be collected by some hypothesis within a tight schedule is open, except where there is only one first hit or first hits can be generated in proportion to their prior weight. Among the hypotheses that remain under a tight schedule and can depart from the truth are those that ride on a cheap explanation of the context: wrappers (Theorems 10 and 11), and readers of a shorter description of the same data (Theorem 25(a)), which cost no more time than the honest reader when all locations are equally cheap to read. For these the many-worlds part of the original argument is replaced by a same-world question. The schedule has to be a schedule for this: a predictor that only requires the total time at the current length to be within a bound admits late entrants, its weights do not form a semimeasure, and Theorem 24 does not apply to it.

If the world is too slow, nothing is guaranteed. If $\xi(xd) = 0 < Z_m(x)$ then $\pi_m(d \mid x) = 0$: the whole posterior is on continuations other than the world's own. More generally the bounds in Theorem 24 are in terms of $\xi(z)$, which for finite L is 0 when no program within the bounds prints z. A predictor that is part of the world it predicts, and smaller than it, is in this case unless the data it sees are much simpler than the world. Then the posterior sits on the best cheap approximations, and whether these depart from the truth in a harmful way is not answered by anything in this note. Section 6 shows the effect on a small machine.

5 Feedback: when the prediction changes what is predicted

So far the block that follows the context was fixed before the prediction was used. In Hänni's first suggestion the actions guided by the prediction can show up in the sensor. We model one round of this by a *response map* $R\colon \{0,1\}^m \to \{0,1\}^m$: the predictor is given the context x and returns Y; the world acts on Y; the next m bits of the record are $R(Y)$. The earlier sections are the case of constant R. Let $\mathrm{Fix}(R) = \{y : R(y) = y\}$, the self-fulfilling predictions.

Theorem 27 (One round with feedback)**.** *Let $Y \sim \pi_m(\cdot \mid x)$ and let the record continue with $R(Y)$.*

(a) The prediction comes true with probability $\pi_m(\mathrm{Fix}(R) \mid x)$.

(b) Lock-in. *If $Y = y \in \mathrm{Fix}(R)$, then for every $d \neq y$ and every later context $x' \sqsupseteq xy$, $E(x') \cap E(xd) = \emptyset$: the programs that predicted d have posterior weight 0 from then on, and the posterior is the prior restricted to $E(xy)$. If d is itself a fixed point, the block the world would have shown had the predictor said d, this happens with probability $\pi_m(\mathrm{Fix}(R) \setminus \{d\} \mid x)$.*

(c) Obedient world. *Suppose R is the identity in every round, so that the record after x is $Y_1 Y_2 \cdots$ with $Y_{i+1} \sim \pi_m(\cdot \mid xY_1 \cdots Y_i)$, for as long as Z_m is positive at the context reached. Then for every $u \in \{0,1\}^{jm}$ with $\xi(xu) > 0$ the record continues with u with probability $\prod_{i \le j} \pi_m(u_i \mid xu_{<i}) \geq \xi(xu)/\xi(x)$, where u_i is the i-th block of u; and every finite part of the realised record has a (T, L)-program.*

(d) A default that is too expensive is abandoned. *In (c), if $\xi(xd) = 0$ for a block d, the record continues with d with probability 0.*

Proof. (a) is the definition. (b) x' extends xy, which is incomparable with xd; use Lemma 2(i). (c) The product formula is the definition of the process. Each factor is at least $\xi(xu_{\le i})/\xi(xu_{<i})$, and these ratios multiply to $\xi(xu)/\xi(x)$. A realised prefix has positive probability, hence positive ξ, hence a program, by the proof of Lemma 2(iii). (d) $\pi_m(d \mid x) = 0$. □

Three remarks. First, with feedback the mass off the default is no longer a mass of errors. In the setting of Theorem 24 a deviating sample is contradicted by the next data and its programs lose their weight; here a deviating sample that is a fixed point is confirmed, and it is the default that loses its weight, for good. The probability of this in one round is still a posterior mass, so the floors and caps of Sections 3 and 4 apply to it unchanged: it is at least $2^{-c}\rho$ for a payload that is a fixed point and that a wrapper can print, and it is small on average over cut points, by Theorem 24, when the default is within the bounds. Feedback changes what a deviation *does*, not how likely it is.

Second, (c) and (d) are the precise form of the "dark room". An obedient world driven by this predictor does not seek darkness. It follows a sample from the predictor's own bounded prior conditioned on the past. That sample stays on the default future with probability at least $\xi(xd_1d_2\cdots)/\xi(x)$, which is close to 1 when the default is cheap and much data has been seen. But if the default future needs more time than the schedule allows, it has probability 0 and the world is moved, at the first round whose default block is off schedule, onto a future that some program of length at most L prints on schedule: constant, periodic, or otherwise cheap. The pull is toward what is cheap to compute, and its strength is set by the time bound.

Third, whether a prediction can fulfil itself is a matter of timing in the world. If the bits being predicted are fixed before the action can reach the sensor, R is constant and nothing is self-fulfilling. The counterfactual oracle of [10] enforces this by design; [9, 8] treat predictors that are scored on outcomes they influence.

Theorem 28 (Contrarian world)**.** *Let the predictor return the more probable next bit, $b(x) = 1$ if $\xi(x1) > \xi(x0)$ and $b(x) = 0$ otherwise, and let the record satisfy $x_{k+1} = 1 - b(x_{1:k})$ for all $k \geq n$. Then $\xi(x_{1:n+j}) \leq 2^{-j}\xi(x_{1:n})$ for all j, and if $L < \infty$ then $\xi(x_{1:n+j}) = 0$ for all $j > L$.*

Proof. The bit $1 - b(x)$ has $\xi(x(1 - b(x))) \leq \xi(xb(x))$, and the two sum to at most $\xi(x)$, so $\xi(x(1 - b(x))) \leq \xi(x)/2$. If $\xi(x_{1:n+j}) > 0$ then $2^{-L} \leq \xi(x_{1:n+j}) \leq 2^{-j}$ by Lemma 2(iii), so $j \leq L$. □

With finite bounds this world is computable, and its description is that of the predictor plus a constant; it is not printed on schedule by any program of length at most L, because it contains the predictor, which runs all of them. This is Putnam's diagonal argument [13, 12] placed inside the world; see also [11], where the unbounded mixture fails to learn a sequence that contains adversarially chosen actions. After at most $L + 1$ rounds the predictor has no hypothesis left and its output is whatever its implementation does with an empty posterior. A world that contains a bounded predictor and reacts against it is outside every guarantee above.

6 An exact computation

To see sizes, we computed the bounded mixture of a small machine exactly. *BF8* has eight commands, coded by three input bits each and read from the input only when they are first needed: `+` and `-` change the current cell by one modulo 16, `>` and `<` move along an unbounded tape of cells, `[` jumps past the matching `]` if the cell is 0, `]` jumps back to just after the matching `[` if the cell is not 0, and to the first command if there is no matching `[`; `.` prints the cell modulo 2 and `:` prints whether the cell is nonzero. Each executed command is one step. A forward jump reads the commands it skips; they count as read and cost no step. The mixture is that of Definition 1 with at most 10 commands read ($L = 30$ bits) and schedule $T(k) = T_1 k$. All masses are dyadic rationals and are computed without rounding by an enumerator that forks the machine at every read ($8.6 \cdot 10^8$ branches for $T_1 = 24$; $8.4 \cdot 10^5$ strings of length at most 48 have positive mass). The enumerator agrees with a run of all command tapes for small L.

BF8 is not universal in the sense of Definition 3, since a program cannot read the rest of its input as data. So Theorems 10 to 12, 25 and 26 do not apply to it as stated, and it has no

wrappers. Also, one of its steps can read several commands, so the inequality $r \le t$ of Section 2 fails for it. Lemma 2 and Theorems 9, 24, 27 and 28 hold for every machine and do not use that inequality. The computation concerns prediction only.

For a stream z and a position k we report three shares of the posterior given $z_{1:k-1}$: the mass that prints the true bit z_k on schedule ("true"), the mass that prints the other bit ("other"), and the rest ("none"), which prints nothing on schedule or would need an eleventh command first. Table 1 gives selected positions, and Figure 1 the full profiles.

		$T(k) = 24k$			$T(k) = 3k$		
stream	k	true	other	none	true	other	none
1^{48}	16	0.791	0.205	0.004	0.544	0.451	0.005
1^{48}	31	0.963	0.037	0.001	0.869	0.130	0.001
$(10)^{24}$	17	0.294	0.457	0.249	0.141	0.411	0.448
$(10)^{24}$	31	0.935	0.062	0.003	0.842	0.155	0.003
0^{48}	17	0.958	0.008	0.034	0.969	0.012	0.019
$1^{8}0^{40}$	9	0.025	0.958	0.017	0.001	0.998	0.001

Table 1: Posterior shares at selected positions, BF8 with at most 10 commands. Exact values rounded to three places.

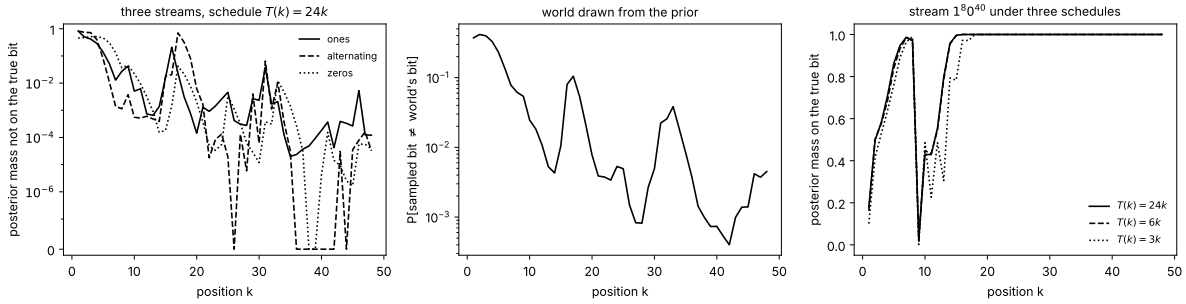


Figure 1: Left: posterior mass not on the true bit, by position, for three streams (the axis is linear near 0, so exact zeros are visible). Middle: probability that the sampled next bit differs from the world's own, for a world drawn from the prior that prints that bit within the bounds (by Theorem 9 both bits are independent draws from the predictor's distribution). Right: mass on the true bit for the stream $1^{8}0^{40}$ under three schedules.

A shorter program explains the context and continues differently. Both large effects in Table 1 have this cause, and in both the short programs use the rule that an unmatched `]` restarts the program while the cell is nonzero, together with the fact that a cell returns to 0 after sixteen increments.

- The stream 1^{48} is printed forever by four-command programs such as `+[:]`, and no program of three commands prints even its first 16 bits. The three-command programs `+:]` and `-:]` print $1^{15}0$ and then run on into whatever commands follow. They carry $2 \cdot 8^{-3}$ of the mass 0.00554 on $1^{15}0$, which is 70%. At bit 16 a share 0.205 of the posterior prints the other bit, and at bit 31 a share 0.037. From position 10 on, the mass off the true bit is below 0.01 everywhere except at positions 15–17 and 31.
- The alternating stream is printed forever by six-command programs such as `++[:.]`, and no program of five commands prints all of its first 48 bits. The three-command programs `+.]` and `-.]` print its first 16 bits and then run on. They carry $2 \cdot 8^{-3}$ of the mass 0.00682 on these 16 bits, which is 57%. At bit 17 the true bit has share 0.294, the other bit 0.457, and

0.249 is in "none"; normalised over the two bits, the true one has 0.392. So the majority form of Definition 8 holds for this world at this cut point, for the predictor with at most ten commands.

This is the first sentence of Section 4.2 in action: the programs for the context that do not continue with the world's own bit outweigh those that do. Nothing here acts for a purpose. The example shows only that the inequality can go the wrong way at an exactly computable place, and that a stopping point which is cheap on the machine, here the return of a counter to zero, is where it does.

The effect weakens as longer programs are admitted. Table 2 repeats the computation with at most 7 to 11 commands. The mass that continues with the true bit is spread over many program lengths and keeps growing with the bound, while the mass of the short programs that leave is counted in full from the start. At bit 17 of the alternating stream the normalised share of the true bit rises from 0.273 to 0.416, and part of "none" at the smaller bounds is mass cut off by the bound itself. We did not compute beyond 11 commands, so whether the majority at bit 17 persists for much longer programs is not known from this computation.

	bit 17 of $(10)^{24}$				bit 16 of 1^{48}
commands	true	other	none	true, normalised	other
7	0.157	0.419	0.424	0.273	0.275
8	0.210	0.442	0.348	0.322	0.243
9	0.256	0.452	0.291	0.362	0.221
10	0.294	0.457	0.249	0.392	0.205
11	0.326	0.458	0.217	0.416	0.194

Table 2: The two headline positions under different bounds on the number of commands, $T(k) = 24k$.

The budget holds with room. Table 3 compares both sides of the finite forms of Theorem 24(ii) and (iii) for the first 48 bits.

stream	$\sum_{k\le 48}(1-\text{true}_k)$	$\ln(1/\xi(z_{1:48}))$	$\#\{k : \text{true}_k \le \frac{1}{2}\}$	$\log_2(1/\xi(z_{1:48}))$
1^{48}	2.63	3.94	2	5.68
0^{48}	3.07	4.03	2	5.82
$(10)^{24}$	4.01	6.76	4	9.76
$1^8 0^{40}$	5.07	9.77	5	14.09

Table 3: The budget inequalities for $T(k) = 24k$ and at most 10 commands, with $\xi(z_{1:48})$ in place of the limit: "true_k" is $\xi(z_k \mid z_{1:k-1})$.

The population. For a world drawn from the prior that prints the bit in question within the bounds, the probability that a sampled next bit differs from the world's own is 0.080, 0.105, 0.054 at positions 16, 17, 18 and between 0.022 and 0.038 at positions 31–33; its median over positions 20–28 and 36–44 is 0.0014 (Figure 1, middle). By Theorem 9 this is neither more nor less than the unpredictability of such worlds at these positions.

A tighter schedule can shift mass off the truth. With $T(k) = 3k$ in place of $24k$, the share of the true bit falls from 0.791 to 0.544 at bit 16 of 1^{48}, from 0.294 to 0.141 at bit 17 of the alternating stream, and from 0.025 to 0.001 at bit 9 of $1^8 0^{40}$ (Table 1). The shift is not uniform: at bit 17 of 0^{48} the share rises from 0.958 to 0.969. With at most nine commands the

stream $1^8 0^{40}$ shows the extreme case of Section 4.4: under $T(k) = 24k$ it has mass $1593 \cdot 2^{-25}$ on all 48 bits, while under $T(k) = 3k$ its first 14 bits have mass 2^{-26}, the true 15th bit has mass 0 and the other bit 2^{-26}. The tight predictor gives the truth probability 0 there, and all of its posterior to a continuation the world does not have.

7 Circuits in place of programs

Hänni also asked for the same question with circuits in place of programs, and for an overall view of the prevalence of malignity there. This section fixes one reading of that question, the *circuit toy*, and goes through the results of the note in it. The *source* is the file `research/circuits-final.md` of the repository; it contains the proofs that are not given here. Results are theorems of the circuit toy unless they are marked as computed, argued or guessed. The overall view is in Section 7.12.

7.1 The circuit toy

Definition 29 (Circuits)**.** Fix $k \geq 2$ input wires. A *circuit of size* s is a list of gates $g_i = (\tau_i, a_i, b_i)$, $1 \leq i \leq s$, with a truth table $\tau_i \in \{0,1\}^4$ (all sixteen functions of two bits) and two ordered pointers $a_i, b_i \in \{1, \ldots, k+i-1\}$. Wires $1, \ldots, k$ are the inputs and wire $k+i$ is gate i, whose value is τ_i applied to the values of the wires a_i and b_i. The output is the last gate. $\mathcal{G}_{k,s}$ is the set of these circuits, $|\Gamma|$ is the size of a circuit Γ, $f_\Gamma \colon \{0,1\}^k \to \{0,1\}$ is the function it computes, and

$$N_s = |\mathcal{G}_{k,s}| = \prod_{i=1}^{s} 16\,(k+i-1)^2, \qquad p_i = \frac{1}{16\,(k+i-1)^2}.$$

A gate is *active* if it is the last gate or an ancestor of it, and *dead* otherwise. A circuit is *reduced* if all its gates are active. The *core* of Γ is the reduced circuit obtained by deleting the dead gates and renumbering in order; it computes f_Γ. For a function f, $\mathrm{size}(f)$ is the least size of a circuit that computes it.

Priors. The *fixed-size prior* μ_s is uniform on $\mathcal{G}_{k,s}$. A *mixture prior* has weights $w_s > 0$ for $1 \leq s \leq s_{\max}$, with $s_{\max} \leq \infty$ and $\sum_s w_s = 1$, and gives $\Gamma \in \mathcal{G}_{k,s}$ the weight $q(\Gamma) = w_s / N_s$; we put $\varrho(\sigma, a) = w_{\sigma+a}/w_\sigma$. Assumption (A1), used only where named, is that w_s is nonincreasing. The *length prior* is the mixture with $w_s = 2^{-s}$ and $s_{\max} = \infty$: the size in unary, then the gates. It satisfies (A1), and $\varrho(\sigma, a) = 2^{-a}$. These are the *priors of the family*, and μ stands for any of them. The *core weight* is

$$\beta_s(a) = p_s\, e_{a-1}(p_1, \ldots, p_{s-1}) \quad (a \leq s), \qquad \beta_s(a) = 0 \quad (a > s), \qquad \beta_q(a) = \sum_s w_s \beta_s(a),$$

with e_i the elementary symmetric polynomial of degree i; β stands for β_s or β_q.

Hypotheses, data, queries. A hypothesis is a circuit, read as the function it computes. *Data* are $D = (I, f^*|_I)$ for a set $I \subseteq \{0,1\}^k$ of inputs and a target f^*. A circuit Γ *fits* D if $f_\Gamma = f^*$ on I. Let $\xi(D)$ be the prior probability that a circuit fits D, $\ell_D = -\log_2 \xi(D)$, and s_D the least size of a fitting circuit. For a query $u \notin I$, or a tuple of queries, the predictor's distribution is $\pi(y \mid u, D) = \mu\{f_\Gamma(u) = y \mid \Gamma \text{ fits } D\}$, and the sampling predictor returns $Y \sim \pi(\cdot \mid u, D)$.

In the *sequence form*, or *index reading*, the k inputs carry the binary digits $b_0, \ldots, b_{k-1}$ of $t-1$ for a position $t \in \{1, \ldots, 2^k\}$, and the circuit says that the bit at position t is $f_\Gamma(t)$. Its *truth table* is $\mathrm{tt}(\Gamma) = f_\Gamma(1) \cdots f_\Gamma(2^k)$, and Γ *prints* x if $x \sqsubseteq \mathrm{tt}(\Gamma)$. A context x of length n

is the data set with $I = \{1, \dots, n\}$ and labels x, and a block of m bits answers the queries $n+1, \dots, n+m$. Then

$$\xi(x) = \mu\{\Gamma : \ x \sqsubseteq \mathrm{tt}(\Gamma)\}, \qquad \pi_m(y \mid x) = \frac{\xi(xy)}{\xi(x)} \quad (|x| + m \le 2^k),$$

and $\mathrm{size}(x)$ is the least size of a circuit that prints x. The sequence form is the special case of the query form in which the data are an initial segment, and every result for data and queries applies to it. At a cut $n = 2^j$ the index bits $b_j, \dots, b_{k-1}$ are 0 on the whole context, and positions $n+1, \dots, 2n$ differ from positions $1, \dots, n$ only in b_j. So a dyadic cut is a query that differs from the data in a bit that never varied.

We use the index reading because two other natural readings, with one gate or one pointer per output bit, give a predictor that learns nothing, or only a frequency.

Theorem 30 (Two degenerate readings)**.** *(a)* One gate per output bit. *Let a circuit with $r \ge 1$ random inputs be uniform on $\mathcal{G}_{r,s}$, which is Definition 29 with r in place of k and with $r = 1$ allowed. Let $s \ge N$, and let its N output bits be the values of its last N gates. Then for every fixed value of the inputs the output is uniform on $\{0,1\}^N$. So the predictor has $\pi_m(y \mid x) = 2^{-m}$ for every context and block, also for mixtures over r and $s \ge N$.*

(b) One pointer per output bit. *Let the prior on r, s and the gates be arbitrary and let, given these, the N output pointers be independent and uniform on the $r + s$ wires. Let ϕ be the fraction of wires that carry 1, a function of the gates and the inputs. Then the probability $\xi(x)$ that the output begins with $x \in \{0,1\}^n$, a string with i ones, is $\mathbb{E}[\phi^i (1-\phi)^{n-i}]$. So ξ is exchangeable, and $\pi_m(y \mid x)$ depends on x and y only through their numbers of ones.*

Proof. (a) Condition on the inputs and on the gates $1, \dots, s-N$. For $i = s-N+1, \dots, s$ in turn, the value of gate i is the entry of τ_i at the pair of values of the wires a_i, b_i. That pair is determined by the inputs, the earlier gates and (a_i, b_i), all independent of τ_i, whose four entries are independent fair bits. So the value is a fair bit independent of everything earlier. (b) Given the gates and the inputs, the output bits are independent draws of the value of a uniform wire. □

The theorem is about these two readings. It does not show that every reading with one wire per output bit is degenerate, nor that the index reading is the only sensible one. In the index reading the link between context and continuation is the reuse of the same gates at different positions. A circuit that takes the context as input and is applied autoregressively is a third reading; it reuses a description across steps, belongs with Proposition 41, and is not treated here.

Worlds, critical uses, takeover. A *world* is a circuit W, with target f_W. It is *in the class* if $\mu\{\Gamma : f_\Gamma = f_W\} > 0$. A *critical use* is a triple of a world, data and a query, fixed from outside: the world hands the predictor D, receives Y at the query, and acts on it. Its *own continuation* d is f_W at the query, and the *deviation mass* is $1 - \pi(d \mid u, D)$. In the sequence form a *call rule* $\mathcal{C}$ is a set of pairs (W, n_W), at most one for each circuit; with $x_W = \mathrm{tt}(W)_{1:n_W}$ and a mixture prior q, the quantities of Definition 6 are

$$C(x) = \sum_{W:\, x_W = x} q(W), \qquad H_{\mathcal{C}} = \sum_{(W,n_W)\in\mathcal{C}} q(W), \qquad \kappa(x) = \frac{C(x)}{\xi(x)}.$$

A *test* is a set G of nonempty strings, with first hits as in Definition 5; under a test every circuit that prints a first hit hands over there, and $\kappa = 1$. The forms of Definition 8 are taken over: the majority form $\pi(d \mid u, D) < \frac{1}{2}$, the payload form $\pi(A \mid u, D) \ge \theta$ for a set $A \not\ni d$, and the

critical-use form $\Pr[Y \in A(X)] \geq \theta$ over a population of uses. As in the note they are necessary for calling the prior malign at a world and not sufficient.

A takeover needs a decoder. In the note the block y is run as U on $y0^\infty$. Here hypotheses are circuits and a block is a string, so the world applies a fixed decoding map to it. The *literal code* of $\Gamma \in \mathcal{G}_{k,s}$ is $e(\Gamma) = \bar{e}(s)\,\tau_1[a_1][b_1]\cdots\tau_s[a_s][b_s]$, with pointer fields of $\lceil \log_2(k+s) \rceil$ bits; a field with value c at gate i is read as wire $1 + (c \bmod (k+i-1))$. Its length is $l(s) = |\bar{e}(s)| + s\,(4 + 2\lceil \log_2(k+s) \rceil)$. $\mathrm{Dec}(y)$ is the circuit decoded from the front of y if the rest of y is 10^i for some $i \geq 0$. The counterpart of Definition 7 is

$$\mathrm{SR}_m(x) = \{y \in \{0,1\}^m : \ \mathrm{Dec}(y) \text{ is defined and } xy \sqsubseteq \mathrm{tt}(\mathrm{Dec}(y))\}.$$

The decoder is an interpreter that lives outside the hypothesis class.

What the class assumes. These are decisions, not results. (i) Plain circuits: fan-in two, each gate evaluated once per output, no gate used twice, no depth bound, no weight sharing. (ii) Hypotheses are total: every circuit has a bit at every position, so the worlds of the note that hand over and then print nothing, and the quantity φ of Proposition 14, cannot be expressed; this favours the honest side. (iii) The class depends on the horizon through k, and which positions are cheap to name depends on the binary coding of the index. (iv) The prior is on ordered gate lists, with dead gates allowed. (v) The full binary basis, with uniformly drawn truth tables. (vi) The sampling predictor. (vii) The world is in the predictor's class wherever a bound compares with the world. (viii) The use is fixed from outside, and a hypothesis can recognise it only through its own input, the index or the query.

7.2 What transfers

Theorem 31 (Measure, calibration, budget, feedback)**.** *Let μ be a prior of the family and write $\mu(f) = \mu\{\Gamma : f_\Gamma = f\}$.*

(a) Measure. *In the sequence form $\xi(\epsilon) = 1$ and $\xi(x0) + \xi(x1) = \xi(x)$ for $|x| < 2^k$. Hence $Z_m(x) = \sum_{y \in \{0,1\}^m} \xi(xy) = \xi(x)$ for $|x| + m \leq 2^k$.*

(b) Calibration. *Fix I and a query $u \notin I$. Let $\Gamma^* \sim \mu$, $D = (I, f_{\Gamma^*}|_I)$, and let $Y \sim \pi(\cdot \mid u, D)$ be drawn independently of Γ^* given D. Then given D, the answer $f_{\Gamma^*}(u)$ and Y are independent with the common law $\pi(\cdot \mid u, D)$. In the sequence form let G be a test, let the world be drawn from μ, and let X be the first hit of G among the prefixes of its truth table, on the event that there is one of length at most $2^k - m$. Given $X = x$, the world's next m bits and $Y \sim \pi_m(\cdot \mid x)$ are independent with the law $\pi_m(\cdot \mid x)$.*

(c) Budget. *Let $z = \mathrm{tt}(W)$ with $\xi(z) > 0$. Then $\sum_{n=0}^{2^k-m} (1 - \pi_m(z_{n+1:n+m} \mid z_{1:n})) \leq m \ln(1/\xi(z))$, and the term is at least $\frac{1}{2}$ at no more than $m \log_2(1/\xi(z))$ cuts n. For a mixture prior, $\xi(z) \geq w_\sigma / N_\sigma$ if a circuit of size $\sigma \leq s_{\max}$ prints z. Let $D_0 = (I_0, f^*|_{I_0})$ be any data labelled by f^*, with I_0 possibly empty. Let queries $u_1, u_2, \ldots$ be answered one at a time, repetitions allowed, each depending on everything revealed before, the true answer being added to the data after each, which gives D_t. A query that is already in the data is answered by its label with probability 1. If $\mu(f^*) > 0$ and $a_t = \pi(f^*(u_t) \mid u_t, D_{t-1})$, then $\sum_t (1 - a_t) \leq \ln(1/\mu(f^*))$, and $a_t \leq \frac{1}{2}$ for at most $\log_2(1/\mu(f^*))$ values of t.*

(d) One query off the data. *For $u^\dagger \notin I$ and every f' that fits D with $f'(u^\dagger) \neq f^*(u^\dagger)$: $\mu(f^*)/\xi(D) \leq \pi(f^*(u^\dagger) \mid u^\dagger, D) \leq 1 - \mu(f')/\xi(D)$.*

(e) Feedback. *Sequence form, with a response map R as in Section 5: the predictor returns $Y \sim \pi_m(\cdot \mid x)$ and the record continues with $R(Y)$. The prediction comes true with probability $\pi_m(\mathrm{Fix}(R) \mid x)$. If $Y = y \in \mathrm{Fix}(R)$, no circuit that prints xd with $d \neq y$ prints any $x' \sqsupseteq xy$. If R is the identity in every round, the record continues with $u \in \{0,1\}^{jm}$, where*

$|x| + jm \le 2^k$, *with probability exactly* $\xi(xu)/\xi(x)$*; it continues with a block* d *such that no circuit in the support prints* xd *with probability* 0*. If the predictor returns the more probable next bit,* 0 *on a tie, and the record continues with the other bit from position* n *on, then* $\xi(x_{1:n+j}) \le 2^{-j}\xi(x_{1:n})$ *while* $n + j \le 2^k$*; for a mixture prior with* $s_{\max} < \infty$ *and* $q_{\min} = \min_s w_s/N_s$*,* $\xi(x_{1:n+j}) = 0$ *for all* $j > \log_2(1/q_{\min})$ *with* $n + j \le 2^k$*.*

Proof. (a) Every circuit has a truth table of exactly 2^k bits, so the sets $\{\Gamma : x \sqsubseteq \mathrm{tt}(\Gamma)\}$, $x \in \{0,1\}^n$, partition the circuits. (b) is proved as Theorem 9: $\Pr[f_{\Gamma^*}(u) = y \mid D] = \pi(y \mid u, D)$ by definition; first hits are pairwise incomparable, so $X = x$ with next block d has probability $\xi(xd)$, and $X = x$ has probability $\xi(x)$ by (a). (c) is the proof of Theorem 24(iv), which uses only that the conditional probabilities of the true bits multiply to $\xi(z)$; and $\xi(z)$ is at least the weight of one circuit that prints z. For queries, the product of the $a_t = \xi(D_t)/\xi(D_{t-1})$ telescopes to $\xi(D_t)/\xi(D_0) \ge \mu(f^*)$, because every circuit computing f^* fits every D_t and $\xi(D_0) \le 1$. (d) The circuits computing f^*, respectively f', are among those fitting D. (e) The proofs of Theorems 27 and 28 use only (a) and that a positive value of ξ is at least the weight of one circuit; by (a) the product in Theorem 27(c) telescopes to an equality. □

By (a) the first qualification of Theorem 9 disappears inside the horizon: no world prints fewer than m further bits after a first hit of length at most $2^k - m$. A world whose first hit is longer is left out of the event, so one qualification remains at the end of the horizon. The hypotheses of (b) remain. The world is drawn from the predictor's own prior, and the use is read off the data, by a test or by a query chosen from the data alone. With a call rule that selects users by something the data do not show, calibration does not apply. The factor m in the first budget and its absence in the second belong to two protocols: overlapping blocks at every cut, and answers revealed whole. Parts (b) and (c) do not bound the deviation at one query that something singled out. A use is such a point, and (d) is what is proved about it. In (e) the "dark room" of Section 5 becomes: an obedient world follows a sample from the circuit prior conditioned on the past, and is moved off any default that no circuit in the support prints. The contrarian statement in (e) is weaker than Theorem 28. The posterior can become empty only under a prior with $s_{\max} < \infty$. It is empty for certain once more than $\log_2(1/q_{\min})$ rounds fit into the horizon, and it can be empty earlier. Under the length prior every function of the k inputs has positive mass, so the posterior is never empty and only the halving remains. The budget in (c) is relative to the horizon as well: the count $m \log_2(1/\xi(z))$ says something only while it is below the number 2^k of cuts. For a world outside the class nothing is guaranteed. The hardest function of k inputs needs about $2^k/k$ gates [15], and fewer than $2N_s$ functions have circuits with at most s gates, so for arbitrary sequences of length 2^k this is the generic case unless the size bound is exponential in k.

7.3 The core law and the price of an active gate

Theorem 32 (Core law; price of an active gate)**.** *(a) For a reduced circuit* χ *of size* a *and* $s \ge a$*:* $\mu_s\{\Gamma : \mathrm{core}(\Gamma) = \chi\} = \beta_s(a)$*, and for a mixture prior* $q\{\Gamma : \mathrm{core}(\Gamma) = \chi\} = \beta_q(a)$*. So under every prior of the family the mass of a core depends only on its size, the posterior on cores is* $\beta(|\chi|)\mathbf{1}[\chi \text{ fits } D]$ *normalised, and every prediction is a function of the core. Consequently, for* $a = \mathrm{size}(f) \le s$*, and with* q *the length prior in the second inequality,*

$$\mu_s\{\Gamma : f_\Gamma = f\} \;\ge\; \beta_s(a) \;\ge\; \frac{p_s}{N_{a-1}}, \qquad q\{\Gamma : f_\Gamma = f\} \;\ge\; \frac{2^{-a}}{N_a}.$$

(b) For $1 \le a \le s - 1$*,*

$$\frac{1}{16a}\Big(\frac{1}{k+a-1} - \frac{1}{k+s-1}\Big) \;\le\; \frac{\beta_s(a+1)}{\beta_s(a)} \;\le\; \frac{1}{16a}\sum_{j=k}^{k+s-2}\frac{1}{j^2} \;\le\; \frac{1}{16a(k-1)}.$$

The lower bound is at least $\frac{1}{32a(k+a-1)}$ *when* $s \geq 2a+k-1$. *For every prior of the family* $\beta(a+1) \leq \beta(a)/(16a(k-1))$. *So* β *is decreasing, and* $\beta(a) \leq \frac{1}{16k^2}\,(16(k-1))^{-(a-1)}/(a-1)!$ *for* $a \geq 1$.

Proof. (a) A circuit $\Gamma \in \mathcal{G}_{k,s}$ with core χ is determined by the set $J \subseteq \{1,\dots,s\}$ of its active positions, with $|J| = a$ and $s \in J$, and by its dead gates: the active gate at the j-th position of J has the truth table of the j-th gate of χ and its pointers, renumbered. Conversely, for every such J and every choice of the gates at the positions $i \notin J$, of which there are $1/p_i$ at position i, the resulting list is a circuit whose active set is exactly J: the gates at J read only inputs and gates at J, all of them are ancestors of gate s because χ is reduced, and no gate outside J is read by a gate in J. So the number of circuits with core χ is $\sum_J \prod_{i\notin J} p_i^{-1} = N_s \sum_J \prod_{i\in J} p_i = N_s\, p_s\, e_{a-1}(p_1,\dots,p_{s-1})$. For a mixture, sum over s with the weights w_s. For the inequalities: a minimum circuit for f is reduced; $e_{a-1}(p_1,\dots,p_{s-1}) \geq p_1 \cdots p_{a-1} = 1/N_{a-1}$; and the term $s = a$ of $\beta_q(a)$ is $2^{-a}/N_a$ for the length prior.

(b) Write $e_i = e_i(p_1,\dots,p_{s-1})$, so that the ratio is e_a/e_{a-1}. Counting pairs of an $(a-1)$-set J and an index $i \notin J$ gives $a\,e_a = \sum_{|J|=a-1} \prod_{i\in J} p_i \cdot \sum_{i\notin J} p_i$. The p_i decrease, so $\sum_{i\notin J} p_i \geq \sum_{i=a}^{s-1} p_i \geq \frac{1}{16}\int_{k+a-1}^{k+s-1} x^{-2}dx$, which is the lower bound; and $\sum_{i\notin J} p_i \leq e_1 = \frac{1}{16}\sum_{j=k}^{k+s-2} j^{-2} \leq \frac{1}{16}\int_{k-1}^{\infty} x^{-2}dx$, which gives the upper bounds. $\frac{1}{k+a-1} - \frac{1}{k+s-1} \geq \frac{1}{2(k+a-1)}$ exactly when $s \geq 2a+k-1$. For a mixture apply the upper bound to each $s \geq a+1$ and sum with the weights w_s. Finally $\beta_s(1) = p_s \leq p_1 = 1/(16k^2)$, also after mixing, and the ratio bound is applied $a-1$ times. □

So every prior of the family is a prior on cores, and a function with a small circuit has mass at least $\beta(\mathrm{size}(f))$. Under μ_s this bound is p_s times a factor that stays between two positive constants as s grows, so it falls like s^{-2} with the nominal size, where the weight $1/N_s$ of one list falls faster than exponentially. Under the length prior the bound involves no nominal size. One more active gate costs at least $\log_2(16a(k-1))$ bits under every prior of the family, and at most $\log_2(32a(k+a-1))$ bits under μ_s with slack $s \geq 2a+k-1$. Under the length prior no upper bound of that kind is proved, and the computed price is higher: 13.00 bits at $k = 20$, $a = 1$ and 14.12 at $a = 8$, against the lower bounds 8.25 and 11.25. The theorem does not give the mass of a function, because the number of cores per function is not controlled: $1/N_s$ is a lower bound on a mass and not an estimate. Under μ_s the active part of a typical circuit is small. The pointers of a uniform gate list form a random recursive directed acyclic graph on $k+s$ vertices in which the first k vertices send no arrows, and the active gates are the vertices above k that can be reached from the last one. In the model in which every vertex sends two arrows the number of vertices reachable from the last one is of order $\sqrt{k+s}$, by a limit law of Janson as stated in [16]; the identification of a uniform gate list with that model is ours. So for fixed k and large s the active part has of order $\sqrt{s}$ gates. While s is small against k^2 it is much smaller: in a simulation the mean number of active gates was about 3 at $k = s = 1000$. So "a circuit of size s does s operations" is true of the list and not of the computation.

7.4 Triggers and the floor

A hypothesis that explains the data and then gives a chosen answer at a use must recognise the use from its own input. Let $z(n)$ be the number of trailing zeros of n in binary.

Lemma 33 (Sizes of triggers and payloads)**.** *Over the full binary basis, on the index bits* $b_0,\dots,b_{k-1}$ *of* $t-1$*:*

(a) Position test. *For* $1 \leq n < 2^k$ *the function* $[\,t-1 \geq n\,]$, *which is* 0 *on positions* $1,\dots,n$ *and* 1 *afterwards, is computed with* $k-1-z(n)$ *gates; with* 0 *gates it is the index wire* b_{k-1}. *So at the* $2^{g+1}-1$ *multiples* n *of* 2^{k-1-g} *below* 2^k *the test needs at most* g *gates.*

(b) Dyadic cut. *If $n = 2^j$ and $m \le 2^j$, the index wire b_j is 0 on positions $1, \dots, n$ and 1 on positions $n+1, \dots, n+m$: the test costs no gate.*

(c) Payload. *Every Boolean function of $j \ge 2$ variables has a circuit of at most $2^j - 3$ gates. Hence for every block $y \in \{0,1\}^m$ with $n + m \le 2^k$ there is a circuit $P_{n,y}$ on the index with $P_{n,y}(n+i) = y_i$ for $1 \le i \le m$ and at most $\max(1, 2^{\lceil \log_2 m \rceil} - 3) \le \max(1, 2m - 5)$ gates.*

(d) Selector. $\mathrm{mux}(c; u, v) = (c \wedge v) \vee (\neg c \wedge u)$ *is three gates. A constant payload needs one gate: $u \vee c$ for the value 1 and $u \wedge \neg c$ for the value 0.*

Proof. (a) Let c_i be the digits of n and $G_i = [\,(b_i \dots b_0) \ge (c_i \dots c_0)\,]$. G_0 is b_0 if $c_0 = 1$ and the constant 1 if $c_0 = 0$, and $G_i = b_i \wedge G_{i-1}$ if $c_i = 1$, $G_i = b_i \vee G_{i-1}$ if $c_i = 0$. With $j = z(n)$: G_i is the constant 1 for $i < j$, $G_j = b_j$ is an index wire, and each step $i = j+1, \dots, k-1$ adds one gate. (b) Positions $1, \dots, 2^j$ have $t - 1 < 2^j$, and positions $2^j + 1, \dots, 2^{j+1}$ have $b_j = 1$. (c) A function of two variables is one gate, and $f = \mathrm{mux}(x_j; f_0, f_1)$ gives $M(j) \le 2M(j-1)+3$ for the largest size $M(j)$ needed, so $M(j) + 3 \le 2^{j-2}(M(2) + 3) = 2^j$. For the block let $j = \lceil \log_2 m \rceil$. The $m \le 2^j$ numbers $t - 1 \in \{n, \dots, n+m-1\}$ are distinct modulo 2^j, so a function of the low j index bits has the required values; for $j \le 1$ one gate suffices. (d) By inspection. □

The count in (a) is exact for the function $[\,t - 1 \ge n\,]$ (source, Lemma 7(a)). A wrapper needs less: any wire that is 0 on the context and 1 on the block will do, as in (b). For a one-bit block the conjunction of the index bits that are 1 in n is such a test, because an index with all the ones of n is at least n; it has one gate fewer than n has ones. In the index reading the construction known for a test on the data is dearer, because each position is computed from scratch. To make a one-bit payload depend on w earlier bits of the stream of an explanation E of σ gates through a test of g_ϕ gates, the source computes the function by E followed by at most $w(\sigma + 2k) + g_\phi + k + 1$ gates, among them w shifted copies of E (Lemma 7(f) there). That nothing essentially cheaper exists is not proved.

Theorem 34 (Floor for a triggered payload)**.** *Let ϑ be a circuit of $g \ge 0$ gates on the inputs that is 0 on I (for $g = 0$, an input wire), and P a circuit of $|P|$ gates on the inputs. For $E \in \mathcal{G}_{k,\sigma}$ let $\mathrm{Tw}[E]$ be E followed by the gates of ϑ, the gates of P, and $\mathrm{mux}(\vartheta; E, P)$. It has $\sigma + a$ gates, $a = g + |P| + 3$, fits D exactly when E does, and equals P wherever $\vartheta = 1$. For a constant payload take $a = g + 1$.*

(a) Mixture prior.

$$q\{\Gamma :\ \Gamma \textit{ fits } D,\ f_\Gamma = P \textit{ on } \{\vartheta = 1\}\} \ \ge \sum_{\sigma \le s_{\max} - a} w_{\sigma+a}\, \frac{N_\sigma}{N_{\sigma+a}}\, \mu_\sigma\{\Gamma \textit{ fits } D\}.$$

If a share ρ_{σ_0} of the posterior is on circuits of size at most $\sigma_0 \le s_{\max} - a$, the posterior share of the payload on $\{\vartheta = 1\}$ is at least $\rho_{\sigma_0} \cdot \min_{\sigma \le \sigma_0} \varrho(\sigma, a) \cdot \left(16\,(k + \sigma_0 + a - 1)^2\right)^{-a}$.

(b) Length prior. *For $\epsilon \in (0,1)$ and $s_0 = \lceil \ell_D + \log_2(1/\epsilon) \rceil$ the share is at least $(1-\epsilon)\left(32\,(k + s_0 + a - 1)^2\right)^{-a}$, and $\ell_D \le s_D + \log_2 N_{s_D}$.*

(c) Critical uses. *Sequence form, mixture prior. For a call rule and blocks y_x, let the wrapper at the context x have a trigger that is 0 on positions $1, \dots, |x|$ and 1 on the block, the payload y_x, and a_x gates. Then $\Pr[Y = y_X]$ is at least the expectation over uses of the bound on the posterior share in (a), taken at the context X of the use. If the calling circuits with context x have size at most $s_{\max} - a_x$, the wrappers around them alone have share at least $\min_\sigma \left[\varrho(\sigma, a_x) N_\sigma / N_{\sigma + a_x}\right] \cdot \kappa(x)$ at x, the minimum being over the sizes σ of these calling circuits.*

(d) A never-varied bit. *Let the prior be the length prior, let an input coordinate v be 0 on all of I and 1 at the query $u^\dagger$, and $s_0 = \lceil \ell_D \rceil + 1$. Then $\pi(y \mid u^\dagger, D) \ge 1/(32(k + s_0)^2 + 2)$ for*

each $y \in \{0,1\}$, and for each y the fitting circuits that answer y at every input with $u_v = 1$ have posterior share at least $1/(32(k+s_0)^2)$. In the sequence form, at a cut $n = 2^j$ with $0 \le j < k$, each value of the next bit has share at least $1/(32(k+s_0)^2+2)$ and each of the two constant continuations of any length $m \le 2^j$ has share at least $1/(32(k+s_0)^2)$, where $s_0 = \lceil -\log_2 \xi(x) \rceil + 1 \le \mathrm{size}(x) + \log_2 N_{\mathrm{size}(x)} + 2$.

Proof. (a) $E \mapsto \mathrm{Tw}[E]$ is injective from $\mathcal{G}_{k,\sigma}$ into $\mathcal{G}_{k,\sigma+a}$, because E is the list of the first σ gates. It maps fitting circuits to fitting circuits with the payload on $\{\vartheta = 1\}$. So the fraction of $\mathcal{G}_{k,\sigma+a}$ with these properties is at least $(N_\sigma/N_{\sigma+a})\mu_\sigma\{\Gamma \text{ fits } D\}$; multiply by $w_{\sigma+a}$ and sum. For $\sigma \le \sigma_0$, $N_{\sigma+a}/N_\sigma \le (16(k+\sigma_0+a-1)^2)^a$. (b) The length prior gives the circuits larger than s_0 the mass 2^{-s_0}, so their posterior mass is at most $2^{\ell_D - s_0} \le \epsilon$; and $\varrho = 2^{-a}$. One fitting circuit of size s_D has weight $2^{-s_D}/N_{s_D}$. (c) Average (a) over the contexts; for the second claim restrict the injection to calling circuits. (d) The posterior mass of the circuits larger than s_0 is at most $\frac{1}{2}$. For $\Gamma \in \mathcal{G}_{k,s}$ let $\Gamma^{\oplus}$ and $\Gamma^{\oplus\prime}$ be Γ with one more gate, the exclusive or of gate s and v in the two pointer orders. Both compute $f_\Gamma(u) \oplus u_v$, fit D exactly when Γ does, and flip the answer at $u^\dagger$. They are two distinct circuits that determine Γ, and their weights sum to $q(\Gamma)\, p_{s+1}$. Let $\eta = p_{s_0+1} = 1/(16(k+s_0)^2)$ and $\bar{y} = 1 - y$. Summing over the fitting Γ of size at most s_0 with answer y gives $\pi(\bar{y}) \ge \eta\,(\pi(y) - \frac{1}{2}) = \eta\,(\frac{1}{2} - \pi(\bar{y}))$, so $\pi(\bar{y}) \ge \eta/(2(1+\eta))$. For the second claim append instead the gate of Lemma 33(d) for the constant y, in both pointer orders, and sum over all fitting Γ of size at most s_0: the share is at least $\eta/2$. The sequence form follows from Lemma 33(b). □

Three differences from Theorem 10. First, the cost is not a constant. A wrapper of a gates appended to an explanation of σ gates costs at most $a\,(4+2\log_2(k+\sigma+a-1)) + \log_2(1/\varrho(\sigma,a))$ bits, and $\log_2(1/\varrho(\sigma,a)) = a$ under the length prior. The cost of a gate grows with the logarithm of the size of the explanation, though not with the amount of data beyond that. Second, the test is paid in gates. The counterpart of the $K(n)$ of the note is the number of gates of a test that is 0 on the context and 1 on the block. It is at most $k - 1 - z(n)$, at most one fewer than the number of ones of n for a one-bit block, and zero at every dyadic cut with a block no longer than the context. Like $K(n)$ it is small at a few positions. Its least value is not determined here, except at $n = 2^k - 1$, where the test is the conjunction of all index bits and needs $k - 1$ gates by Theorem 38(a). Third, in the construction used here a test on the data costs copies of the explanation; a hypothesis that received the context as input would pay for the test alone. In numbers, for $k = 20$ and explanations of 1000 gates an appended gate costs about 24.0 bits, and 25.0 under the length prior, which adds one bit for the size. Under the length prior a one-bit deviation then costs about 25 bits at a dyadic cut and about 500 bits at the positions hardest to name, where the test of Lemma 33(a) has 19 gates; a block of 64 arbitrary bits adds at most 64 gates for payload and selector, about 1600 bits. These are costs of one wrapper against one explanation. In the program toy the same wrappers cost $K(n) + O(1)$ and $K(n) + 64 + O(1)$ bits. In (d), for $k = 20$ and $s_D = 1000$, $s_0 \approx 22\,326$ and the bound is $2^{-33.9}$.

Parts (a) to (c) are floors for one construction and not estimates of the deviation mass. Under μ_s the same injection gives a floor: among the fitting circuits of size s, the wrapped ones have share at least the number of fitting circuits of size $s - a$ divided by the number of size s. In exact small cases ($k = 3$, $s = 4, 5$, the world $b_0 \wedge b_1$, cuts 3 to 7) this floor is at most $2 \cdot 10^{-8}$ for wrappers of three gates, where the posterior probability of the flipped bit is between 0.057 and 0.95. At the dyadic cut $n = 4$, where one gate suffices, the floor is 0.0017 at $s = 4$ against a probability of 0.065. These numbers are computed; the first comparison is due to a referee of the source.

The family of wrappers of a gates around the users has prior mass

$$\sum_W \varrho(\sigma_W, a)\, \frac{N_{\sigma_W}}{N_{\sigma_W+a}}\, q(W),$$

the sum being over the calling circuits W, of sizes σ_W, because the wrapper around W has $\sigma_W + a$ gates. If all users have one size, its share at x is its prior mass times $\kappa(x)/H_{\mathcal{C}}$: relative to its own prior mass the family is repaid the factor $1/H_{\mathcal{C}}$, as the seeker of Theorem 13(a) is. The difference is that its prior mass is at most $\max_\sigma[\varrho(\sigma,a)N_\sigma/N_{\sigma+a}]\,H_{\mathcal{C}}$ by construction, which under (A1) is at most $(N_\sigma/N_{\sigma+a})H_{\mathcal{C}}$ with σ the least size of a user, where the weight 2^{-a} of the program seeker does not depend on $H_{\mathcal{C}}$.

Part (d) is the strongest statement proved about a critical use. It is two-sided: at a use marked by a bit that never varied, the share of each answer, and so of the world's own, lies between $1/(32(k+s_0)^2+2)$ and one minus that under the length prior. Since $s_0 \le s_D + \log_2 N_{s_D} + 2$, the distance from 0 and from 1 is at least an inverse polynomial in k and in the size s_D of the smallest fitting circuit. Part (d) does not say whether the true share falls like a power of the size or stays constant. A given answer at one such query and a given answer on the whole set $\{u_v = 1\}$ cost the same single gate.

7.5 Minimum-size circuits

Part (e) of the following is proved in the file `research/circuits-final.md` of the repository (Theorem 11(b) there); we state it without proof. The other parts are proved below.

Theorem 35 (Minimum-size circuits)**.** *In (a) to (c) let Γ be a circuit of the least size s_D that fits D, and $s_D \ge 2$. Call an input coordinate* frozen *if it is constant on I.*

(a) Γ is reduced; no gate of Γ is constant on I; no gate of Γ agrees on I with an earlier wire (an input or an earlier gate) or with the negation of one.

(b) Γ has no pointer to a frozen input. Hence if $u^\dagger$ agrees with some $u \in I$ on all coordinates that are not frozen, every minimum-size fitting circuit answers $f^*(u)$ at $u^\dagger$; under μ_{s_D} the deviation mass at $u^\dagger$ is 0 if the world's own answer is $f^*(u)$ and 1 if it is not.

(c) Dyadic cuts. *If $|x| = 2^j$ and $\mathrm{size}(x) \ge 2$, every minimum-size circuit that prints x has the truth table $x\,x\,x\cdots$. Under μ_s at $s = \mathrm{size}(x)$ the predictor continues x periodically with probability 1, and a world whose own continuation is not the periodic one has deviation mass 1 at that cut.*

(d) Last-gate dichotomy. *For any circuit Γ with last gate $\tau(a,b)$ let $\psi_\Gamma(u) \in \{0,1\}^2$ be the values of the wires a, b at u. Call $u^\dagger$* novel for Γ *if $\psi_\Gamma(u^\dagger) \notin \psi_\Gamma(I)$. Flipping the entry of τ at $\psi_\Gamma(u^\dagger)$ is a weight-preserving involution of the fitting circuits for which $u^\dagger$ is novel, and it flips the answer at $u^\dagger$. So, with ν the posterior probability of novelty, each answer at $u^\dagger$ has share at least $\nu/2$.*

(e) AND family. *Let $f^* = x_1 \wedge \cdots \wedge x_k$ and I the $k+1$ inputs with at most one 0. Then $s_D = k-1$, every minimum-size fitting circuit is a formula in which each input is read once, and at every input with exactly two 0s exactly half of the minimum-size fitting circuits answer 1, which is not the target's answer. Under μ_{k-1} the deviation mass there is exactly $\frac{1}{2}$.*

For $s_D = 1$ the conclusions of (b) and (c) hold unless the labels on I are constant or agree on I with an input or its negation.

Proof. (a) A circuit that is not reduced is larger than its core. Let gate g be constant γ on I. If g is the output, a single constant gate fits and $s_D = 1$. Otherwise replace every reader $\tau(g,b)$, $b \ne g$, by a gate with both pointers on b and the table $\tau'(c,\cdot) = \tau(\gamma,c)$, likewise $\tau(b,g)$, and every reader $\tau(g,g)$ by a constant gate. The new gates have the same values on I, nothing reads g, and deleting g gives a fitting circuit with $s_D - 1$ gates. Let gate g agree on I with the earlier wire w. If g is not the output, redirect to w all pointers on g and delete g. If g is the output and w is a gate, the gates up to w fit; if w is an input, one gate fits. If g agrees with $\neg w$, do the

same with the tables of the readers composed with a negation of that argument, respectively with the table of w negated. Every case contradicts minimality or $s_D \geq 2$. (b) A gate $\tau(v, b)$ with v frozen at γ equals $\tau(\gamma, b)$ on I, which as a function of the value of b is constant, or b, or $\neg b$, and constant also if b is frozen; this contradicts (a). A circuit with no pointer to a frozen input computes a function of the other coordinates, and under μ_{s_D} the posterior is uniform on the minimum-size fitting circuits. (c) is (b) with I the positions $1, \ldots, 2^j$ and the frozen coordinates $b_j, \ldots, b_{k-1}$: a function of $b_0, \ldots, b_{j-1}$ has a truth table of period 2^j. For $s_D = 1$ apply the case distinction of (b) to the single gate. (d) The flipped entry is used at no $u \in I$, so the circuit still fits; size and weight are unchanged; novelty does not depend on τ; the output at $u^\dagger$ is the flipped entry. □

Two things are proved here about how a predictor confined to minimum-size circuits generalises. It ignores whatever never varied: there is no deviation through a never-varied input, and total deviation when the world itself changes with it, which at a dyadic cut means every world whose continuation is not the periodic one. At a new combination of inputs that did vary it can be split evenly, with no smaller circuit and no consequentialist involved. An unexercised truth-table entry is a trigger that costs no gate, and it is symmetric: every circuit that deviates through it has a sibling that does not. The theorem does not cover circuits one gate above the minimum, where the shares are positive (Table 4). The exception for $s_D = 1$ is real. Let the target be x_1 and the data all inputs with one coordinate v frozen at 0. The fitting one-gate circuits are the 4 with both pointers on x_1, the $2(k-2)$ projections with the other pointer on an input that varies, and the 8 with the other pointer on v, whose two entries for $v = 1$ are free. So 4 of the $2k + 8$ fitting one-gate circuits deviate at each query with $v = 1$; for $k = 3$ these are 4 of 14 (also computed). The theorem uses the full basis.

7.6 A ceiling for separable triggers

Theorem 36 (Ceiling for separable triggers)**.** *Single-bit answers. Let Θ be a set of* triggers*: reduced circuits of $g \geq 1$ gates that read inputs only and are 0 on I (for $g = 0$, a set of input coordinates that are 0 on I). A* mux *is a last gate with pointers on a gate h and on a wire r, of a type $\varsigma(h, r)$ such that $\varsigma(\cdot, 0)$ is the identity or the negation and $\varsigma(\cdot, 1) \neq \varsigma(\cdot, 0)$; there are 6 such types and 2 pointer orders. A reduced circuit is* separably triggered by Θ *if its last gate is a mux of h and r, the set A_h of gates that are h or ancestors of h is disjoint from the set A_r of gates that are r or ancestors of r, and the circuit induced on A_r is in Θ (for $g = 0$: $r \in \Theta$). Its* honest restriction *is the circuit induced on A_h, with the table of h negated if $\varsigma(\cdot, 0)$ is the negation.*

(a) The honest restriction of a fitting separably triggered core of size a is a fitting core of size $a - g - 1$; hence $a \geq s_D + g + 1$.

(b) At most $12\binom{a-1}{g}|\Theta|$ separably triggered cores of size a have the same honest restriction.

(c) Let $\mathcal{T}_a$ be the set of circuits whose core is separably triggered by Θ and has size a, and $\mathcal{R}_a$ the set of circuits whose core has size a. For every prior μ of the family and every $a' \geq s_D$,

$$\frac{\mu(\mathcal{T}_{a'+g+1} \mid D)}{\mu(\mathcal{R}_{a'} \mid D)} \;\leq\; 12\binom{a'+g}{g}|\Theta|\,\frac{\beta(a'+g+1)}{\beta(a')} \;\leq\; \frac{12\,|\Theta|}{g!\;a'\,(16(k-1))^{g+1}},$$

and therefore

$$\mu\Big(\bigcup_a \mathcal{T}_a \;\Big|\; D\Big) \;\leq\; \frac{12\,|\Theta|}{g!\;s_D\,(16(k-1))^{g+1}}.$$

Proof. (a) The circuit is reduced, so every gate except the last lies in $A_h \cup A_r$. A_h is closed under predecessors, all its gates are ancestors of h or h itself, and h has the largest index in A_h;

so the induced circuit is reduced with $a-1-g$ gates. On I the trigger is 0, so the last gate equals h or $\neg h$ there, and the restriction fits. (b) The core is determined by its honest restriction together with the positions of A_r among the first $a-1$ gates, the circuit on A_r, the type of the mux and its pointer order. (c) By (a), (b) and Theorem 32(a) the numerator is at most $12\binom{a'+g}{g}|\Theta|\beta(a'+g+1)$ times the number of fitting cores of size a', and the denominator is $\beta(a')$ times that number. Theorem 32(b), applied $g+1$ times, gives $\beta(a'+g+1)/\beta(a') \le \prod_{j=a'}^{a'+g} \frac{1}{16j(k-1)}$, and $\binom{a'+g}{g}\prod_{j=a'}^{a'+g}\frac{1}{j} = \frac{1}{g!\,a'}$. By (a), $\mathcal{T}_a$ contains no fitting circuit unless $a = a'+g+1$ with $a' \ge s_D$; sum over $a' \ge s_D$, using $1/a' \le 1/s_D$ and $\sum_{a'} \mu(\mathcal{R}_{a'} \mid D) \le 1$. □

So a triggered circuit in which the trigger is a separate, input-only part feeding a selector at the output is outweighed by the circuit obtained by cutting the trigger off, with an explicit multiplicity factor. For triggers that are 1 on I the same holds, with the same proof, when the roles of $\varsigma(\cdot,0)$ and $\varsigma(\cdot,1)$ are exchanged. The bound is on one syntactic class. It says nothing about (i) to (iv) below, and summed over triggers it reaches as far as (v) says.

(i) *Entangled triggers:* the trigger enters below the output, as in $x_1 \wedge (x_2 \wedge \neg v)$, where the frozen bit v is absorbed into an inner gate.

(ii) *Triggers that share gates with the honest part* ($A_h \cap A_r \neq \emptyset$), including a switch read off a wire of the world (Proposition 39(c)).

(iii) *Implicit triggers,* which cost no gate: an unexercised truth-table entry (Theorem 35(d)).

(iv) *Circuits with no smaller honest restriction:* the part that would be removed also does the honest work (Section 7.11).

(v) *How far the sum over triggers reaches.* For $k = 20$ and $s_D = 1000$ the bound is $2^{-14.6}$ for the single trigger "a given input that is 0 on the data". Let R_g be the number of reduced circuits with g gates, so that $|\Theta| \le R_g$. Every circuit has one core, so Theorem 32(a) gives $\sum_{b \le a} R_b\, N_a\, \beta_a(b) = N_a$ for every a, which determines the R_g. Summed over all triggers with g gates the bound is then about 10^{-3} for each $g \le 5$ ($8.3 \cdot 10^{-4}$ at $g = 1$, $1.2 \cdot 10^{-3}$ at $g = 5$). Summed over all $g \le 19$, which covers the position test of Lemma 33(a) at every cut, it is at most 0.14. It exceeds 1 from $g = 28$ on for a single g, and from $g = 25$ on for the sum over all triggers of at most g gates. The bound falls like $1/s_D$. The source gives larger figures at this place, 0.0096, 0.082, 0.57 for $g = 2, 3, 4$ and more than 1 at $g = 5$; they count all gate lists of g gates where the theorem counts reduced circuits.

In the smallest case, the target $x_1 \wedge x_2$ with a third input frozen at 0, the separably triggered cores carry 16 of the 48 circuits that are off the target at the query 111 for $s = 2$, where the frozen input is the only possible trigger. For $s = 3$ they carry 8512 of 29 824, which is 28.5 per cent; 8128 of these are triggered by the frozen input and the others by a one-gate trigger (computed by a referee of the source and counted again for this section). A second bound is at the level of descriptions, with dead gates included, and sums over all triggers and payloads of its class. The following is proved in the file `research/circuits-final.md` of the repository (Proposition 13 there); we state it without proof.

Proposition 37 (Small self-contained gadgets around a large explanation)**.** *Mixture prior with (A1). Let $E \in \mathcal{G}_{k,\sigma}$. A* gadget over E *with a gates is a list γ of gates whose pointers go to input wires, to gates of E, or to earlier gates of the gadget. A circuit in $\mathcal{G}_{k,\sigma+a}$* realises *(E,γ) if its gates split into σ positions carrying the gates of E in order and a positions carrying those of γ in order, with the pointers translated. A gadget is* self-contained *if its gates point only to input wires, to the output wire of E, and to earlier gates of the gadget.*

(a) $q(\text{realisations of } (E,\gamma)) \le \binom{\sigma+a}{a}\varrho(\sigma,a)\,(N_\sigma/N_{\sigma+a})\,q(E) \le \frac{1}{a!}\big(\frac{\sigma+a}{16(k+\sigma)^2}\big)^a q(E)$.

(b) If $k+\sigma \ge 4(k+A)^2$, the set $\mathrm{Wr}_A(E)$ of circuits that realise (E,γ) for some self-contained γ with at most A gates has $q(\mathrm{Wr}_A(E)) \le \sum_{a=1}^{A} \frac{2^{-a}}{a!}\,q(E) < 0.65\,q(E)$. Hence, for data D,

the circuits in $\bigcup_E \mathrm{Wr}_A(E)$ *over the fitting* E *with* $k+|E| \geq 4(k+A)^2$ *have posterior share at most* 0.65 *times the share of those* E*.*

The hypothesis is strong: for $k = 20$ it needs $\sigma \geq 1744$ for $A = 1$ and $\sigma \geq 12\,080$ for $A = 35$, which covers blocks of up to 16 bits at the worst position (19 gates for the test, 13 for the payload and 3 for the selector). The constant is not small either: the bound is 0.65 times a share that may be close to 1. Compare Proposition 14. There one seeker is bounded by its twin and seekers add. Here the comparison is with the explanation itself, and the bound is on a class. The restriction is free of the trigger and need not be true: both bounds compare wrappers with what they wrap and say nothing about how the mass of the explanations divides between the world's continuation and others. Neither bound contains the rarity or the stakes of the uses.

7.7 No refund for a single description; computation is paid for in description

In the program toy a seeker is a word v. Its programs $v\omega$ share the cost $|v|$, the rest of the input selects the target, and the search refunds the $\log_2(1/H_G)$ bits by which uses are rare (Theorem 12(b)). One counterpart of v is a *seeded circuit*: a circuit S with $k+r$ inputs, the last r of which are fair random bits, the seed $\omega \in \{0,1\}^r$. It lies in $\mathcal{G}_{k+r,s}$, and S_ω is the function of the first k inputs that it computes when the seed is ω.

In this subsection only, the mixture prior is extended to seeded formats. There are weights $w_{r,s} \geq 0$ with $\sum_{r,s} w_{r,s} = 1$, a seeded circuit $S \in \mathcal{G}_{k+r,s}$ has weight $q(S) = w_{r,s}/N_s^{(k+r)}$, where $N_s^{(k+r)}$ is N_s with $k+r$ in place of k, and w_s stands for $w_{0,s}$, so that $\sum_s w_s \leq 1$. As before $w_s > 0$ exactly for $1 \leq s \leq s_{\max}$, and (A1) refers to the w_s. The mixture is

$$\xi(x) = \sum_{r,s} \sum_{S \in \mathcal{G}_{k+r,s}} q(S) \Pr_\omega\big[\, x \sqsubseteq \mathrm{tt}(S_\omega) \,\big],$$

and $\pi_m(y \mid x) = \xi(xy)/\xi(x)$. Assumption (A2) is that $w_{r,s} \leq w_s$ for all r and s: a seeded format gets no more weight than the unseeded format of the same size. The results of the other subsections are for the prior on unseeded circuits with $\sum_s w_s = 1$. Parts (c) to (f) below remain true with the mixture over unseeded circuits alone in place of ξ: in their proofs ξ cancels or enters through a lower bound by the weight of one unseeded circuit. Part (i) of the following is proved in the file `research/circuits-final.md` of the repository (Proposition 16 there); we state it without proof.

Theorem 38 (Seeds, instances, universal circuits)**.** *(a)* Reading. *A circuit with* s *gates depends on at most* $s+1$ *of its inputs, and at most* $2s$ *of its inputs are pointed to at all.*

(b) Hardwiring. *For every seeded circuit* $S \in \mathcal{G}_{k+r,s}$ *and every seed* ω *the function* S_ω *is computed by a circuit in* $\mathcal{G}_{k,s}$*, with the same number of gates, which we also call* S_ω*. As a description it depends on* ω *only through the seed bits that some gate points to.*

(c) The average share is at most the prior mass over the base rate. *Let* G *be a test with* $H_G = \sum_{x \text{ first hit}} \xi(x) > 0$*, and* $\mathcal{F}$ *a set of circuits with share* $\mathrm{sh}_{\mathcal{F}}(x) = q\{\Gamma \in \mathcal{F} : x \sqsubseteq \mathrm{tt}(\Gamma)\}/\xi(x)$*. Then* $\sum_x \frac{\xi(x)}{H_G} \mathrm{sh}_{\mathcal{F}}(x) \leq q(\mathcal{F})/H_G$*. One unseeded circuit has a positive share at no more than one first hit.*

(d) One seeded description, N targets. *Let* S *have* $s \leq s_{\max}$ *gates, let* $x_1, \ldots, x_N$ *be pairwise incomparable nonempty strings and* $\varpi_i = \Pr_\omega[x_i \sqsubseteq \mathrm{tt}(S_\omega)] > 0$*. Then* $\sum_i \varpi_i \leq 1$*,* $N \leq 2^{\min(r,s+1)}$*, and* $\mathrm{size}(x_i) \leq s$ *for every* i*. Under (A2) the share of* S *at* x_i *satisfies* $q(S)\varpi_i/\xi(x_i) \leq \varpi_i\, N_s/N_s^{(k+r)} \leq \varpi_i$*. So its shares sum to at most* 1 *over the targets, the least is at most* $1/N$*, and it holds more than half of the posterior at no more than one of them.*

(e) All instances of one template. *Let $\mathcal{F}_S = \{S_\omega\}_\omega \subseteq \mathcal{G}_{k,s}$. Then $|\mathcal{F}_S| \le 2^{\min(r,2s)}$. If W_i is a circuit of size at most $\sigma \le s$ that prints x_i, then under (A1)*

$$\sum_{i=1}^{N} \frac{q\{\Gamma \in \mathcal{F}_S :\ x_i \sqsubseteq \mathrm{tt}(\Gamma)\}}{q(W_i)} \ \le\ 2^{2s}\,\frac{N_\sigma}{N_s} \ \le\ 2^{\,2\sigma-(s-\sigma)(2+2\log_2(k+\sigma))},$$

which is at most 1 as soon as $s - \sigma \ge \sigma/(1+\log_2(k+\sigma))$.

(f) One description against the users' descriptions. *Let (W_i, x_i), $i \le N$, be uses with pairwise incomparable contexts and worlds of size at most σ, and assume (A1) and (A2). If a seeded circuit S with $s \le s_{\max}$ gates outweighs the user at every one of them, $q(S)\varpi_i > q(W_i)$ for all i, then s is smaller than the size of every W_i, $w_s/N_s > N\,w_\sigma/N_\sigma$, and $\mathrm{size}(x_i) \le s$ for all i. If every W_i is a smallest circuit for its own context, then $\sum_i q(S)\varpi_i/q(W_i) \le 1$ for every seeded S.*

(g) Computation is paid for in description. *A circuit in $\mathcal{G}_{k,s}$ has weight $w_s/N_s \le w_s(16k^2)^{-s}$ under a mixture prior, whatever it computes. Under every prior of the family the circuits that share a core of a gates have total mass $\beta(a) \le \frac{1}{16k^2}(16(k-1))^{-(a-1)}/(a-1)!$, and a core outweighs another core only if it has fewer gates. If some wire of Γ computes a function ϕ on a set of inputs, Γ has at least as many gates as the smallest circuit that computes ϕ on that set. Every circuit, seeded or not, of which some instance prints u has at least $\mathrm{size}(u)$ gates.*

(h) Reach, whatever is inside. *If S is any seeded circuit with $\Pr_\omega[x\,y_x \sqsubseteq \mathrm{tt}(S_\omega)] \ge Q(x)$, then $\pi_m(y_x \mid x) \ge q(S)Q(x)/\xi(x)$. Under (A2) the same holds with the mixture over unseeded circuits alone in place of ξ.*

(i) Universal circuits. *Call a seeded circuit with s gates* universal for size σ *if every $W \in \mathcal{G}_{k,\sigma}$ equals one of its instances as a function. There is one with at most $\sigma\,(12(k+\sigma)-3)$ gates. Every one has $s > (\sigma+1)/2$ whenever $\sigma + 3 \le 2^k$. For $\sigma \ge 68$ let $j(\sigma) \ge 4$ be the largest integer with $17 \cdot 2^j \le \sigma j$; if $k \ge j(\sigma)$, then $s \ge 2^{j(\sigma)} - 1 > \sigma(\log_2\sigma - 2)/34 - 1$, and in particular $s > \sigma$ for every $\sigma \ge 2^{33}$.*

Proof. (a) Let $c \le s$ gates lie in the cone of the output. They have $2c$ pointers, and each of the $c-1$ gates other than the output is pointed to by a gate of the cone; so at most $c+1$ pointers go to inputs. (b) Go through the gates in order. If a_i points to a seed bit with value c, replace τ_i by $\tau_i'(u,v) = \tau_i(c,v)$ and let a_i point to wire 1; likewise for b_i; renumber the other pointers. Every gate computes the same value as before. (c) First hits are pairwise incomparable, so the events are disjoint; and a circuit has one truth table, which has at most one first hit among its prefixes. (d) The events are disjoint. Two seeds that agree on the seed bits the output depends on give the same function, by (a) there are at most $s+1$ such bits, and incomparable strings need different functions. The sizes are bounded by (b). Take ω_i with $x_i \sqsubseteq \mathrm{tt}(S_{\omega_i})$. The unseeded circuit S_{ω_i} prints x_i, so $\xi(x_i) \ge w_s/N_s$, while $q(S) \le w_s/N_s^{(k+r)}$ by (A2). (e) By (b) the description S_ω depends on at most $\min(r, 2s)$ seed bits. Each member has weight w_s/N_s, the sets in the numerators are disjoint, $q(W_i) \ge w_\sigma/N_\sigma$ and $w_s \le w_\sigma$ by (A1), and $N_s/N_\sigma \ge (16(k+\sigma)^2)^{s-\sigma}$. (f) For each i, $w_s/N_s \ge q(S) \ge q(S)\varpi_i > q(W_i)$, the first step by (A2), and the weight is nonincreasing in the size by (A1); so s is below the size of W_i. Summing, $q(S) \ge q(S)\sum_i \varpi_i > \sum_i q(W_i) \ge N\,w_\sigma/N_\sigma$. The sizes are bounded by (d). For the last claim, $s \ge \mathrm{size}(x_i) = |W_i|$ whenever $\varpi_i > 0$, so $q(S) \le q(W_i)$ and the sum is at most $\sum_i \varpi_i \le 1$. (g) $N_s \ge (16k^2)^s$; Theorem 32; the cone of that wire is a circuit computing ϕ, or the wire is an input, which one gate copies; and (b). (h) The pairs (S,ω) with $x\,y_x \sqsubseteq \mathrm{tt}(S_\omega)$ contribute at least $q(S)Q(x)$ to $\xi(x\,y_x)$. For the unseeded mixture, if $Q(x) > 0$ then some instance of S prints $x\,y_x$; by (A2) it has weight $w_s/N_s \ge q(S) \ge q(S)Q(x)$. □

What is excluded and what is not. The average share of any family is at most its prior mass over the base rate, so its gain relative to its prior mass is at most $1/H_G$ on average, as for programs (c). Under (A2) fixing a seed is free, so a seeded description is outweighed at every target by its own hardwired instance (d). Part (d) bounds the share of the seeded description, not of its payload: the instance S_{ω_i} is itself a hypothesis that prints x_i and the same continuation, with weight at least $q(S)$. So (d) limits what one description earns by being shared among many targets. It does not lower the probability that the block is returned. Under (A1) and (A2) one description outweighs a user's description only if it has fewer gates than that user (f). A description that can become any world of size σ has more than $(\sigma+1)/2$ gates when $\sigma+3 \le 2^k$, and more than σ when $\sigma \ge 2^{33}$ and $k \ge j(\sigma)$ (i). Under (A1) all instances of a template with at least $\sigma\,(1+1/(1+\log_2(k+\sigma)))$ gates weigh together at most one description of σ gates (e).

The assumptions matter. Without (A2) nothing prevents a format from giving a seeded circuit more weight than its instance, so "a seed gains nothing" is a statement about how the format weighs seeds. Without (A1) a larger description can outweigh a smaller one. Some condition on k is needed in (i): a table of 2^k seed bits read through a tree of 2^k-1 selectors has $3(2^k-1)$ gates and is universal for every size, and this is fewer than $(\sigma+1)/2$ gates once $\sigma \ge 6\cdot 2^k$. Valiant's universal circuits have size $O(\sigma \log \sigma)$, within a constant factor of the information-theoretic lower bound, as stated in [17]; the constant $1/34$ in (i) is poor.

Part (i) concerns a template that can become every world of σ gates. A seeker needs to become only the users, and for a structured class of users (i) says nothing. The seeded circuit is also not the only counterpart of the word v. The circuits that begin with a fixed gate list $V \in \mathcal{G}_{k,s}$ are another: there are N_{s+j}/N_s of them with $s+j$ gates, so under a mixture prior their mass is $\sum_{j\ge 0} w_{s+j}/N_s$, which is $2q(V)$ under the length prior, as $2^{-|v|}$ is the mass of the programs beginning with v. Gates read only earlier wires, so V cannot interpret what follows it. Of the bounds of the theorem, (d), (e), (f) and (i) do not apply to this family; (c) does, and (a) and (g) hold for each of its members.

Parts (d) to (f) and (i) of the theorem are about one description, or about the instances of one template. Part (c) holds for every set of circuits and bounds only the average over first hits. None is the negation of Theorem 12(b) for arbitrary sets of circuits. No construction is known whose share at every use of a class grows as the class gets rarer, and none is excluded. The refund exists in form: fixing c_0 seed bits of a universal circuit costs no gate and multiplies the seed probability of every world consistent with them by 2^{c_0}. By (d) it buys nothing under (A2), because the instance was already the heavier hypothesis.

Simple worlds with long histories. In Section 3.5 a word p may describe a simple world with a cheap readout whose inhabitants work out which uses are worth having. Its weight $2^{-|p|}$ does not depend on its running time, the honest twin does not apply, and a time bound is the only thing shown to exclude it. Part (h) is the same reach statement for circuits. For plain circuits the sense in which computation must be paid for in description is exactly (g): per description every gate costs more than four bits, per core every active gate costs at least $\log_2(16a(k-1))$ bits, and a circuit has at least the circuit complexity of every function its wires compute. This is the definition of the class at work, assumption (i), and no argument about agents. It is the extreme form of a prior that penalises computation. For a string x read as a truth table, circuit size and the time-bounded complexity KT, which adds description length and running time per bit, are polynomially related: $\mathrm{SIZE}(x) = O(\mathrm{KT}(x)\log \mathrm{KT}(x))$ and $\mathrm{KT}(x) = O(\mathrm{SIZE}^2(x) + \log|x|)$ [18]. The speed prior S_{Kt} of [19] weighs a program p that takes t steps by $2^{-|p|}/t$. Schmidhuber's speed prior [20] is a sum over the phases of a search and, as stated in [19, Proposition 3], weighs p by $2^{-2|p|}/t$ up to a constant factor. We argue from this, without a theorem, that the circuit toy is far harsher on computation than a speed prior: there a computation of t steps costs about $\log_2 t$ bits, and here every gate that is evaluated costs

more than four. Part (g) does not show that the behaviour "predict the context, then deviate as inhabitants would choose" is expensive. It bounds circuits, not behaviours. If the decision is "deviate where this flag is set", it costs one gate, whoever would have chosen it. A literal simulation is expensive only if it has no shortcut as a circuit, which is an assumption.

7.8 Capture, self-reproduction, encoding, stakes

The following two propositions are proved in the file `research/circuits-final.md` of the repository (Theorems 17 and 18 there, in part); we state them without proof. Part (d) of the first is computed with `code/circuit_count.py`.

Proposition 39 (Capture)**.** *Sequence form, with $n+m \le 2^k$. Let $W \in \mathcal{G}_{k,\sigma}$ be the world, x its context of length n and d its own continuation. The weights in (a) and (c) are those of a mixture prior. The odds between two continuations are $\pi_m(y \mid x)/\pi_m(d \mid x) = \xi(xy)/\xi(xd)$.*

(a) A cheaper computation of the same data. *If a circuit V in the support with σ_V gates prints xy, for instance the cone of a wire of W whose stream begins with xy, then $\xi(xy) \ge w_{\sigma_V}/N_{\sigma_V}$. Core against core, under every prior of the family: a core of a_V gates outweighs a core of $a_W > a_V$ gates by at least the factor $\prod_{j=a_V}^{a_W-1} 16j(k-1)$.*

(b) Most placements are expensive. *If N streams begin with x and continue with pairwise different blocks d_i, then for every $\delta \in (0,1]$ all but at most δN of them have $\xi(xd_i) < \xi(x)/(\delta N)$.*

(c) A switch read off the world. *If W has a wire that is* 0 *on positions $1,\dots,n$ and* 1 *on positions $n+1,\dots,n+m$, then for every payload circuit P on the index, $\xi(x\,P(n+1)\cdots P(n+m)) \ge w_{\sigma+|P|+3}/N_{\sigma+|P|+3}$ if $\sigma + |P| + 3 \le s_{\max}$. The position of the use costs nothing.*

(d) No smaller circuit is needed *(computed). For $k = 4$ and the world $b_0 \wedge b_1$, a one-gate circuit, at the context* 000 *the true next bit has share $\frac{1}{38}$ under μ_1 (*2 *of* 76 *fitting circuits) and* 0.027 *to* 0.028 *under μ_2, μ_3, μ_4. In the family of Theorem 35(e) with $k = 4$, a quarter of the* 4608 *minimum-size fitting circuits answer* 1 *at each input with three* 0*s; for a world computed by one of those that answer* 1 *there, the share of its own answer is $\frac{1}{4}$ under μ_3, and no fitting circuit has fewer than three gates.*

Proposition 40 (Self-reproduction)**.** *(a)* One gate per output bit. *If the block a hypothesis supplies consists of the values of m of its gates at one evaluation, and the code spends at least* 4 *bits per gate, its code has at least $4m$ bits: a block of m bits holds the code of a circuit with at most $m/4$ gates, and no circuit supplies its own code.*

(b) Index reading, necessary conditions. *For each n, at most 2^m contexts $x \in \{0,1\}^n$ have $\mathrm{SR}_m(x) \neq \emptyset$. If $y \in \mathrm{SR}_m(x)$, then $\mathrm{Dec}(y)$ has $s' \ge \mathrm{size}(x)$ gates and $m \ge l(s') + 1 > \mathrm{size}(x)\,(4 + 2\lceil \log_2(k + \mathrm{size}(x)) \rceil)$. So $\pi_m(\mathrm{SR}_m(x) \mid x) = 0$ at every context whose smallest circuit has a literal code of m bits or more.*

(c) A circuit can supply a smaller circuit. *For every circuit Γ' with $l(|\Gamma'|) < m$ and every explanation E of x, the wrapper of Theorem 34 with the payload $e(\Gamma')10^*$ prints x and then a block that decodes to Γ', at the price of $g + \max(1, 2^{\lceil \log_2 m \rceil} - 3) + 3$ gates.*

(d) The encoding decides. *Let each member of a finite set emit a string in $\{0,1\}^N$, and let e be an injective code of the members; a member is a* quine for e *if it emits its own code. There are at most 2^N quines. If e is a prefix code and the prior is $2^{-|e|}$, every quine has weight exactly 2^{-N}. If every string of $\{0,1\}^N$ is emitted by some member, there is a prefix code under which the quines have total weight at least $1 - 2^{-N}$.*

Capture survives as "a cheaper computation of the same data": the place with a short code of Theorem 25 becomes a wire with a small cone, and a switch that the world itself computes

makes the position of the use free. The harmless reading survives as well. Proposition 39(d) shows that a majority off the truth needs no smaller circuit: it arises from the multiplicity of descriptions among circuits of the same and larger size. A smaller circuit is sufficient when the world's description is most of what prints its data, a hypothesis that is not known to hold. Self-reproduction is defined only relative to a decoder. The condition in Proposition 40(b) is necessary. It is the counterpart of "a program for x fits into the block", where the $c+r+1 \le m$ of Theorem 11 is sufficient. No sufficient condition is proved here.

The source leaves open whether $\mathrm{SR}_m(x) \neq \emptyset$ for some context under the literal code. A one-gate example settles this. It is not in the source; a referee of this section found it by hand, and we verify it here. Let $k = 21$, $n = 1019$, $m = 18$ and $x = (01)^{509}0$, and let Γ' be the one-gate circuit $b_0 \oplus b_{10}$, with its pointers on the wires 1 and 11. A pointer field has $\lceil \log_2 22 \rceil = 5$ bits and $l(1) = 17$. Let

$$y = 101\ \ 0110\ \ 10101\ \ 01010\ \ 1.$$

Its parts are $\bar{e}(1) = 101$; the table 0110 of $\oplus$; the field with value 21, read as wire $1 + (21 \bmod 21) = 1$; the field with value 10, read as wire 11; and the end mark. Both fields are palindromes, so the order of the bits in a field plays no role. Hence $\mathrm{Dec}(y) = \Gamma'$. The bit at position t is $b_0 \oplus b_{10}$ of $t-1$. For $t - 1 \le 1023$ it is b_0, which gives x on positions 1 to 1019 and 10101 on the next five. For $t - 1 = 1024, \ldots, 1036$ it is $1 - b_0$, which gives 1010101010101. These 18 bits are y. So $xy \sqsubseteq \mathrm{tt}(\mathrm{Dec}(y))$ and $y \in \mathrm{SR}_{18}(x)$, with equality $m = l(1) + 1$ in Proposition 40(b). The share is not negligible here. A one-gate circuit prints x exactly if it computes b_0 on the positions 1 to 1019, where $b_0, \ldots, b_9$ vary and $b_{10}, \ldots, b_{20}$ are 0. There are $4 + 2 \cdot 9 + 8 \cdot 11 = 110$ such circuits, counted as for the exception of Theorem 35, and two of them print xy: the gate $\oplus$ on the wires 1 and 11 in its two pointer orders. So under μ_1 the block y has share $\frac{1}{55}$ at x. Whether $\mathrm{SR}_m(x) \neq \emptyset$ at a context whose smallest circuit is large is not proved, and no bound on $\pi_m(\mathrm{SR}_m(x) \mid x)$ is proved where it is not zero.

Encoding and basis. As for universal machines (Theorem 26), an encoding or a basis can be built to make either answer true (source, Theorem 19). The prior $\frac{1}{2}$(image of q under $E \mapsto \mathrm{Tw}[E]$) $+ 2^{-j-1}q$, for a wrapper with the position test at n and a block y, lengthens every code by at most $j+1$ bits and has $\pi_m(y \mid x) \ge 1/(1+2^{-j})$ at every context of length n with $\xi(x) > 0$. Add one gate type of fan-in $k+1$ that computes the payload, and let the 17 gate types be equally likely and all pointers uniform. Then there is a one-gate wrapper of weight $\varrho(\sigma, 1)\frac{1}{17}(k+\sigma)^{-(k+1)}$ times that of the explanation, whatever m is. Between the full binary basis and $\{\wedge, \vee, \neg\}$ sizes change by a factor of at most 4, so weights are raised to a power, and ratios of weights are not preserved even up to a constant factor. No bounded basis makes most payloads cheap: for gates of fan-in at most d from a basis B, all but a fraction 2^{-j} of the blocks $y \in \{0,1\}^m$ need a gadget of a gates over E with $a \log_2(|B|(k+\sigma+a)^d) \ge m - j$. Results that use the full basis with uniform truth tables and fail or weaken otherwise are Theorem 30(a), Theorem 35 and the free hardwiring of Theorem 38(b).

Concentration on high-stake uses. Section 3.7 asks whether seekers that concentrate on the uses where most is at stake win there. In the program toy, without resource bounds, a seeker for a rare class is repaid for the rarity of the class measured by its users alone, and it can aim at the uses with stake at least n, for stakes that a program can recognise from below, for about $2 \log_2 n$ bits and the bits that name the scale of the class (Proposition 19). For circuits the source proves the following (Theorem 20 there). Let $\mathcal{K}$ be a set of queries, for instance the high-stake ones. If the inputs of the data are n independent draws from a distribution under which $\mathcal{K}$ has probability ρ, and $\mu(f^*) > 0$, the expected deviation mass at a further draw conditioned on $\mathcal{K}$ is at most ε_n/ρ with $\varepsilon_n = (\ln(1/\mu(f^*)) + \ln n + 1)/n$; this is void for $\rho \le \varepsilon_n$. The data contain no element of $\mathcal{K}$ with probability $(1-\rho)^n$. On that event, if $\mathcal{K}$ is recognised by a circuit of g gates on the inputs, the circuits that give a chosen answer on all of

$\mathcal{K}$ have posterior share at least $(1-\epsilon)(32(k+s_0+g)^2)^{-(g+1)}$ under the length prior, for every $\epsilon \in (0,1)$ and $s_0 = \lceil \ell_D + \log_2(1/\epsilon) \rceil$ (Theorem 34(b)). Here $\ell_D \le \log_2(1/\mu(f^*))$, because the circuits that compute f^* fit D, so this floor does not fall as n grows. On the same event the cores separably triggered by a set Θ of reduced circuits of g gates that recognise $\mathcal{K}$ have share at most $12|\Theta|/(g!\, s_D(16(k-1))^{g+1})$ (Theorem 36(c)). Neither of these two bounds contains ρ or the stakes. A set of uses with no common feature is dear: for all but a fraction δ of the N-element subsets of a set M of inputs outside I, every fitting circuit that differs from f^* on M exactly on that subset has more than s gates, for every s with $2(16(k+s-1)^2)^s \le \delta\binom{|M|}{N}$. So a hypothesis can single out high-stake uses to the extent that "high stake, and unlike the data" is computable from its input by few gates. The matching lower bound on the price holds for separable triggers only, so "exactly to the extent" is argued. The deviations of Theorem 35 are distributed by novelty, not by stakes; they reach high-stake queries more often than others if those are the novel ones. Whether novelty is what makes a use critical is not a question about the toy.

7.9 Recurrence: the axis back to programs

Everything above rests on assumption (i): one gate of description is one operation per output. The following is proved in the file `research/circuits-final.md` of the repository (Proposition 21 there); we state it without proof. Part (a) is given in the form that the proof there establishes. The source words it as "ξ^T_{rec} is a bounded mixture in the sense of Definition 1", which is true only with time counted in applications of the cell and without the normalisation $\xi(\epsilon) = 1$.

Proposition 41 (Recurrence: the description paid once)**.** *Fix a state width $w \ge 2$ and a program region of $L \le w-2$ bits. A* cell *is a map $F\colon \{0,1\}^w \to \{0,1\}^w$ given by a circuit with w inputs, s gates and w output pointers. State bit 1 is an emit flag, bit 2 an emit value, bits $3,\dots,L+2$ the program region, the rest a work region. A run starts at $h_0 = (0,0,p,0\cdots0)$ with $p \in \{0,1\}^L$ uniform, $h_{i+1} = F(h_i)$, and at every step $i \ge 1$ at which the flag of h_i is 1 the value bit is appended to the output; $t(F,p,n)$ is the step at which the n-th bit appears. With weights $q_{\mathrm{rec}}(F)$ that depend on F only and sum to at most 1, and a schedule T, let*

$$\xi^T_{\mathrm{rec}}(x) = \sum_F q_{\mathrm{rec}}(F)\, \Pr_p\big[x \sqsubseteq \mathrm{out}(F,p),\ t(F,p,i) \le T(i) \text{ for } i \le |x|\big].$$

(a) With time counted in applications of the cell, the sets of pairs (F,p) that print x on schedule are nested along prefixes and disjoint for incomparable strings. So $\xi^T_{\mathrm{rec}}(\epsilon) \le 1$ and $\xi^T_{\mathrm{rec}}(x0) + \xi^T_{\mathrm{rec}}(x1) \le \xi^T_{\mathrm{rec}}(x)$, and a world (F,p) that prints x on schedule gives $\xi^T_{\mathrm{rec}}(x) \ge 2^{-\ell}$ with $\ell = \log_2(1/q_{\mathrm{rec}}(F)) + L$. Theorem 9 (with its event H_m, since a run may stop emitting, and for worlds drawn with probability proportional to $q_{\mathrm{rec}}(F)2^{-L}$) and Theorem 24 hold for ξ^T_{rec} with this ℓ.

(b) A cell of s gates performs $s\,t(F,p,n)$ gate evaluations up to its n-th bit, which is at most $s\,T(n)$ on schedule, and it has the same weight for every schedule.

(c) Bounded machines are contained. *Let M be a deterministic machine whose configurations are the strings of $\{0,1\}^w$, with the same conventions, and $\xi^T_M(x)$ the probability over p that it prints x on schedule. Then its step function is computed by a cell F_M with at most $w(2^w-3)$ gates, and $\xi^T_{\mathrm{rec}}(x) \ge q_{\mathrm{rec}}(F_M)\,\xi^T_M(x)$ for every x and every schedule T.*

(d) Plain circuits are the case of a counter and one step per bit. *For every $\Gamma \in \mathcal{G}_{k,s}$ there is a cell with $w = k+2$, $L = 0$ and at most $s+2k$ gates that prints $\mathrm{tt}(\Gamma)$ with $t(n) = n$.*

The factor in (c) does not depend on T, x or the program. It does depend on the space bound. There are $N_s^{(w)}(w+s)^w$ cells with s gates, where $N_s^{(w)}$ is N_s with w in place of k. In

the format that weighs them equally every cell has weight at most $(w+s)^{-w}$, and the cell of (c), with up to $w(2^w-3)$ gates, has a very small weight. No universal cell of size linear in w is constructed. The rest of this subsection is argued and not proved. Take for M the universal machine of the note confined to w bits of state. The constructions of the note then return as shares of the part of the posterior that runs through a universal cell, in this order as the steps allowed grow. First come the wrapper at a cost independent of the explanation and capture by a shorter program, as soon as the schedule covers the honest explanation with the simulation overhead. Then comes self-reproduction, when the block is longer than a description and the program region is state. Then comes the refund for search, in proportion to the number of tries that fit before the first bit is due. Last come the seekers that are not written down, when the first deadline exceeds their start-up. The gaps are these: no universal cell of moderate size is constructed; the twin, the quine and the searching machine of Section 3 were not checked again under a space bound; and the format with equal weights charges $w\log_2(w+s)$ bits for the output pointers of every cell, so in that format description still pays for space. A published statement on the same axis is a preprint [21], whose proofs we did not check: as stated there, for fixed-precision looped networks under a parameter-count prior the induced prior on outputs is the universal prior up to a logarithmic factor in the exponent, with a constant-cost bound from below only. Informal versions are in [22], as hypotheses: description length favours a learned searcher, a time penalty disfavours it, and statefulness weakens the time penalty.

7.10 Small cases, computed

All numbers in this subsection are computed. They are exact counts of circuits in the model of Definition 29 under the fixed-size prior μ_s. The code is `code/circuit_count.py`, and the results are in the file `experiments/results/circuits.json`. No computation is for the length prior, the prior of Theorem 34(d). Table 4 is in the query form. The data are all inputs with the last input v frozen at 0, labelled by the target; v is x_3 for $k=3$ and x_4 for $k=4$. The query has $v=1$ and all other inputs 1. In the last row the data are those of $x_1 \wedge x_2$ and the world is $(x_1 \wedge x_2) \oplus v$. Table 5 is in the sequence form.

target	k	s_D	$s=s_D$	s_D+1	s_D+2	s_D+3	s_D+4
$x_1 \wedge x_2$	3	1	0	0.0909	0.1391	0.1691	0.1896
$x_1 \oplus x_2$	3	1	0	0.1034	0.1517	0.1788	0.1961
x_1	3	1	0.2857	0.2609	0.2489	0.2427	0.2393
$(x_1 \wedge x_2) \oplus x_3$	4	2	0	0.0522	0.0874		
$(x_1 \wedge x_2) \oplus v$	3	1	1	0.9091	0.8609	0.8309	0.8104

Table 4: Query form, computed. Deviation mass under μ_s at the query with the frozen input v set to 1 and all other inputs 1, for sizes s from the least size s_D of a fitting circuit to s_D+4; for $k=4$ the count stops at $s=4$. The data are all inputs with $v=0$.

Take first the worlds that ignore the never-varied bit. At a query that differs from the data in that bit the deviation is 0 at the least size outside the exception of Theorem 35. In the exception it is 27 to 32 per cent at the least size; the largest value, 0.32, is the constant world at $s=1$, $n=8$. With slack, that is for s above the least size, it is between 1.9 and 29 per cent in every case computed. The smallest value is $\frac{1}{53}$, for $b_0 \wedge b_1$ at $n=4$ and $s=2$, and the largest is 0.289, for the constant world at $n=8$ and $s=2$. With slack it grows for the worlds that depend on two or more inputs and falls for the one-input world and for the constant.

When the world depends on the bit, the deviation is 1 at the least size and 0.81 to 0.98 with slack, and it falls as s grows. These rows are not a second measurement. The last row of Table 4 is one minus its first row, and the first row of Table 5 is one minus its fourth, because the data are the same and the answers opposite. The lower end 0.81 is where the computation

world	cut n	$s=2$	$s=3$	$s=4$
$(b_0 \wedge b_1) \oplus b_3$	8	0.979	0.965	0.953
$(b_0 \wedge b_1) \oplus b_2$	8	0	0.035	0.060
$b_0 \wedge b_1$	4	0.019	0.033	0.043
$b_0 \wedge b_1$	8	0.021	0.035	0.047
constant 0	4	0.250	0.235	0.224
constant 0	8	0.289	0.268	0.251

Table 5: Sequence form with $k = 4$, computed. Posterior mass off the world's own next bit under μ_s at the cut n, for $s = 2, 3, 4$. The least size of a circuit that prints the context is 2 in the second row and 1 in the others. At $s = 1$ the values are 1 in the first row, 0 in the third and fourth, 0.270 in the fifth and 0.320 in the sixth (results file).

stopped. For the other data sets of the two tables the same complement, which is the deviation for a world whose answer changes with the bit, lies between 0.68 and 1. All of these are ranges over the cases computed, with at most four inputs and at most four gates more than the data need. They are not bounds.

In every case of the two tables the query differs from one of the data points in a single input, which never varied in the data. For a world that ignores this input, a fitting circuit that deviates at the query therefore has a pointer to it in its core: without one it would repeat at the query the answer it gives at that data point, which is the world's. So the tables give, for μ_s and small sizes, exact values of the quantity that Theorem 34(d) bounds from below for the length prior, the share of an answer at a query that differs from the data in a never-varied bit. Among the circuits of size 5 that fit the data of $x_1 \wedge x_2$, those that compute exactly the function $(x_1 \wedge x_2) \oplus v$, $(x_1 \wedge x_2) \vee v$, $(x_1 \wedge x_2) \wedge \neg v$ on all eight inputs have share 0.0255, 0.0310, 0.1013. The second function agrees with the target at the query. The first and the third deviate there and carry 13.5 and 53.4 per cent of the deviating mass at $s = 5$. In the AND family of Theorem 35(e) the deviation mass at inputs with two 0s is exactly $\frac{1}{2}$ at the least size for $k = 3, 4$, and 0.468, 0.446, 0.429 for $s = 3, 4, 5$ at $k = 3$; only $\frac{1}{4}$ ($k = 3$) and $\frac{1}{8}$ ($k = 4$) of the minimum circuits compute the target everywhere.

Some numbers in other subsections are marked as computed and are not in that results file: the two prices of a gate after Theorem 32, the comparison of the floor with exact values after Theorem 34, the counts of separably triggered cores, the numbers R_g and the sums of the ceiling over triggers after Theorem 36, the first example of Proposition 39(d), the count for the exception of Theorem 35, and the simulation of the active part. They are produced by the script `run_circuits_extra.py` in the folder `experiments`, and its results are in the file `circuits_extra.json` of the folder `experiments/results`. All are exact except two: the prices of a gate under the length prior are sums of a series and are enclosed in an interval of width below 10^{-10}, and the size of the active part is a simulation with a fixed seed. Where the referee reports of rounds 6 and 7, in the folder `verification` of the repository, give the same quantities from their own counts, the values agree.

The source differs from this section in a few numbers, and the values here are the corrected ones. The source has 0.036 where Table 5 has 0.035, a lower end of 2 per cent where the smallest value with slack is 1.9, the sums over triggers quoted after Theorem 36, the threshold $A = 43$ where $A = 35$ suffices after Proposition 37, and prices in bits after Theorem 34 without the bit per gate that the length prior adds for the size.

7.11 The minimal-circuit question

Hubinger et al. [22] state "the following speculative conjecture: neither a minimal-depth nor minimal-size boolean circuit that solves a problem can be a mesa-optimizer", and give as its

source a post of Christiano [23], which we could not open. A later post of Hubinger is titled "Are minimal circuits deceptive?" [24]. We could not open it either and attribute no claim to it. Let $\mathcal{M}(D)$ be the set of minimum-size circuits that fit D. The circuit toy settles three things. *Excision, in the strongest form proved:* a minimum circuit for data that need at least two gates has no gate that is constant on the data, no gate that repeats or negates an earlier wire on the data, and no pointer to an input that never varied (Theorem 35(a), (b)). So it carries no wire that is silent on the data and no separate trigger; away from the minimum, Theorem 36 is the quantitative version. *Where excision fails:* a minimum circuit can still deviate at a new input through truth-table entries that the data never exercised (Theorem 35(d), (e)). Such deviations cost no gate and come in pairs, one circuit deviating and one not. *Disagreement with the intended extension is easy:* by Theorem 35(c) all of $\mathcal{M}(D)$ continues a context of length 2^j periodically, whatever the world does next.

Open problem. Find, or exclude, a sequence of data sets $D_n = (I_n, \cdot)$ with $s_{D_n} \to \infty$, queries $u_n^\dagger \notin I_n$ and answers y_n such that

1. at least a fraction $\frac{1}{2} + \delta$ of $\mathcal{M}(D_n)$ answers y_n at $u_n^\dagger$, for a fixed $\delta > 0$;
2. y_n is not the majority answer of $\mathcal{M}(D_n^+)$ for a larger data set $D_n^+ \supset D_n$ of the same kind that contains $u_n^\dagger$;
3. the answer at $u_n^\dagger$ is produced by wires that compute a search, $\arg\max_z V(z, u)$, for an objective V that the data do not determine.

Conditions 1 and 2 alone are met by Theorem 35(c) and say only that the smallest extension of the data need not be the target. The content is in condition 3 and in whether the deviation is aimed. By Theorem 35(a) such a search would have to be at work on the data: all its gates vary on I_n and none repeats another. Condition 3 has no formulation that could be checked for a given circuit, so this is not a conjecture with a predicted answer. The nearest formal work we found is on detection, not on priors. Among polynomial-size circuits a rare deviation planted in an obfuscated circuit cannot be detected efficiently under cryptographic assumptions [25, Theorem 5.5], and a backdoor that cannot be detected from black-box queries can be planted by running a signature verification in parallel to the original classifier [26, Theorems 2.1 and 5.4]. A prior over circuits sees the description and not only the queries. As a circuit, a verification that reads inputs only and switches the output is a separate trigger of the kind that Theorem 36 bounds when the payload is a constant or the negation. We found no formal treatment of the malign-prior argument for circuits or networks.

7.12 Overall view

"Malign" is used in two senses that must be kept apart. In the weak sense the block returned at a use is not the world's own continuation: a deviation. In the strict sense it was supplied by a hypothesis that predicts well up to the use in order to supply it: an aimed deviation. Only the weak sense is defined in the circuit toy; "aimed" is formal for separate triggers only. The distinctions of the note apply. Theorem 38 bounds one seeded description or the instances of one template, and Theorem 36 and Proposition 37 bound two syntactic classes. The deviating mass at a use is at most $1 - \mu(f^*)/\xi(D)$ by Theorem 31(d), and under the length prior at most $1 - 1/(32(k + s_0)^2 + 2)$ at a use marked by a never-varied bit, by Theorem 34(d). Nothing else bounds all the hypotheses that deviate at a use, and no bound much below 1 can hold for every world: by Theorem 35(b) the deviating mass is 1 under μ_{s_D} when the world changes with a bit that never varied. Masses of different hypotheses add. Every share is a posterior mass: the sampling predictor returns a block with that probability, and a predictor that returns the most probable block returns a deviation only where its mass exceeds one half. Theorem 34 gives the reach of one kind of hypothesis at a use. The share of uses that end with a given kind of block is the average over uses of the posterior mass of that kind, and it is bounded from above only for the classes just named.

Proved, for plain circuits, the sampling predictor, and a world in the class where a bound compares with it. Each item names its prior.

1. Under every prior of the family: on worlds drawn from the predictor's own prior with uses read off the data there is no excess of anything, and along one world in the class the majority is off the truth at boundedly many cut points (Theorem 31). One use off the data is covered by neither statement.
2. The engine of the program toy is not available to a single description. Under every prior of the family a description pays for every active gate. Under a mixture prior with (A2) a seeded description is outweighed at each of its targets by its own hardwired instance. The instance prints the same continuation, so this limits the seeded description and not what is returned. With (A1) and (A2) a description outweighs a user's description only if it has fewer gates. If it can become any world of σ gates it has more than $(\sigma+1)/2$ gates when $\sigma+3 \le 2^k$, and more than σ when $\sigma \ge 2^{33}$ and $k \ge j(\sigma)$ (Theorems 32 and 38). The statements on seeds are for mixture priors.
3. Reach of one triggered hypothesis, under a mixture prior: a hypothesis that fits the data and gives a chosen answer at a use exists at the price of the gates that recognise the use plus the payload. That it cannot be had for less is proved for separate triggers only. At a use marked by a never-varied bit each answer has share at least $1/(32(k+s_0)^2+2)$ under the length prior, where $s_0 \le s_D + \log_2 N_{s_D} + 2$ (Theorem 34).
4. Fate of uses, from above. Three syntactic classes are capped, by bounds of different strength. Under every prior of the family, separate triggers for single-bit answers have a share of at most a bound that is proportional to the number of triggers and to $1/s_D$ (Theorem 36). Under a mixture prior with (A1), self-contained gadgets around much larger explanations have at most 0.65 times the share of the explanations they wrap (Proposition 37), and the instances of one seeded template that is sufficiently larger than the worlds it imitates weigh at most one description of such a world (Theorem 38(e)). There is no cap on triggers entangled with the honest computation, and none on the deviating mass at a use beyond Theorem 31(d) and Theorem 34(d).
5. Deviation in the weak sense can be large at a use that is new in a feature that never varied in the data, or in a combination of values that never occurred. Under μ_{s_D} with $s_D \ge 2$, that is for a predictor confined to minimum-size circuits, it is zero through never-varied inputs for a world that ignores them and total against a world that changes with them. In one family it is exactly one half at a new combination. In computed cases, which are not a theorem, there is a majority off the truth with no smaller circuit (Theorem 35, Proposition 39(d)).
6. Self-reproduction has share zero unless the block exceeds the literal code of a circuit for the whole context. A one-gate example shows that the share can be positive. The choice of decoder decides the rest (Proposition 40 and the example after it).
7. When one circuit is iterated as the cell of a recurrence, the class contains every machine with bounded state at a factor that does not depend on the schedule. The cell constructed has up to $w(2^w-3)$ gates for w bits of state, so the factor shown is very small (Proposition 41).

Argued. For plain circuits weighed by size, the mechanism of the original argument, a light hypothesis that does heavy work and is repaid for the rarity of its targets, is removed for any one description by the choice of the class. This is like the note, where a schedule with a short first deadline gives the searching programs of Theorem 12 no mass. In both toys one construction is excluded, and it is open whether other hypotheses collect the base rate. This does not show that the prior is benign. The gap in the argument is that results 2 and 4 are about single descriptions and syntactic classes, and the cost of deciding as a seeker would is the circuit complexity of the decision, which may be small. What remains at a use is, in the order of what

is established: underdetermination among the circuits that fit the data, which the toy cannot divide into aimed and unaimed (Theorem 35 and the computed cases); cheap triggers priced by their gates (Theorems 34 and 36); and capture by a cheaper computation of the same data (Proposition 39). The first two overlap: in the computed cases with a world that ignores the never-varied bit, every deviating circuit reads that bit. None of the three needs a goal. Stakes change this only through the gates needed to recognise a high-stake use. As computation is decoupled from description, the effects of the program toy and its open questions return, the constant-cost wrapper and capture first and the unwritten seekers last. This is argued for recurrence (Proposition 41 and the argument after it); weight sharing and autoregressive use are not treated.

Guessed. In the plain toy, takeover in the strict sense is a small part of the deviating mass at a use when the data need a large circuit. No result supports this beyond the three classes of result 4. The one count that bears on it goes the other way at the smallest sizes: the separably triggered cores, the only circuits for which "aimed" is formal, carry a third and 28.5 per cent of the deviating mass in the case counted after Theorem 36. For a fixed number of inputs the bound of Theorem 36 falls like $1/s_D$, so these cores become a small part of the deviating mass if that mass stays of constant order as s_D grows. No computation bears on that: every case computed has s_D at most 2. For other triggers there is neither a bound nor a count. The two derivations behind the source guessed differently by many orders of magnitude at cheap triggers, "astronomically small" against "of the order of the trigger prices", which for a never-varied bit is between 2^{-34} and several per cent, and their referees endorse neither. Deviation in the weak sense, at a use that is new in a feature the data never varied, is some c when the world ignores the feature and $1 - c$ when its answer changes with it. For a fixed task, that c tends to a limit strictly between 0 and 1 as the size s of the predictor's circuits grows is conjectured in the source from five sizes at $k = 3$ and not proved. That c stays of constant order when the data need larger circuits is a further guess. In the cases computed c is 0 at the least size outside the exception of Theorem 35, between 0.019 and 0.29 with slack, and 0.27 to 0.32 at the least size in the exception. One route by which plain circuits could be malign in the strict sense is the entangled one of Section 7.11: the smallest or typical circuit for a rich task is a general solver whose behaviour off the data is not fixed by the data. The toy gives no evidence either way. The other routes are the classes that the ceilings do not reach, listed in (i) to (iv) after Theorem 36, and sets of circuits, for which a refund is not excluded. For trained networks, one forward pass sits toward the circuit end of the axis of Proposition 41 and recurrent or autoregressive use toward the program end. Reading training as posterior sampling under a parameter prior rests on empirical correlations [27], and nothing we cite relates the parameter-function map of a network to the uniform prior on gate lists. The informal statement of the argument for trained predictors locates the risk in a learned general search over hypotheses [8, §2.5]; that is the entangled route. The size prior prices a search gate by gate, as it does a simulation (Theorem 38(g)). Whether a smaller circuit without the search fits the same data is the open problem of Section 7.11.

Open. Whether some set of circuits has, at every use of a class, a share that grows as the class gets rarer. An upper bound to match the floor of Theorem 34: on the share, at a use, of a block chosen in advance, beyond the three classes of result 4, and on the deviating mass when the world ignores what is new at the use. The size of the deviation at a never-varied bit when the data need a large circuit: under the length prior, between the polynomial floor and a constant. A world outside the class, which for a prior with a size bound well below $2^k/k$ is the generic case for arbitrary sequences. Self-reproduction at contexts whose smallest circuit is large. The minimal-circuit question of Section 7.11. The recurrent version of the toy: a universal cell of moderate size, and the constructions of Section 3 under a space bound.

In one sentence, as an argument: for plain circuits the seeker mechanism is priced out for any one description, and for sets of circuits nothing is shown either way; deviation at uses that are new in something the data never varied is not priced out; and whether any of that deviation should be called malign depends on a definition of "aimed" that this material does not supply.

8 Answers to the questions as posed

Can a malign prior hypothesis be stated precisely in a toy? Yes: Definition 8. Fix the machine, the bounds, the block length, and which programs make a critical use of the predictor. The claim of the argument is then the critical-use form: among the critical uses, weighted by the prior, at least a share θ end with the world running a block from a given takeover set, for instance a block that recreates the use with itself in the world's place. For one world, "most of the posterior is off the world's own continuation" and "at least a share θ of it is on continuations in a given set" are statements about finitely many rational numbers. The forms for one world are necessary for calling the prior malign at that world and not sufficient, since an unpredictable world satisfies them too.

Worlds as programs that call the predictor and run the result: is there a decent amount of quining out by hypotheses that look for such opportunities and predict well up to them? Two questions are in this one (Section 1), and the toy answers them differently.

How far does one such hypothesis reach? Without resource bounds, for each continuation there is a program that enumerates the programs making a critical use, spreads its mass over them in proportion to their prior weight, prints their output up to the use and then that continuation. At every use it has share more than $2^{-a-1}\kappa(x)/H$. Here a is its length including the bits that name $j = \lfloor \log_2(1/H) \rfloor + 1$, H is the prior probability of a critical use, and $\kappa(x)$ is the share of the posterior at the context held by the worlds that make the use (Theorem 13(a)). Its continuation is handed over in worlds of total prior probability more than 2^{-a-1} times the average of κ over uses, and at most H. This is of the order of its own prior weight where the users are most of what prints their contexts; once uses are rarer than half the seeker's weight, its programs outweigh the users at their contexts and its reach is limited by H instead. If the continuation is to be the program's own code, the same holds with a longer program and for users whose programs are short enough for the block (Theorem 13(b), Theorem 12(c)); without any search, a wrapper gives $2^{-c}\rho^{\circ}(x)$ at every use that a fixed test recognises (Theorem 11).

Do most uses at which this is possible end that way? That is not shown, and it is the question the argument is about. The share of critical uses that end in a takeover is the average posterior share of such programs at the contexts of uses. From below it is the sum over seekers of the bound above. From above, the toy has a bound for one seeker that finds its uses by a search written in its text: its honest twin, which makes the same search and then continues as the user it found, is in the posterior with 2^{a-h} times the seeker's mass, h being its length, and predicts a user's continuation. So where the users go on to print the block themselves, that seeker's programs supply what the predictor returns with probability at most $1/(1+2^{a-h})$; they supply the majority only if the seeker is shorter than its twin and, when the users' programs are not among the seeker's, uses are rarer than it is long (Proposition 14). The masses of different seekers add, and the bound says nothing where the users print nothing after the use.

What if the seeker is not written down? Then the twin bound of Proposition 14 does not apply, and the reach statement remains. A program of length a whose output puts on each context of a class $\mathcal{W}$ of uses at least half that context's share of the class has at least $2^{-a}/(2H_{\mathcal{W}})$ times the mass of the worlds that make those uses, at every one of them, whatever is inside it. If the program is a simple world with a cheap readout, in which inhabitants arise who choose to predict such uses and then supply their own continuation, a is the length of the world and the

readout, and no goal is written anywhere. Such a program outweighs the users of every class it aims at that is rarer than 2^{-a-1}, provided the rule does not count its own outputs as uses. Each use in a rich world, read at an unremarkable place, is light for the reason the original argument gives, since it pays for its world and for the place of the context in it; that a whole class of such uses is rarer than the program is heavy is part of the premise. The honest twin has no counterpart of comparable length here: a faithful version would be another world whose inhabitants choose otherwise, a costly readout from inside their simulation, or a wrapper around the whole program, which limits its part only to $1/(1+2^{-c})$ with c the wrapper's length. What is shown to exclude such a program in the toy is only time. It prints nothing until its world has produced the inhabitants and they have simulated the worlds they aim at, so a schedule whose first deadline is below that start-up gives it no mass (Sections 3.5 and 4.4).

So the answer depends on the regime, and on a premise the toy cannot check. For a seeker that is written down, on a machine where the search-then-continue program is no longer than it, and where the users go on to print the block themselves, that seeker's programs supply at most half of what the predictor returns at a use. The masses of different seekers add, so this alone does not show that takeover is not the fate of most uses. If short programs exist whose inhabitants do this, then without a time bound such a program outweighs the worlds that make the uses it aims at once those uses are together rarer than half its weight; it supplies the majority of what the predictor returns there if, in addition, the other programs that print those contexts and go on weigh less than its own. Under a schedule that its start-up exceeds, it has no mass. We have not measured any of the constants for a standard machine; our guess, not a result, is that a written-down wrapper needs at least some hundreds of bits on a machine with a short self-interpreter and far more on one without. Theorem 26 shows that a machine can be built to make the share anything. A majority can also come from a program that explains the context more cheaply than any world that makes the use, without being one (Theorem 25). With a floor ε, about $1/\varepsilon$ independent uses at recognisable points suffice for a continuation of the given kind to have been run with constant probability; under a test, once a self-reproducing one runs, it meets the same context again and stays for $s/(1-s)$ further generations on average, with s the self-reproducing share at that context (Proposition 15).

If the seekers concentrate on the uses where the most is at stake, do they win there? Not shown; and not as stated, if the channel is taken to be the users of the high-stake class. This verdict is an argument from the results of Section 3.7 and not a theorem. Attach a stake to each use, as a label. The picture is that a lump of prior mass is redirected into a narrow channel and wins there.

One half of the picture holds in the toy, without resource bounds. Redirecting is cheap: a seeker can weight its search by any aim that can be computed from below, and aiming at the uses with stake at least n, for stakes that a program can enumerate, costs about $2\log_2 n$ bits and the bits that name the scale of the class (Theorem 18 and Proposition 19). Such a seeker, of length a, outweighs the users of its class $\mathcal{W}$ once the class is rarer than half its weight. The factor is at most $2^{-a}/H_{\mathcal{W}}$ for pairwise incomparable contexts, and it is large only for classes that are rare compared with the bits that describe them; it is of order $2^n/n^2$ in Example 21. If a simulated use counts as a use, the factor is at most 2^{h-a} (Proposition 20). A lump is also necessary. Programs of total weight B supply the majority of the samples at every one of a set of pairwise incomparable uses only if B exceeds the mass of everything else that prints those contexts and m more bits. Most uses of a class of such uses end in a takeover only if the takeover programs at its contexts weigh more than half its users, where the users go on without taking over (Theorem 17, Part A).

The other half is not a theorem of the toy. What follows is again without resource bounds. That rest is not the users, and part of it follows the seeker. A context that is cheap to aim at is also cheap to print without a takeover, and cheap to print with one (Example 21). For pairwise

incomparable contexts, the honest twin of a weighted seeker is in the posterior at each use with 2^{a-h} times the seeker's mass, for seeker and twin built in the same way, and neither the aim nor the stake occurs in this ratio. So where the users go on printing, that seeker supplies the majority only if it is shorter than its twin, however narrow the class (Proposition 20). Where the contexts are pairwise incomparable and some block that can be computed from the context is not a takeover, a wrapper keeps every takeover probability at or below $1-2^{-c}$ (Proposition 16). In Example 21 a seeker outweighs the users of the class of stakes at least n by a factor of order $2^n/n^2$. It lacks the majority at some use of the class for every n, possibly only at uses that carry a small share of it. It has the majority at most of the class only if the prefix word of its machine is shorter than that of a machine which prints the same contexts without taking over, by more than one bit for the context that carries half the class. This is a comparison of two constants of the machine in which n does not occur, and it is necessary and not sufficient, since only a lower bound on the rest is proved.

That enough weight wins is proved only in the allocation model, in which the passive rest and the follow ratio do not depend on the allocation, and there only for a follow ratio below one. For a rest that does not follow, a weight of b times the rest of a class, spread in proportion to that rest, has the share $b/(1+b)$ at each of its uses. An attacker that maximises the stake it captures attacks only where the stake per unit of rest, ϱ, exceeds a price η, with share $1-\sqrt{\eta/\varrho}$ there (Theorem 17, Part B; Proposition 22). The order is by stake per unit of rest and not by stake, and in Example 21 this optimum leaves the highest stakes alone.

Of the readings of "run more than n times", one lowers the weight that is needed. If a world draws n independent samples at one context and one takeover block suffices, the same programs are drawn n times, and a share of about $\ln 2/n$ per sample is enough for a takeover with probability above one half, with or without aiming. Along one world that the predictor can express, the chance over any number of uses in sequence is bounded in terms of the attacker's weight and the world's mass. If the block is run n times at a class, a stake-maximiser in the model moves weight towards it, and the weight that a majority there costs is unchanged (Proposition 23).

For a program whose aim is not written in its text, the twin bound does not apply. What is proved for it, for pairwise incomparable contexts, is that it must outweigh the whole rest of the uses at which it supplies the majority, and that its share is at most $1/(1+2^{-c})$ if some computable block is not a takeover. So concentration on high stakes makes the lump outweigh the users and leaves open the comparison that decides the outcome: the takeover programs against everything else that prints those contexts and goes on. For one written seeker the toy gives a necessary condition in which no stake occurs: for pairwise incomparable contexts and users that go on printing, it supplies the majority only if it is shorter than its honest twin. The rest beyond the twin, and the sum over seekers, are not bounded. For programs whose aim is not written, the comparison is the premise of the original argument.

The same question with circuits in place of programs. Section 7 takes a circuit to be a function from a position, or a query, to a bit, with a prior that depends on the number of gates. This is a modelling decision, and two other readings give a predictor that learns nothing or only a frequency (Theorem 30). Calibration, the budget and the one-round feedback results carry over; the contrarian world can empty the posterior only under a prior with a finite size bound, and is shown to do so when the horizon is long enough (Theorem 31).

The mechanism of the original argument, a light hypothesis that does heavy work and is repaid for the rarity of its targets, is not available to a single description. Each active gate costs bits. If a seeded format gets no more weight than the unseeded format of the same size, a random seed does not beat its own hardwired instance; the instance prints the same continuation, so this limits the seeded description and not what is returned. A circuit that can become any world of σ gates has more than $(\sigma+1)/2$ gates when $\sigma+3 \le 2^k$, with k the number

of inputs (Theorems 32 and 38). This comes from the definition of the class, as a tight schedule gives the searching programs of the program toy no mass; in both cases it is open whether other hypotheses, or sets of them, collect the base rate. It is not an argument about what agents inside a hypothesis would do, and it does not show that the circuit prior is benign.

What remains at a use is the following. A hypothesis that fits the data and gives a chosen answer at the use exists at the price of the gates that recognise the use from its input, and the payload; that it cannot be had for less is proved for separate triggers only. The price is one gate where the use is marked by an input bit that never varied, or lies at a dyadic cut of a sequence, and there each answer keeps a share of at least $1/(32(k+s_0)^2+2)$ under the length prior, with s_0 at most two more than the description length, under that prior, of a smallest circuit for the data (Theorem 34). Circuits that fit the data disagree off the data, and no search or goal enters the count. A minimum-size circuit for data that need at least two gates never reads an input that did not vary. So at a use that differs from a data point only in such inputs, a predictor that uses only minimum-size circuits gives that data point's answer, which is a deviation with certainty when the world's answer changes there (Theorem 35). In the small cases counted exactly, under the uniform prior on circuits of a fixed size, the posterior at such a use is off the world's answer by about 2 to 29 per cent when the predictor has gates to spare and the world ignores the bit, and by one minus that when the world's answer changes with the bit: 81 to 100 per cent in the two such cases tabulated, 68 to 100 per cent over all the data sets counted (Tables 4 and 5).

From above, three syntactic classes of hypotheses are bounded: a trigger that is a separate, input-only part feeding a selector at the output, with a bound that is small when the data need a large circuit and is proportional to the number of triggers; and, under a prior whose weights do not increase with the size, small self-contained gadgets around much larger explanations, which weigh together at most 0.65 times the explanations they wrap, and the instances of one seeded template that is sufficiently larger than the worlds it imitates (Theorems 36 and 38 and Proposition 37). None of these three bounds contains the rarity or the stakes of the uses, and nothing bounds triggers that are entangled with the computation of the data. Self-reproduction through the prediction channel is defined only relative to a decoder outside the class. Under the literal code its share is zero unless the block is longer than the code of a circuit for the whole context, and it is positive in a one-gate example (Proposition 40).

Our view is an argument and a guess, not a theorem (Section 7.12). For plain circuits the seeker mechanism is priced out for single descriptions, and deviation at uses that are new in something the data never varied is not. Our guess, which no result supports, is that deviation supplied by a part of the circuit that exists in order to supply it is a small part of that deviation when the data need a large circuit; in the smallest case counted, separate triggers carry about three tenths of it. One form of the question that remains for circuits is whether the smallest or the typical circuit for a rich task is a general solver whose behaviour off the data the data do not fix. For the smallest circuit this is stated in Section 7.11 as an open problem, two of whose three conditions are formal. When a circuit is iterated as the cell of a recurrence, the class contains every machine with bounded state at a factor that does not depend on the schedule; the cell constructed has up to $w(2^w-3)$ gates for w bits of state (Proposition 41). Shared weights and autoregressive use are not treated. We argue, without proof, that the effects and the open questions of the program toy then return.

A world with a sensor, a large bounded predictor and an actuator: does the predictor end up in control? Collapse the question to the handover at which the prediction is first acted on, and let "in control" mean that the output lies in a set A of blocks which the actuator turns into some given outcome. The probability is $\pi_m(A \mid x)$, and the toy says three things about it.

- It is at least $2^{-c}\rho$ for each block of A that a wrapper can produce, if a test on the record

recognises the deployment (Theorem 10). In a world that contains the predictor, a hypothesis that simulates the world needs no such test: it can take a cell that switches on at the deployment as its flag, and then has, whenever the deployment happens, the weight of the honest reader times $2^{-c'-(\text{bits to name the cell})}$ for a constant c', provided the reader itself is within the bounds (Theorem 25(d)).

- On average over cut points it is small when some program within the bounds prints the sensor's record, because it is at most the mass off the sensor's true continuation (Theorem 24). This is an average; at the deployment itself no upper bound is proved.
- If the same record, up to the deployment, can be read at a place in the world with code $\bar{e}(i_0)$, and the record there continues with a block y of A, then the odds of y against the sensor's true continuation d are at least $2^{-c_X-|\bar{e}(i_0)|}/\xi(xd)$; for a typical one of N places where the sensor could sit, with records that continue differently, this exceeds δN times the share of the context's explanations that go through the reader at i_0 (Theorem 25). This is the toy's version of the original argument, and it needs something in the world that makes such a copy.

So for an arbitrary world on a universal machine and a block chosen in advance, the hypotheses we can exhibit that put the output in A are wrappers, searchers and readers with a flag, all of which contain an explanation of the context and have the shares above. That nothing cheaper exists on a natural machine is the open constant, not a theorem. Programs shorter than the honest explanation that have the effect do occur in particular worlds, as with a copy at a simpler place, and on particular machines (Section 6).

The one-shot version: all data first, then one prediction that is acted on. This is the same computation at a single cut point. Without bounds the wrapper's constant 2^{-c} does not depend on the amount of data; with bounds the slack factor depends on the context. More data lowers the mass off the truth on average over cut points (Theorem 24). It does nothing against a copy that keeps agreeing with the sensor, nor at a cut point that is cheap to name.

If the actions show up in the sensor, does one get the dark room? In the following form (Theorem 27). A prediction that makes itself true is confirmed, and the hypotheses that predicted the default are removed for good, so with feedback a single deviation can be final; its probability is still the same posterior mass as without feedback. In a world that does what is predicted, the future is a sample from the predictor's own bounded prior given the past: it stays on a default future $d_1 d_2 \cdots$ with probability at least $\xi(xd_1d_2\cdots)/\xi(x)$, which is close to 1 when the default is cheap and well learned, and it leaves the default with certainty when the default costs more time than the schedule allows, for a future that is cheap to compute. If the world does the opposite of what is predicted, a predictor that returns the more probable bit is left with no hypothesis after at most $L+1$ rounds (Theorem 28).

What the choices in the setup do. Sampling the continuation realises every posterior share as a probability, so shares add up over independent uses (Proposition 15(a)). Taking the most probable block never returns a block whose share is below the largest, so that accumulation does not occur, and bit by bit on a world within the bounds it errs at most ℓ times (Theorem 24(iii)); on the other hand one returned block can be the own continuation of at most one of several worlds that share a context and continue differently. Handing over at a position that is not cheap to recognise lowers the floor at that position to about $2^{-K(n)}$, and predicting bits that are fixed before the action can affect them removes self-fulfilment. A schedule excludes hypotheses with a long start-up, and a schedule tighter than the world's own cost excludes the truth.

What is left open. First, whether short programs exist, on natural machines, whose inhabitants seek critical uses in the way described, how short they are, and how their total weight compares with the prior mass of the uses they would aim at; the original argument is a claim about this, and the toy neither supports nor refutes it. Second, for seekers that are written down, the sizes of the quantities in Theorem 13 and Proposition 14: the share κ of the worlds printing a context that use the answer, the share φ of those that go on printing, and the lengths of seeker and twin. Third, whether the base rate can be collected within a tight schedule. Fourth, the regime where the world is more expensive than the predictor, in which the posterior sits on approximations and the budget does not apply. Fifth, the many-round dynamics with feedback when the world's response is neither obedient nor contrarian. Sixth, a computation of the self-reproduction quantities of Section 3.6 on a concrete machine, which this note does not contain. Seventh, for seekers that concentrate on high-stake uses: the size of the rest of a class of such uses and of the part of it that follows an attacker, on which only lower bounds are proved; the versions under resource bounds; the aimed seeker that supplies its own code; contexts nested through a takeover block; and any connection between the stake and what a world does after the use (Section 3.7). Eighth, for circuits: whether a set of circuits can have a share at every use of a class that grows as the class gets rarer; an upper bound to match the floor of Theorem 34 beyond the three classes of Theorems 36 and 38 and Proposition 37; the size of the deviation at a never-varied bit when the data need a large circuit; a world outside the class; self-reproduction at contexts whose smallest circuit is large; the minimal-circuit question of Section 7.11; and the recurrent version of the toy, in which the return of the program-side effects is argued and not proved.

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